

Assessment Scheme

Class tests Best two of three.

Weight 10 + 10

Midterm 20

Semester 60

Generating Functions for Sequences

Suppose you have a sequence $\{a_n : n \geq 0\}$.

Define

$$A(s) = \sum_{n=0}^{\infty} a_n s^n.$$

The radius of convergence is

$$R = \frac{1}{\limsup_{n \rightarrow \infty} |a_n|^{1/n}}.$$

Note: if $|a_n| \leq Mr^n$ for some $M > 0$, $r > 0$, then $R \geq 1/r$.

Example

Example:

$A_n : n \geq 0$

$$A(s) = \frac{1}{1-s}, \quad |s| < 1$$

Remark:

$$A(s) = \frac{1}{(1-s)^2}, \quad |s| < 1$$

Probability Generating Function

X taking values in $\{0, 1, \dots\}$

$$P_X(s) = \sum_{m=0}^{\infty} P(X = m) s^m$$

— generating function of $\{P(X = m)\}$

Properties of PGF

$P_X(s)$ is infinitely differentiable on $|s| < 1$.

$$P_X^{(k)}(s) = \sum_{m=k}^{\infty} m(m-1)\cdots(m-k+1)p_m s^{m-k}$$

Put $s=0$:

$$P_X^{(k)}(0) = k! p_k$$

Distribution of X is determined by P_X .

Abel's Lemma

[label=)]

1. If $\sum a_k < \infty$, then

$$\lim_{s \rightarrow 1} A(s) = \sum_{n=0}^{\infty} a_n$$

2. If $a_n \geq 0$, then

$$\lim_{s \rightarrow 1} A(s) = \sum_{n=0}^{\infty} a_n$$

PGF Properties and Moments

$$\lim_{s \rightarrow 1} P_X(s) = \sum_{m=0}^{\infty} p_m = 1, \quad P_X(1) = 1.$$

$$E(X(X-1)(X-2)\cdots(X-k+1)) = \sum_{m=0}^{\infty} m(m-1)(m-2)\cdots(m-k+1)p_m,$$

$$E(X) = \lim_{s \rightarrow 1} P'_X(s) = P'_X(1).$$

Concentrate on $[0,1]$

$$P'(s) = \sum_{m=0}^{\infty} m p_m s^{m-1}$$

Increasing (strictly if $p_0 < 1$).

$$P''(s) = \sum_{m=0}^{\infty} m(m-1)p_m s^{m-2}$$

is convex (strictly if $p_0 < 1$).

Tail Probabilities and Generating Functions

$$q_m = P(X > m)$$

$$Q(s) = \sum_{m=0}^{\infty} q_m s^m$$

$$\begin{aligned} q_m - q_{m+1} &= P(X > m) - P(X > m+1) \\ &= P(X = m+1) \quad \forall m \geq 0. \end{aligned}$$

$$\begin{aligned} q_{m+1}^{(S)} - q_m^{(S)} &= P_{m+1} \\ &= \sum P_{m+1} \quad (\text{sum over } m) \end{aligned}$$

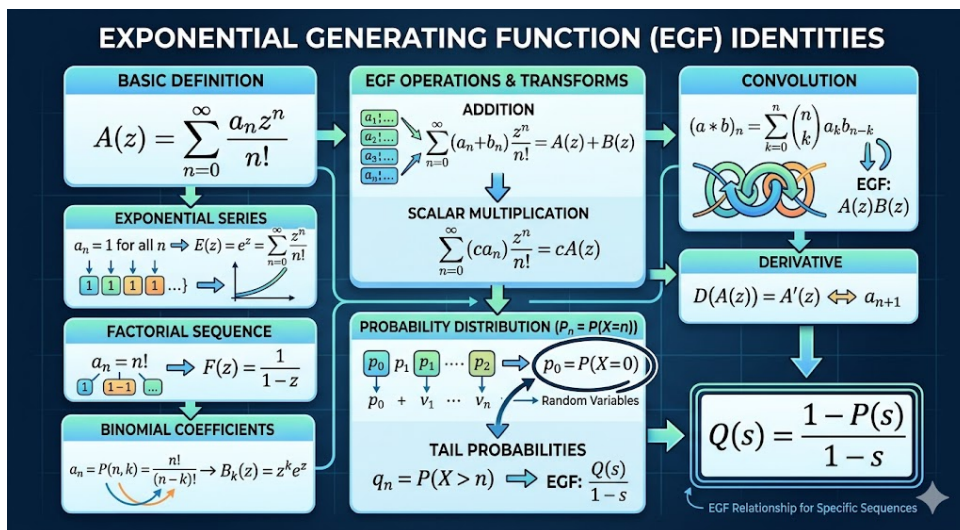
Relations for Generating Functions

$$\sum_{m=0}^{\infty} a_m \frac{s^m}{m!} - \sum_{m=0}^{\infty} a_m \frac{s^m}{m!} = \sum_{m=0}^{\infty} b_m \frac{s^m}{m!}$$

$$sQ(s) - (Q(s) - 1) = P(s) - p_0$$

$$Q(s)(s-1) = P(s) - p_0$$

$$Q(s) = \frac{1 - P(s)}{1 - s}$$



$$Q(s) = P'(c_s), \quad c_s \in (s, 1).$$

As $s \uparrow 1$, we have $c_s \uparrow 1$. Hence,

$$\lim_{s \uparrow 1} Q(s) = \lim_{s \uparrow 1} P'(c_s) = P'(1) = \mathbb{E}(X).$$

Therefore,

$$Q(1) = \lim_{s \uparrow 1} Q(s) = \sum_{n=0}^{\infty} q_n.$$

So,

$$\mathbb{E}(X) = \sum_{n=0}^{\infty} q_n = \sum_{n=0}^{\infty} \mathbb{P}(X > n).$$

Generating Function

For a sequence $\{a_n\}$, the generating function is

$$A(s) = \sum_{n=0}^{\infty} a_n s^n.$$

Probability Generating Function

For a random variable X ,

$$a_n = \mathbb{P}(X = n).$$

Then the probability generating function is

$$P_X(s) = \sum_{n=0}^{\infty} \mathbb{P}(X = n) s^n.$$

Example

A coin is tossed n times. Find the probability that the number of heads is even.

Denote

$$E_i = \begin{cases} 1, & \text{if the } i\text{th coin toss is H,} \\ 0, & \text{otherwise,} \end{cases} \quad 1 \leq i \leq n,$$

with $\mathbb{P}(E_i = 1) = p$.

Define

$$S_n = \sum_{i=1}^n E_i, \quad S_0 = 0.$$

Let

$$P_n = \mathbb{P}(S_n \text{ is even}).$$

$$p_0 = 1.$$

$$p_1 = \mathbb{P}(E_1 \text{ is even}) = \mathbb{P}(E_1 = 0) = 1 - p.$$

$$\begin{aligned}
p_2 &= \mathbb{P}(E_1 + E_2 \text{ is even}) \\
&= \mathbb{P}(E_1 = 0, E_2 = 0) + \mathbb{P}(E_1 = 1, E_2 = 1) \\
&= \mathbb{P}(E_1 = 0) \mathbb{P}(E_2 = 0) + \mathbb{P}(E_1 = 1) \mathbb{P}(E_2 = 1) \\
&= (1 - p)^2 + p^2.
\end{aligned}$$

Recurrence Relation

$$S_{n+1} = E_1 + \cdots + E_{n+1} = S_n + E_{n+1}.$$

$$p_0 = 1.$$

$$\begin{aligned}
p_{n+1} &= \mathbb{P}(S_{n+1} \text{ is even}) \\
&= \mathbb{P}(S_{n+1} \text{ is even}, E_1 = 1) + \mathbb{P}(S_{n+1} \text{ is even}, E_1 = 0) \\
&= \mathbb{P}(S_{n+1} - S_1 \text{ is odd}, E_1 = 1) + \mathbb{P}(S_{n+1} - S_1 \text{ is even}, E_1 = 0). \\
p_{n+1} &= \mathbb{P}(S_{n+1} - S_1 \text{ is odd}) \mathbb{P}(E_1 = 1) + \mathbb{P}(S_{n+1} - S_1 \text{ is even}) \mathbb{P}(E_1 = 0) \\
&= \mathbb{P}(S_n \text{ is odd}) \mathbb{P}(E_1 = 1) + \mathbb{P}(S_n \text{ is even}) \mathbb{P}(E_1 = 0) \\
&= (1 - p_n)p + p_n(1 - p).
\end{aligned}$$

For $n = 0$,

$$p_1 = (1 - p_0)p + p_0(1 - p) = 1 - p.$$

$$S_n = E_1 + \cdots + E_n, \quad S_{n+1} - S_1 = E_2 + \cdots + E_{n+1}.$$

So $S_{n+1} - S_1$ has the same distribution as S_n .

$$X \stackrel{d}{=} Y \implies \mathbb{P}(X \in A) = \mathbb{P}(Y \in A), \quad \forall A.$$

If $E_1, \dots, E_{n+1}$ are i.i.d., then

$$E_1 + \cdots + E_n \stackrel{d}{=} E_2 + \cdots + E_{n+1}.$$

More generally, for a function $f : \mathbb{R}^k \rightarrow \mathbb{R}$,

$$X \stackrel{d}{=} Y \implies f(X) \stackrel{d}{=} f(Y).$$

Example: $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(u, v) = u + v$.

$$p_{n+1} = (1 - p_n)p + p_n(1 - p), \quad \forall n \geq 0.$$

Define the generating function

$$P(s) = \sum_{n=0}^{\infty} p_n s^n.$$

Then

$$\sum_{n=0}^{\infty} p_{n+1} s^{n+1} = \sum_{n=0}^{\infty} \left((1-p_n)p + p_n(1-p) \right) s^{n+1}.$$

So,

$$\begin{aligned} P(s) - 1 &= ps \sum_{n=0}^{\infty} s^n - ps \sum_{n=0}^{\infty} p_n s^n + (1-p)s \sum_{n=0}^{\infty} p_n s^n \\ &= \frac{ps}{1-s} - spP(s) + (1-p)sP(s). \end{aligned}$$

From

$$P(s) - 1 = \frac{ps}{1-s} - spP(s) + (1-p)sP(s),$$

we get

$$P(s) = \frac{ps - s + 1}{(1-s)(1-s+2ps)}.$$

If $p = \frac{1}{2}$,

$$P(s) = \frac{1 - \frac{s}{2}}{1-s} = \sum_{n=0}^{\infty} s^n - \frac{1}{2} \sum_{n=0}^{\infty} s^{n+1} = 1 + \sum_{n=1}^{\infty} \frac{1}{2} s^n.$$

Hence $p_0 = 1$ and $p_n = \frac{1}{2}$ for all $n \geq 1$. For $p \neq \frac{1}{2}$,

$$P(s) = \frac{1 - (1-p)s}{(1-s)(1+s(2p-1))} = \frac{A}{1-s} + \frac{B}{1+s(2p-1)}.$$

Compare coefficients:

$$1 - (1-p)s = A(1+s(2p-1)) + B(1-s).$$

So

$$1 = A + B, \quad p - 1 = A(2p-1) - B.$$

Hence $A = \frac{1}{2}$ and $B = \frac{1}{2}$.

Using

$$\frac{1}{1-\alpha s} = \sum_{n=0}^{\infty} \alpha^n s^n,$$

we get

$$P(s) = \sum_{n=0}^{\infty} A s^n + \sum_{n=0}^{\infty} B(1-2p)^n s^n.$$

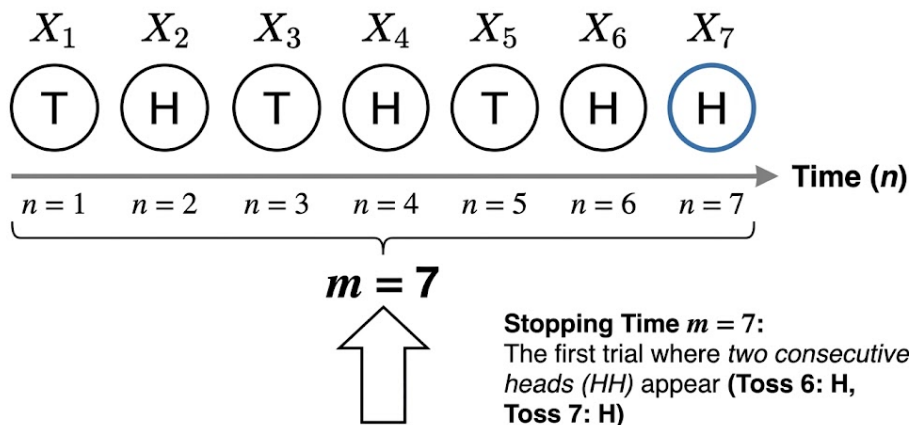
Therefore

$$p_n = A + B(1-2p)^n = \frac{1}{2} + \frac{1}{2}(1-2p)^n \xrightarrow{n \rightarrow \infty} \frac{1}{2}.$$

Coin Toss Until Two Consecutive Heads

Ex 2: Toss a coin until two consecutive Heads appears. Let p_n be the probability that the experiment stops after n tosses.

Coin Toss Sequence (Stopping Time for HH)



After n tosses:

$$\begin{aligned}
 E_n &= \text{Exp. stops at } n, \quad \text{fix } n \geq 3 \\
 p_0 &= 0, \quad p_1 = 0, \quad p_2 = p^2 \\
 P(E_n) &= P(E_n, E_1 = 1) + P(E_n, E_1 = 0)
 \end{aligned}$$

Probability Recurrence Relations

$$\begin{aligned}
 P(E_m) &= P(\varepsilon_1 = 0)P(E_{m-1}) \\
 &\quad + P(E_m, \varepsilon_1 = 1, \varepsilon_2 = 1) \\
 &\quad + P(E_m, \varepsilon_1 = 1, \varepsilon_2 = 0)
 \end{aligned}$$

$$P(E_m) = (1-p)p_{m-1} + p(1-p)P(E_{m-2})$$

$$p_m = (1-p)p_{m-1} + p(1-p)p_{m-2}, \quad m \geq 3$$

$$\sum_{k=0}^{\infty} p_k = 1, \quad p_k = 1 \quad (k=1), \quad \text{as } n \rightarrow \infty$$

$$P(T < \infty) \rightarrow 1 \implies P(T > m) \rightarrow 0, \quad P(T = +\infty) = 0$$

Probability Generating Functions

$$P(s) = \sum_{m=2}^{\infty} p_m s^m$$

$$\begin{aligned}
P(1) &= p_2 + \sum_{m=3}^{\infty} p_m \\
&= p_2 s^2 + \sum_{m=3}^{\infty} p_m s^m \\
&= p_2 s^2 + \sum_{m=3}^{\infty} (1-p)p_{m-1} s^m + p(1-p) \sum_{m=3}^{\infty} p_{m-2} s^m \\
&= p_2 s^2 + (1-p)s \sum_{m=3}^{\infty} p_m s^{m-1} + (1-p)p s^2 \sum_{m=3}^{\infty} p_m s^{m-2} \\
&= p_2 s^2 + (1-p)s P(s) + (1-p)p s^2 P(s) \\
\sum_{n=0}^{\infty} p_n &= P(1) = 1 \\
A(s) &= \sum_{n=0}^{\infty} a_n s^n, \quad A(1) = \lim_{s \rightarrow 1} A(s) = \sum_{n=0}^{\infty} a_n = 1
\end{aligned}$$

Waiting Times for HH and HT Patterns

Let T_{HH} be the number of tosses to get 2 consecutive H.

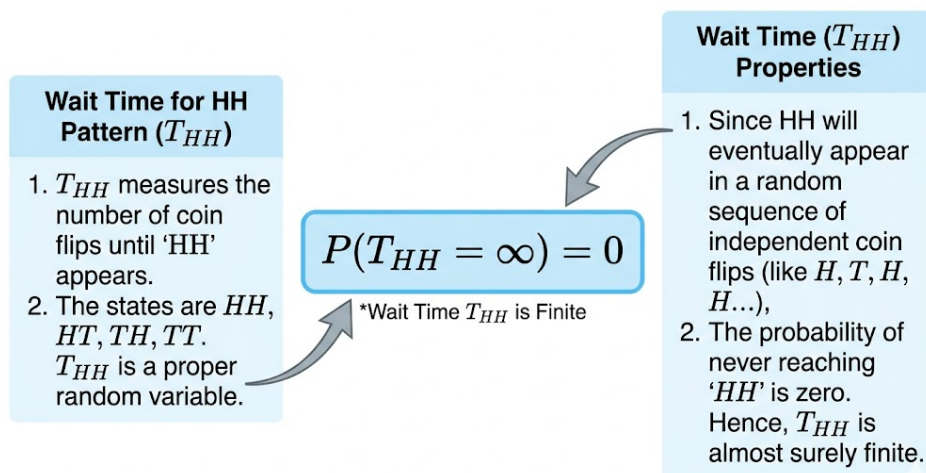
Then,

$$P(T_{HH} = n) = p_n$$

and T is a proper r.v.

Calculate

$$E(T_{HH}) = P()$$



Let T_{HT} be the number of tosses required to get T followed by H for the first time.

Expected Waiting Time for Heads

Left:

$$E(T_H) = 6$$

Imp

$$P(T_H = m)$$

Right:

$$E(T_H) = 4$$

(with sequence of heads and final T, $m \geq 3$)

$$= P(T_H = m, E = 0) + P(T_H = m, E = 1)$$

$$= q \cdot q^{m-1} + p^m q(1 - p)$$

where $q = (1 - p)$.

Probability Generating Functions

Gen. fn. : For X , non-neg int valued r.v.

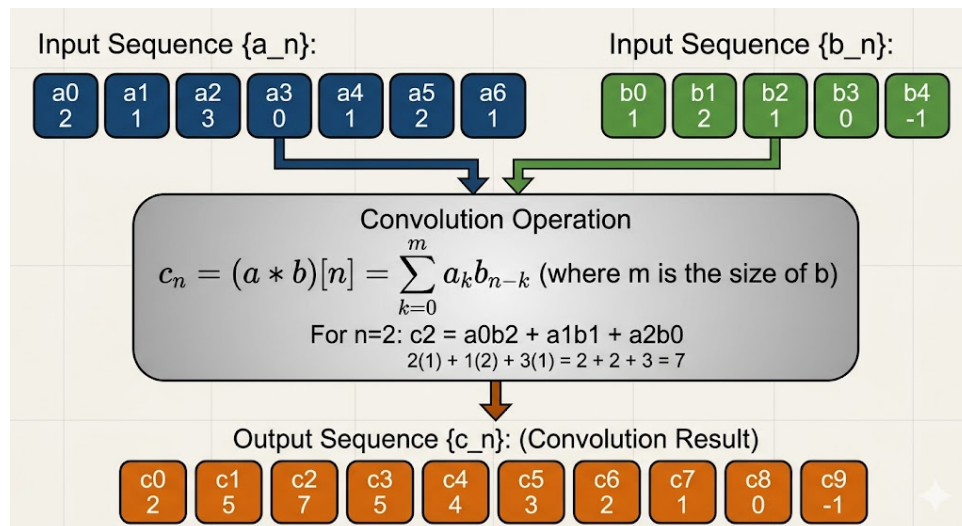
$$\begin{aligned} P_X(s) &= \sum_{n=0}^{\infty} p_n s^n = \sum_{n=0}^{\infty} P(X = n) s^n \\ &= E(s^X), \quad |s| < 1. \end{aligned}$$

X, Y indept.,

$$\begin{aligned} P_{X+Y}(s) &= E(s^{X+Y}) = E(s^X s^Y) \\ &= E(s^X) E(s^Y) \quad (\rightarrow \text{ind.}) \\ &= P_X(s) P_Y(s) \end{aligned}$$

Convolution of Sequences

$$\begin{aligned} P(X + Y = n) &= \sum_{k=0}^m P(X = k, Y = n - k) \\ &= \sum_{k=0}^m P(X = k) P(Y = n - k) \\ &= \sum_{k=0}^m P(Y = k) P(X = n - k) \end{aligned}$$



$\{c_n\}$ is the convolution of

Theorem on Generating Functions

Thm: Let $A(s)$ & $B(s)$ be the generating functions of $\{a_n\}$ and $\{b_n\}$ resp. Assume that the radius of convergence is at least R for both $A(s)$ & $B(s)$. Then, the convolution sequence $\{c_n\}$ has a radius of convergence at least R and the generating function is given by $A(s)B(s)$.

Proof: Bound on $|c_n|$

Pf: Fix $\epsilon > 0$, then

$$\limsup_{n \rightarrow \infty} |a_n|^{1/n} \leq \frac{1}{R} + \epsilon$$

$$\limsup_{n \rightarrow \infty} |b_n|^{1/n} \leq \frac{1}{R} + \epsilon$$

where $A, B > 0$ is a constant. Hence $\exists N$ such that $\forall n \geq N$,

$$|a_n| \leq A \left(\frac{1}{R} + \epsilon \right)^n, \quad |b_n| \leq B \left(\frac{1}{R} + \epsilon \right)^n.$$

Then

$$\begin{aligned} |c_n| &= \left| \sum_{k=0}^n a_k b_{n-k} \right| \leq \sum_{k=0}^n |a_k| |b_{n-k}| \leq \sum_{k=0}^n A \left(\frac{1}{R} + \epsilon \right)^k \cdot B \left(\frac{1}{R} + \epsilon \right)^{n-k} \\ &\leq AB \left(\frac{1}{R} + \epsilon \right)^n (n+1). \end{aligned}$$

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$$|c_n|^{1/n} \leq A^{1/n} B^{1/n} (n+1)^{1/n} \left(\frac{1}{R} + \epsilon \right).$$

$$\implies \limsup_{n \rightarrow \infty} |c_n|^{1/n} \leq \left(\frac{1}{R} + \varepsilon \right).$$

Since $\varepsilon > 0$ is arbitrary,

$$\limsup_{n \rightarrow \infty} |c_n|^{1/n} \leq \frac{1}{R}.$$

Compare the coefficients of $C(s)$ and $A(s)B(s)$ on $|s| < R$:

$$\left(\sum_{k \geq 0} a_k s^k \right) \left(\sum_{k \geq 0} b_k s^k \right) = \sum_{n \geq 0} c_n s^n, \quad c_n = \sum_{k=0}^n a_k b_{n-k}.$$

Example

$a_0 = 0$, $a_1 = 1$. For $k \geq 2$,

$$a_k = a_0 k + a_1 (k-1) + \cdots + a_{k-1}.$$

Find a_n .

Let $b_n = n$ and define

$$c_n = \sum_{k=0}^n b_k a_{n-k}.$$

This is the convolution of $\{b_k\}$ and $\{a_k\}$.

$$c_0 = b_0 a_0 = 0, \quad c_1 = b_0 a_1 + b_1 a_0 = 0.$$

$$C(s) = \sum_{m=0}^{\infty} c_m s^m.$$

Since $a_m = c_m$ for $m \geq 2$,

$$A(s)B(s) = \sum_{m=0}^{\infty} c_m s^m = \sum_{m=2}^{\infty} a_m s^m.$$

But

$$\sum_{m=2}^{\infty} a_m s^m = A(s) - a_1 s = A(s) - s.$$

Therefore,

$$A(s)B(s) = A(s) - s \implies A(s)(1 - B(s)) = s \implies A(s) = \frac{s}{1 - B(s)}.$$

$$P_X(s) P_Y(s) = P_{X+Y}(s).$$

If $X \sim \text{Ber}(p)$, then

$$P_X(s) = q + ps.$$

If $Y \sim \text{Bin}(m, p)$, then

$$P_Y(s) = \left(P_X(s)\right)^m = (q + ps)^m.$$

Let $\{X_i\}$ be i.i.d. with generating function $P(s)$, and let N be another independent r.v. having generating function $G(s)$. Define the random sum

$$S_N = \begin{cases} 0, & \text{if } N = 0, \\ \sum_{i=1}^m X_i, & \text{if } N = m \geq 1. \end{cases}$$

$$\mathbb{E}(s^{S_N}) = \sum_{n=0}^{\infty} \mathbb{E}(s^{S_N} \mathbf{1}_{(N=n)}) = \sum_{n=0}^{\infty} \mathbb{E}(s^{X_1+\dots+X_n} \mathbf{1}_{(N=n)}) = \sum_{n=0}^{\infty} \mathbb{E}(s^{X_1+\dots+X_n}) \mathbb{P}(N = n).$$

Equivalently,

$$\mathbb{E}(s^{S_N}) = \mathbb{E}(\mathbf{1}_{(N=0)}) + \sum_{n=1}^{\infty} \mathbb{E}(s^{X_1+\dots+X_n}) \mathbb{P}(N = n).$$

$$\mathbb{E}(s^{S_N}) = \mathbb{P}(N = 0) + \sum_{n=1}^{\infty} \mathbb{E}(s^{X_1+\dots+X_n}) \mathbb{P}(N = n) = \mathbb{P}(N = 0) + \sum_{n=1}^{\infty} (P(s))^n \mathbb{P}(N = n) \quad (\text{use } P(1) = 1).$$

$$P_{S_N}(s) = G(P(s)).$$

$$P'_{S_N}(s) = G'(P(s)) P'(s).$$

Let $s \uparrow 1$, then

$$\mathbb{E}(S_N) = \mathbb{E}(N) \mathbb{E}(X_1),$$

which is Wald's equation.

Fixed Points of Probability Generating Functions

We study fixed points of probability generating functions in $[0, 1]$.

Let $f : [0, 1] \rightarrow [0, 1]$. A point $x \in [0, 1]$ is called a fixed point of f if

$$f(x) = x.$$

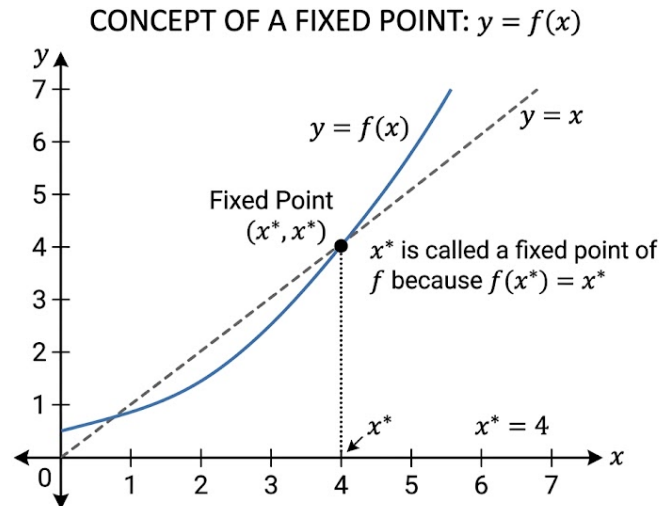
For a probability generating function

$$P(s) = \sum_{k=0}^{\infty} p_k s^k,$$

we always have

$$P(1) = \sum_{k=0}^{\infty} p_k = 1.$$

Hence $s = 1$ is always a fixed point of P .



Convexity of a PGF

First exclude the trivial case

$$p_1 = 1.$$

In that case,

$$P(s) = s,$$

so every $s \in [0, 1]$ is a fixed point.

Now assume

$$p_1 < 1.$$

Since

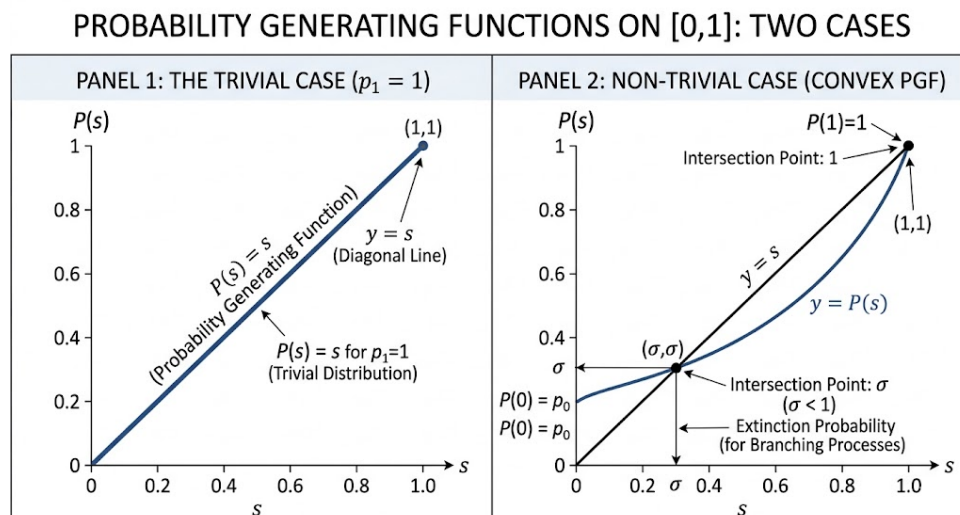
$$P''(s) = \sum_{k=2}^{\infty} k(k-1)p_k s^{k-2} \geq 0,$$

the function P is convex on $[0, 1]$. Also,

$$P(1) = 1.$$

Thus fixed points of P are exactly the intersections of the curves

$$y = P(s) \quad \text{and} \quad y = s.$$



Case 1: One Fixed Point

Assume

$$P(s) > s \quad \text{for all } s \in [0, 1).$$

Since $P(1) = 1$, this gives

$$1 - P(s) < 1 - s, \quad 0 \leq s < 1.$$

Hence

$$\frac{1 - P(s)}{1 - s} < 1.$$

By the mean value theorem, for each $s \in [0, 1)$ there exists $\sigma_s \in (s, 1)$ such that

$$\frac{P(1) - P(s)}{1 - s} = P'(\sigma_s).$$

Therefore

$$P'(\sigma_s) < 1.$$

Letting $s \uparrow 1$ gives $\sigma_s \uparrow 1$, and hence

$$P'(1) \leq 1.$$

Since

$$P'(1) = \mathbb{E}(X),$$

the one-fixed-point case implies

$$\mathbb{E}(X) \leq 1.$$

Case 2: More Than One Fixed Point

Suppose first that

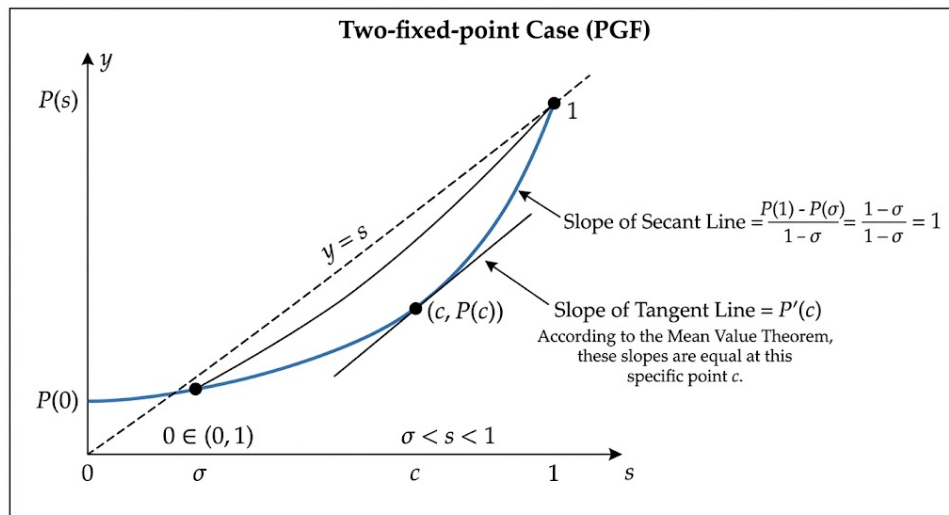
$$p_0 = 0.$$

Then

$$P(0) = p_0 = 0,$$

so 0 is a fixed point. Since $P(1) = 1$, the pgf has at least two fixed points. If $p_1 < 1$, then there is some $k \geq 2$ with $p_k > 0$, and therefore

$$\mathbb{E}(X) = \sum_{k=0}^{\infty} k p_k > \sum_{k=0}^{\infty} p_k = 1.$$



Now suppose there is a fixed point

$$\sigma \in (0, 1), \quad P(\sigma) = \sigma.$$

Then

$$1 - P(\sigma) = 1 - \sigma.$$

By the mean value theorem, for some $c \in (\sigma, 1)$,

$$\frac{P(1) - P(\sigma)}{1 - \sigma} = P'(c).$$

Since $P(1) = 1$ and $P(\sigma) = \sigma$, this gives

$$P'(c) = 1.$$

The next step is to show that P' is strictly increasing. Then

$$P'(1) > P'(c) = 1,$$

so

$$\mathbb{E}(X) = P'(1) > 1.$$

Strictly Increasing $P'(s)$

Need to show that $P'(s)$ is strictly increasing. If not, $P'(s) = p_1$.

$$\Rightarrow P(s) = p_0 + p_1 s$$

This implies that two lines $y = \lambda$, $y = p_0 + p_1 s$ (with $p_k = 0$) have exactly two intersections, not possible. Hence P' is strictly increasing.

$$P'(1) > P'(0) = 1$$

Fixed Point Theorem

Theorem. For $p < 1$, $P(s)$ has exactly one fixed point if and only if $\mathbb{E}(X) \leq 1$.

Galton-Watson Branching Process

Branching Process (Galton-Watson Process)

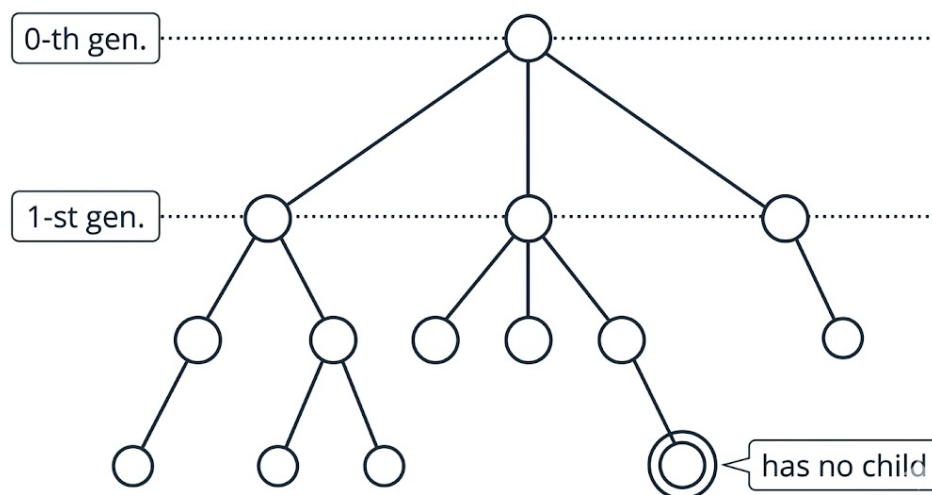
Z_n = total size of the n th generation.

Offspring distribution:

$$P(\xi = k) = p_k, \quad k = 0, 1, \dots$$

Offspring distribution.

At each generation, every individual generates offspring according to the offspring distribution given by ξ .



Example values:

- $\xi_1^{(0)} = 3$
- $\xi_1^{(1)} = 2$
- $\xi_2^{(1)} = 1$
- $\xi_3^{(1)} = 3$
- $\xi_1^{(2)} = 0$

Branching Process Notation

Let $\xi_i^{(m)} : m \geq 0, i \geq 1$ denote the number of offsprings born to the i th member of the m th generation.

The random variables $\{\xi_i^{(m)} : m \geq 0, i \geq 1\}$ are i.i.d.

Define Z_n as the size of the n th generation. In this example $Z_0 = 1, Z_1 = 3, Z_2 = 6, Z_3 = 8, \dots$

Family Extinction

Note.

$$Z_{m+1} = \sum_{i=1}^{Z_m} \xi_i^{(m)}$$

Question: Does this family become extinct w.p. 1?

i.e., for some $n, Z_n = 0$.

$$A = \{Z_n = 0 \text{ for some } n \geq 1\}$$

$$T = \inf\{n \geq 1 : Z_n = 0\}.$$

If $P(A) < 1$, then $P(T = \infty) > 0$.

Note. T = time of extinction of the family.

Branching Processes and Generating Functions

$$Z_{n+1} = \sum_{i=1}^{Z_n} \xi_i^{(n)}$$

Let P_n be the generating function of Z_n :

$$P_n(s) = \sum_{k=0}^{\infty} \mathbb{P}(Z_n = k) s^k$$

Let G be the pgf of ξ , i.e.,

$$G(s) = \sum_{k=0}^{\infty} p_k s^k$$

Then

$$P_{n+1}(s) = P_n(G(s))$$

For the expectation,

$$\mathbb{E}(Z_{n+1}) = \mathbb{E}(Z_n) \mathbb{E}(\xi) = \mathbb{E}(Z_{n-1}) (\mathbb{E}(\xi))^2 = \dots = \mathbb{E}(Z_0) (\mathbb{E}(\xi))^n$$

Left side notes:

- X_i gen. fn. R
- N gen. fn. Q
- $S_N = \sum_{i=1}^N X_i$
- $P_{S_N}(s) = Q(R(s))$

Moments of Z_n

Let $m = E(\xi)$.

Then,

$$E(Z_{n+1}) = m$$

Assume $V(\xi) = \sigma^2$. Compute $\text{Var}(Z_n)$.

In general, expression of P_n is difficult to obtain.

Iterated Probability Generating Functions

$$P_{m+1}(s) = P_m(G(s)) = P_{m-1}(G(G(s)))$$

$$\left\{ \begin{array}{l} P_0(s) = \sum_{k=0}^{\infty} P(Z_0=k) s^k \\ \quad = P(Z_0=0)s^0 + P(Z_0=1)s^1 + P(Z_0=2)s^2 + \dots \end{array} \right. \quad \begin{array}{l} \text{Assuming } P(Z_0=1)=1 \\ \quad = 0s^0 + 1s^1 + 0s^2 + \dots = 0 + 1 \cdot s + 0 + \dots \\ \quad = 1 \cdot s \\ \quad = s \end{array}$$

Probability $P(Z_0=k)$ is 1 for $k=1$ and 0 otherwise

Result: $P_0(s) = s$

$$P_0(s) = P_{m-2}(G(G(G(s)))) = \dots = P_0(G(G(\dots G(s) \dots)))$$

$$= G(G(\dots G(s) \dots)) \quad (m+1)\text{-fold composition}$$

$$= G(G(\dots G(s) \dots)) \quad (m+1)\text{-fold convolution}$$

m-Fold Composition

For any $m \geq 1$,

$$\begin{aligned} P_m(s) &= G\left(G\left(\cdots G(s) \cdots\right)\right) && \text{(m-fold composition)} \\ &= G\left(P_{m-1}(s)\right) \end{aligned}$$

$$\boxed{P_m(s) = G\left(P_{m-1}(s)\right) \quad \forall m \geq 1.}$$

Event A and Limiting Probability

B_n event

- $B_m \subset B_{n+1}$
- $P(\cup_n B_n) \geq \lim_{n \rightarrow \infty} P(B_n)$

$$A = \{Z_n = 0 \text{ for some } n\} = \bigcup_{n=1}^{\infty} A_n$$

$$A_n \subset A_{n+1}$$

$$P = P(A) = P\left(\bigcup_{n=1}^{\infty} A_n\right) = \lim_{n \rightarrow \infty} P(A_n) = \lim_{n \rightarrow \infty} P_n(0)$$

$$p = \lim_{n \rightarrow \infty} P_n(0)$$

$$A_n = \{Z_n = 0\}, \quad P(A_n) = P(Z_n = 0) = P_n(0)$$

$$\lim_{n \rightarrow \infty} P(A_n) = \lim_{n \rightarrow \infty} P_n(0)$$

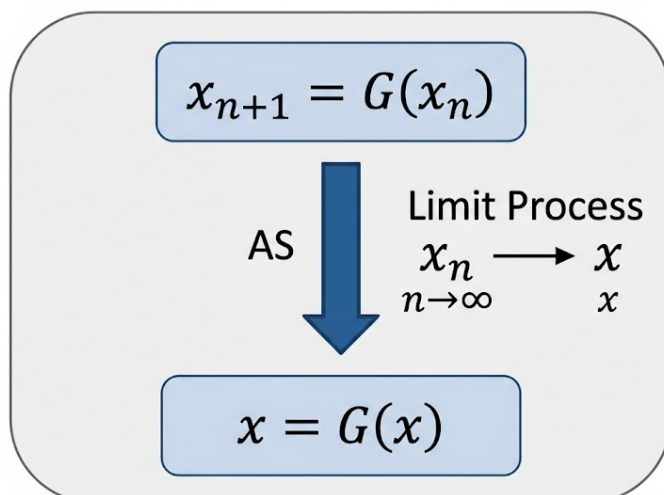
Fixed Point of Iteration

Note: $P_{n+1}(0) = G(P_n(0))$

Taking limit as $n \rightarrow \infty$,

$$\rho = G(\rho).$$

Thus, ρ is a fixed point of G .



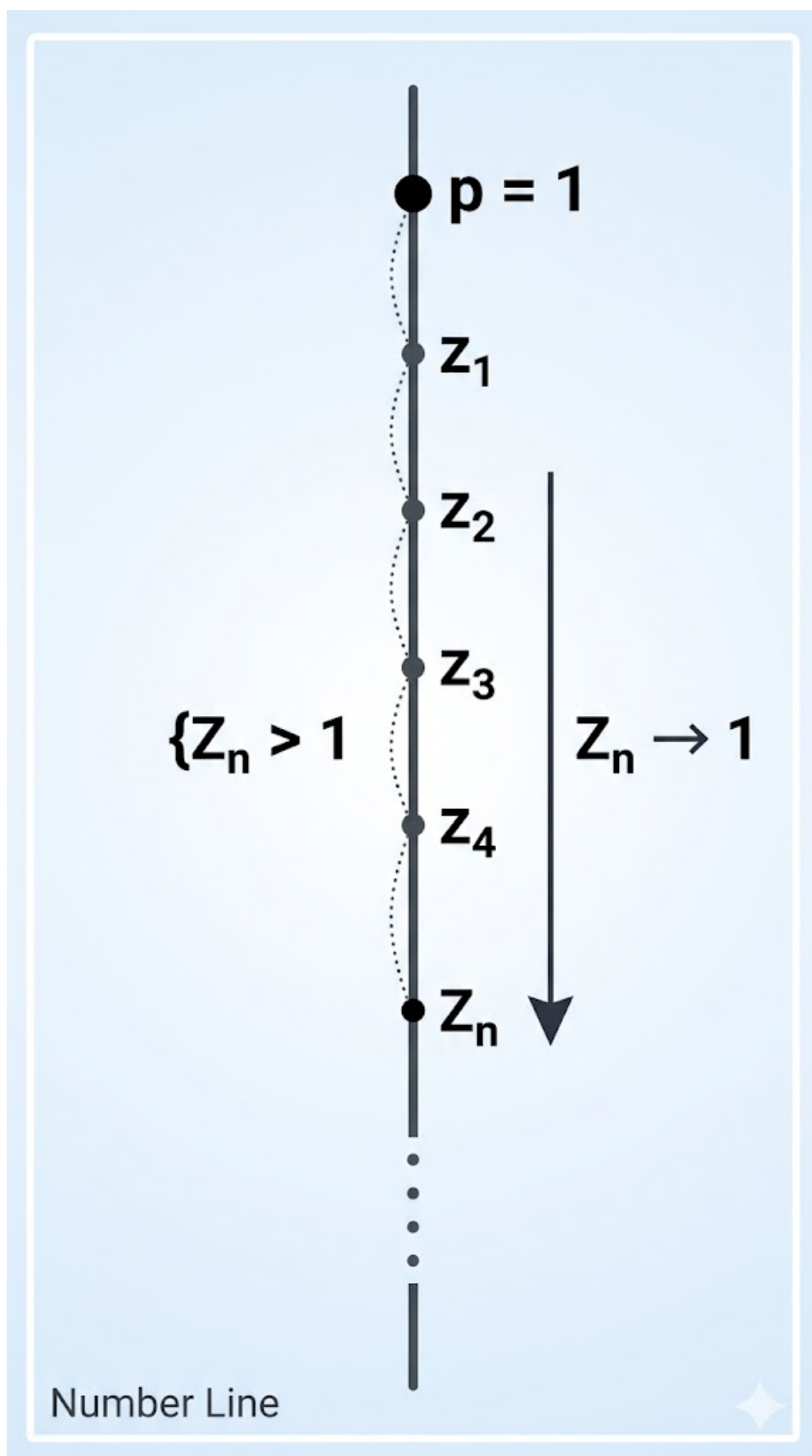
Fixed Point Analysis

Assume $p \neq 1$.

$G'(s) \geq 0$ on $[0, 1]$, i.e., G is non-decreasing.

Let τ be any fixed point of G in $[0, 1]$, i.e., $G(\tau) = \tau$.

$$\begin{aligned} \tau \geq 0 &\Rightarrow G(\tau) \geq G(0) \quad \text{for } \tau \in [0, 1] \\ &\Rightarrow \tau \geq G(0). \end{aligned}$$



Induction Proof for Iterated Composition

$$G(T) \geq G(G(0))$$

$$T \geq G(G(0))$$

By induction,

$$T \geq \underbrace{G(G(\dots G(0) \dots))}_{m\text{-fold composition}}.$$

Taking the limit,

$$T \geq \lim_{n \rightarrow \infty} P_n(0) = \rho.$$

(m -fold conn. Hence $u > 1$.)

Fixed Point Theorem

Theorem. The smallest fixed point p of G in $[0, 1]$ is given by

$\frac{Z_m}{Z_{>0}}$. G has one fixed point, namely $1 = 0$, if $G'(1) = m \leq 1$.

G has two fixed points $0, 1$ if $m = G'(1) > 1$.

Dice Rolling Outcomes

Start with a single die.

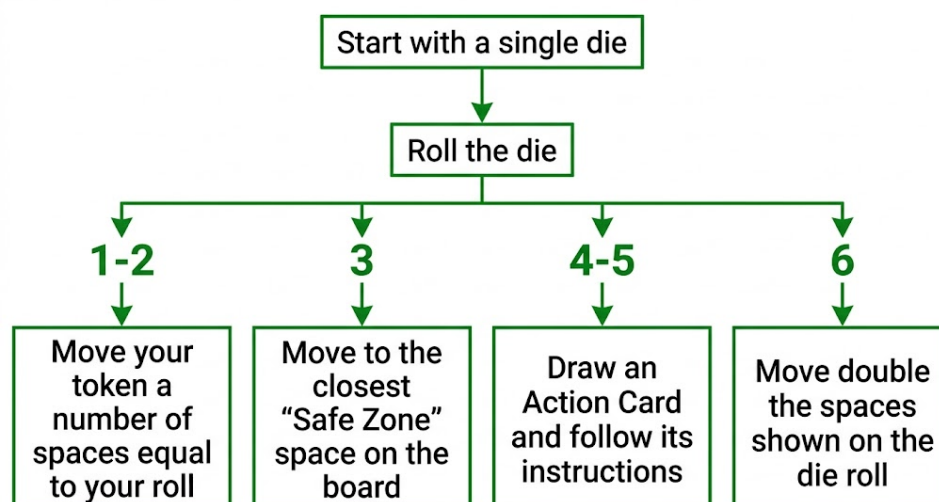
Roll the die. If it shows:

1, 2 The die is taken back.

3 The die is given back.

4, 5 The die is given back & another die is also given.

6 Two more dice are given.



Random Walks

Experiment: On $\mathbb{Z}$. Toss a fair coin (prob. of H. is $1/2$). If H turns up, move one step to right otherwise move one step to the left.

Keep on doing it.

Define

$$\varepsilon_i = \begin{cases} +1 & \text{w.p. } 1/2 \\ -1 & \text{w.p. } 1/2 \end{cases}$$

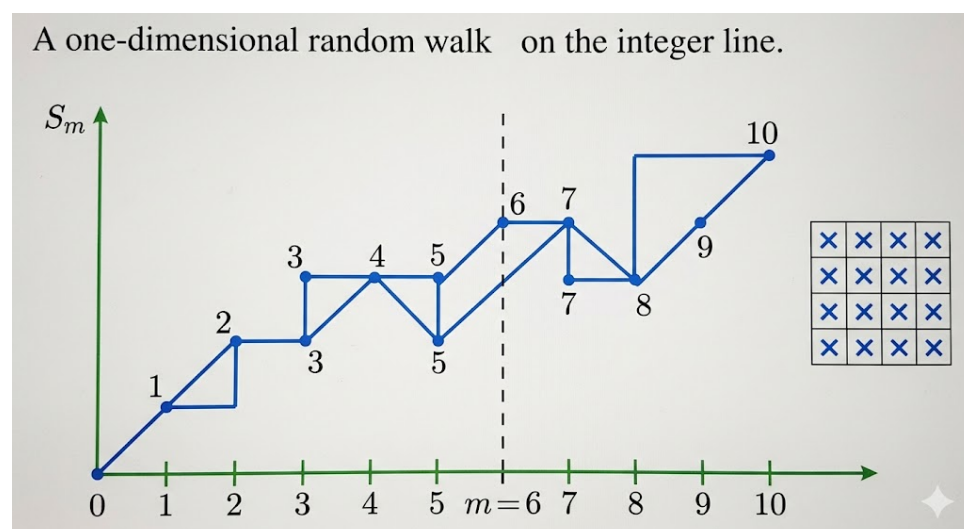
represents the random step at the i th toss.

Random Walk Position

$$S_0 = a \in \mathbb{Z}$$

$$S_m = a + \sum_{i=1}^m \varepsilon_i, \text{ position of the walker at time } m.$$

A one-dimensional random walk on the integer line.



$\mathbb{Z}, E$

$$\epsilon_1 = 1, \epsilon_2 = 1$$

$$\epsilon_3 = -1, \epsilon_4 = 1$$

$$\epsilon_i = \begin{cases} +e_i & \text{w.p. } 1/4 \\ -e_i & \text{w.p. } 1/4 \end{cases}$$

$$\epsilon_1 = (1, 0), \epsilon_2 = (0, 1)$$

Recurrence of Simple Symmetric Random Walk

Q. For a simple symmetric random walk starting at the origin, does it come back to the origin at some time with probability 1?

Let B_n ($n \geq 1$) be the events satisfying $B_n \cap B_m = \emptyset$ for $n \neq m$. Then

$$P\left(\bigcup_{n=1}^{\infty} B_n\right) = \sum_{n=1}^{\infty} P(B_n).$$

Let $S_0 = 0$ and define the event

$$A = \{S_n = 0 \text{ for some } n \geq 1\}.$$

We want to show that $P(A) = 1$.

Definition. Let

$$A_n = \{S_n = 0\}.$$

Then

$$A = \bigcup_{n=1}^{\infty} A_n.$$

Note. $A_n = \emptyset$ if n is odd.

Defining Disjoint Sets

Define

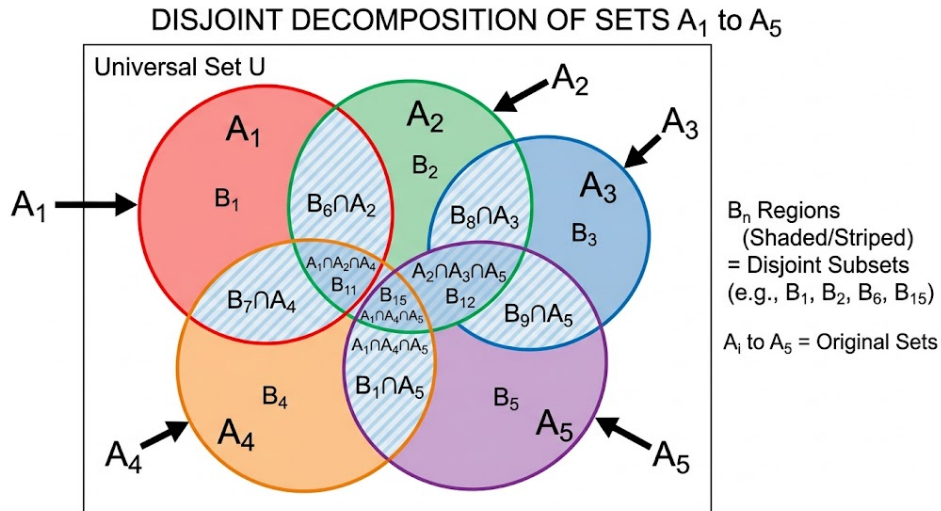
$$B_1 = A_1 \quad \text{for } n \geq 2$$

$$B_m = A_m \setminus \left(\bigcup_{k=1}^{m-1} A_k\right)$$

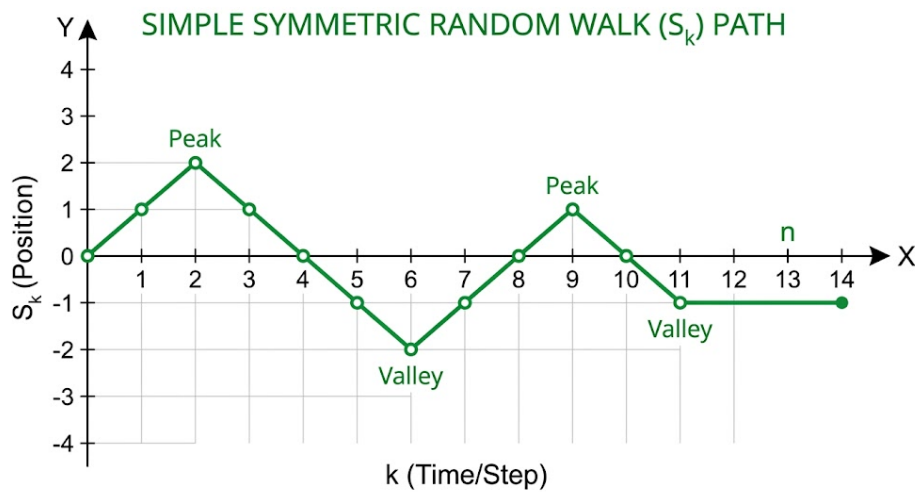
Note, $B_n \cap B_m = \emptyset$.

$$\bigcup_{k=1}^m A_k = \bigcup_{k=1}^m B_k \quad \forall n \geq 1.$$

$$\bigcup_{k=1}^{\infty} A_k = \bigcup_{k=1}^{\infty} B_k.$$



Random Walk First Return



$$B_n = \{S_n = 0, S_k \neq 0 \text{ for } k = 1, \dots, n-1\}$$

$$= \{\text{R.W. path at 0 for the first time at } n\}$$

Notation

Notation:

$$u_m = P(S_m = 0)$$

$$f_m = P(S_n \geq 0, S_k \neq 0 \text{ for } k = 1, \dots, n-1)$$

$$\begin{aligned}
 P(A) &= P\left(\bigcup_{n=1}^{\infty} A_n\right) \\
 &= P\left(\bigcup_{n=1}^{\infty} B_n\right) \\
 &= \sum_{n=1}^{\infty} P(B_n) = \sum_{n=1}^{\infty} f_n = \sum_{n=1}^{\infty} f_{2n}
 \end{aligned}$$

$$u_n = f_n = 0 \quad \text{for } n \text{ odd.}$$

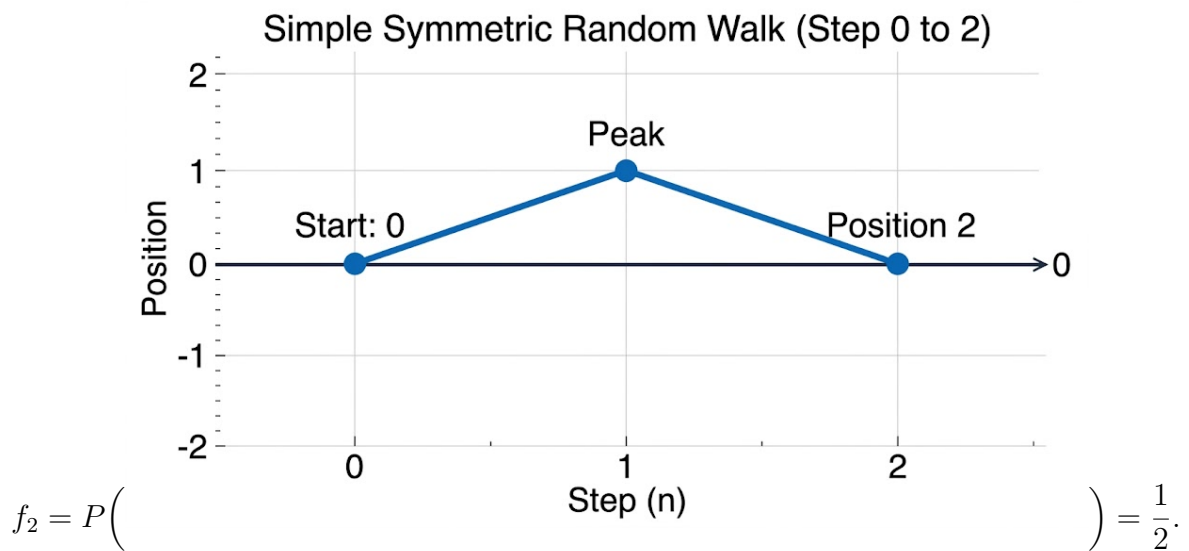
First Return Probabilities

Proposition: For $m \geq 1$.

$$f_{2m} = u_{2m-2} - u_{2m}$$

Notation: $u_0 = P(S_0 = 0) \geq 1$, $f_0 = 0$.

Calculation for $n = 1$

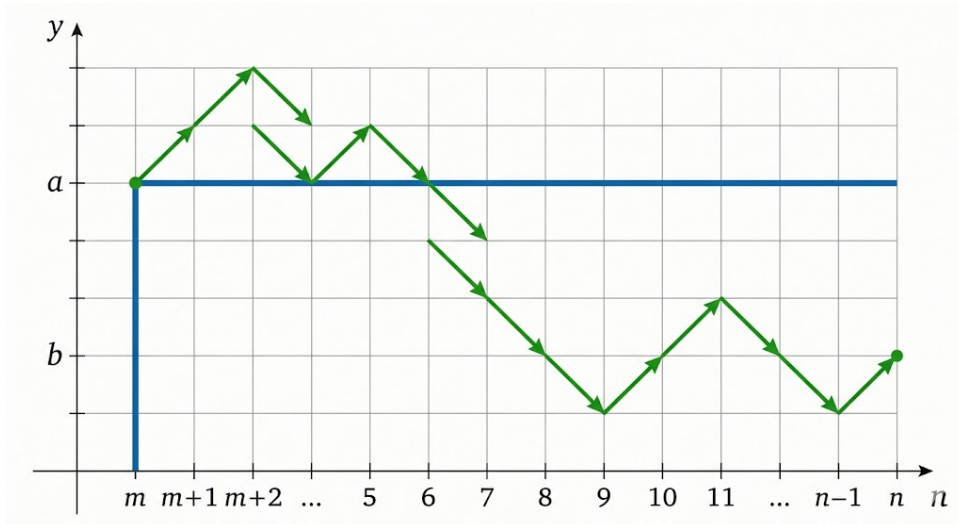


$$f_2 = \frac{1}{2} = 1 - \frac{1}{2} = u_0 - u_2.$$

Path Definition

Def.: A path start from (m, a) and ending at (n, b) is a finite sequence $\delta_m, \delta_{m+1}, \dots, \delta_n$ s.t.

$$\begin{aligned} \delta_m &= a, & \delta_n &= b, \\ |\delta_k - \delta_{k-1}| &= 1 & \text{for } k &= m+1, \dots, n. \end{aligned}$$

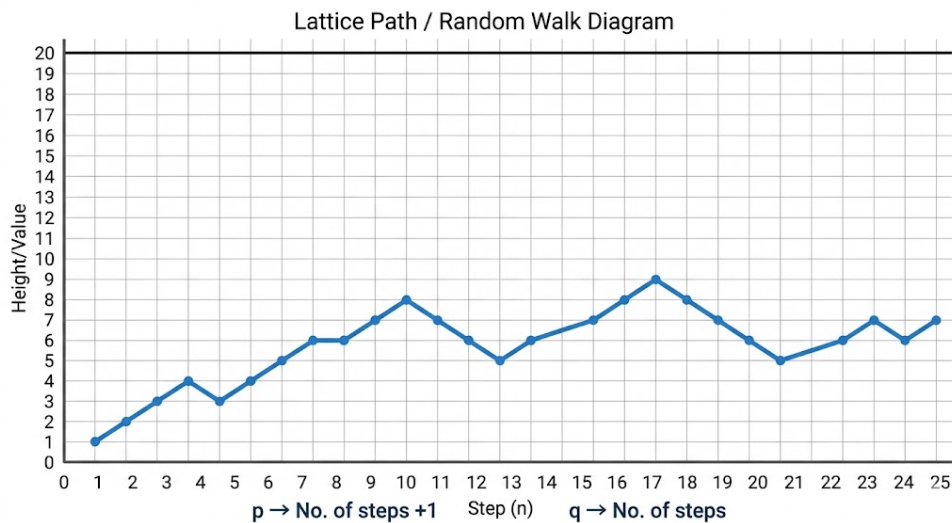


Lattice Path Notation

Nat. : $N_{(n,b)}^{(m,a)}$ is the set of all paths st. from (m, a) and ending at (n, b) . When $m = a = 0$, we drop the superscript.

Note,

$$N_{(n,b)}^{(m,a)} = N_{(n-m,b-a)}$$



Number of Paths and Event Probability

$$|N_{m,a}| = \begin{cases} 0 & \text{if } m+a \text{ is odd} \\ \binom{m}{\frac{m+a}{2}} & \text{o.w.} \end{cases}$$

Any event B which involves $E_1, \dots, E_n$ has the prob.

$$P(B) = \frac{1}{2^m} \#(\text{No. of paths satisfying the cond. of } B)$$

Conditions:

$$\begin{aligned} p+q &> n, \\ p-q &= a, \\ p &= \frac{m+a}{2}, \\ q &= \frac{n-a}{2}. \end{aligned}$$

Stirling's Approximation and Generating Functions

Find the generating function of $\{u_n\}$

$$\begin{aligned} u_{2n} &= \frac{1}{2^{2n}} \binom{2n}{n} \\ u_{2n} \sqrt{\pi n} &= \frac{1}{2^{2n}} \cdot \frac{(2n)!}{n! n!} \sqrt{\pi n} \\ &= \frac{(2n)!}{2^{2n} n! n!} \cdot \frac{\sqrt{\pi n} n^n e^{-n} n^n e^{-n}}{(\sqrt{2\pi n} n^n e^{-n})^2} \\ &\rightarrow 1 \quad \text{as } n \rightarrow \infty. \end{aligned}$$

Stirling's approximation

$$n! \sim \sqrt{2\pi n} n^n e^{-n}$$

We say $a_n \sim b_n$ if

$$\frac{a_n}{b_n} \rightarrow 1 \quad \text{as } n \rightarrow \infty.$$

Limits and Infinite Sums

$$\begin{aligned} u_{2n} \sqrt{\pi n} &\rightarrow 1 \quad \text{as } n \rightarrow \infty. \\ u_{2n} &\sim \frac{1}{\sqrt{\pi n}}. \end{aligned}$$

$$u_{2n} \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

$$\sum_{n=0}^{\infty} \nu_{2n} = \infty$$

Ex:

$$f_{2n} \sim \frac{1}{\sqrt{4\pi n^3}}$$

$$P(A) = \sum_{n \geq 1} f_{2n}.$$

$$\sum_{n=1}^N f_{2n} = \sum_{n=1}^N (\nu_{2n-2} - \nu_{2n}) = \nu_0 - \nu_{2N}.$$

$$\sum_{n=1}^{\infty} f_{2n} = \lim_{N \rightarrow \infty} (1 - \nu_{2N}) = 1.$$

Return Time Definition

Define:

$$T = \inf\{n \geq 1 : S_n = 0\}$$

$$P(T = n) = \mathbb{P}(S_n = 0, S_k \neq 0 \text{ for } k = 1, \dots, n-1) = f_n.$$

Boxed properties:

$$(T^{1/2-\varepsilon}) < \infty \quad \forall \varepsilon > 0.$$

$$T < \infty \quad \text{a.s.}$$

$$(T) = \sum_{n=0}^{\infty} \mathbb{P}(T > n) = \sum_{n=0}^{\infty} u_{2n}.$$

$$\mathbb{P}(T > n) = \mathbb{P}\left(\bigcup_{k=n+1}^{\infty} B_k\right) = \sum_{k=n+1}^{\infty} \mathbb{P}(B_k) = \sum_{k=n+1}^{\infty} f_k.$$

Infinite Sum Derivation

$$\begin{aligned}
 \sum_{k=2n}^{\infty} f_k &= f_{2n} + f_{2n+2} + f_{2n+4} + \cdots \\
 &= \lim_{N \rightarrow \infty} \sum_{k=n}^{m+N} f_{2k} \\
 &= \lim_{N \rightarrow \infty} \sum_{k=n}^{m+N} (u_{2k-2} - u_{2k}) \\
 &= u_{2n-2} - \lim_{N \rightarrow \infty} u_{2n+2N} \\
 &= u_{2n-2} - 0 \\
 &= u_{2n-2}.
 \end{aligned}$$

Random Walk and Gambler's Ruin

Random Walk.

$$S_0 = a, \quad S_n = \sum_{i=1}^n \varepsilon_i$$

where

$$\varepsilon_i = \begin{cases} +1 & \text{w.p. } \frac{1}{2} \\ -1 & \text{w.p. } \frac{1}{2} \end{cases}$$

Gambler's Ruin Problem

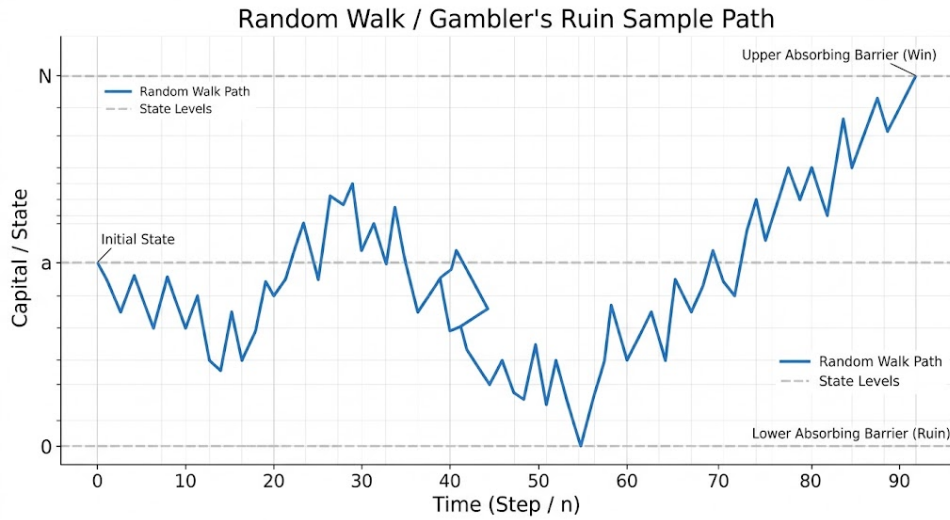
Consider two gamblers A & B with initial capital a & b . Each time a fair coin is tossed, if H turns up, then A gets Rs 1 from B, otherwise B gets Rs 1 from A.

Gambler's Ruin Capital Process

Total capital $N = a + b$. Let S_n denote the capital of A after n tosses. Then $S_0 = 0$ and

$$S_m = a + \sum_{i=1}^m E_i.$$

Capital of B is $N - S_m$.



The probability that A reaches N before 0 is

$$P(S_m = N \text{ for some } m > 1 \text{ before } S_m = 0 \text{ for some } m \geq 1) = P(T_N < T_0),$$

where T_i is the time to hit level i .

Basic Renewal Equation

$P(\text{RW returns to the origin starting from the origin}) = 1$.

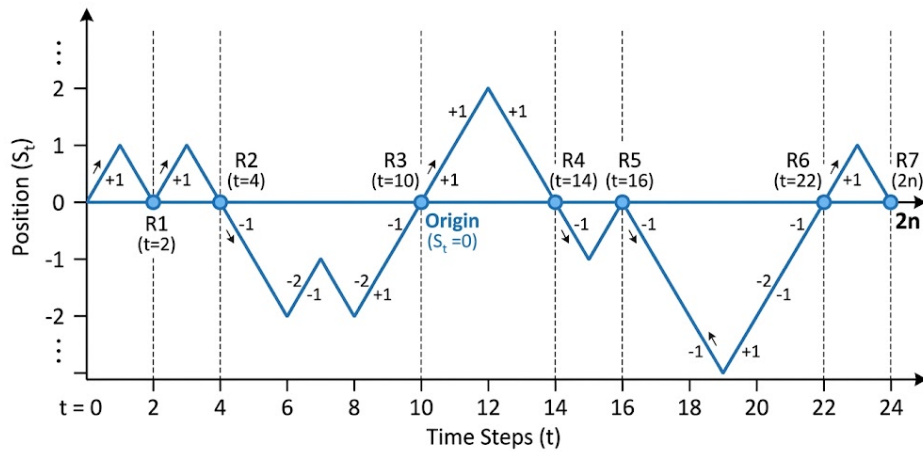
Basic renewal equation:

$$f_{2n} = u_{2n-2} - u_{2n}$$

$$u_{2n} = P(S_{2n} = 0) = f_{2n} + \dots$$

$$\begin{aligned} \nu_{2n} &= P(S_{2n} = 0) \\ &= P(S_{2n} = 0, S_k \neq 0 \text{ for } 1, 2, \dots, (2n-1)) \\ &\quad + P(S_{2n} = 0, S_{2k} = 0 \text{ for some } k = 1, \dots, n-1). \end{aligned}$$

SYMMETRIC RANDOM WALK ON THE INTEGER LINE: MULTIPLE RETURNS TO THE ORIGIN



First Return Time in Random Walk

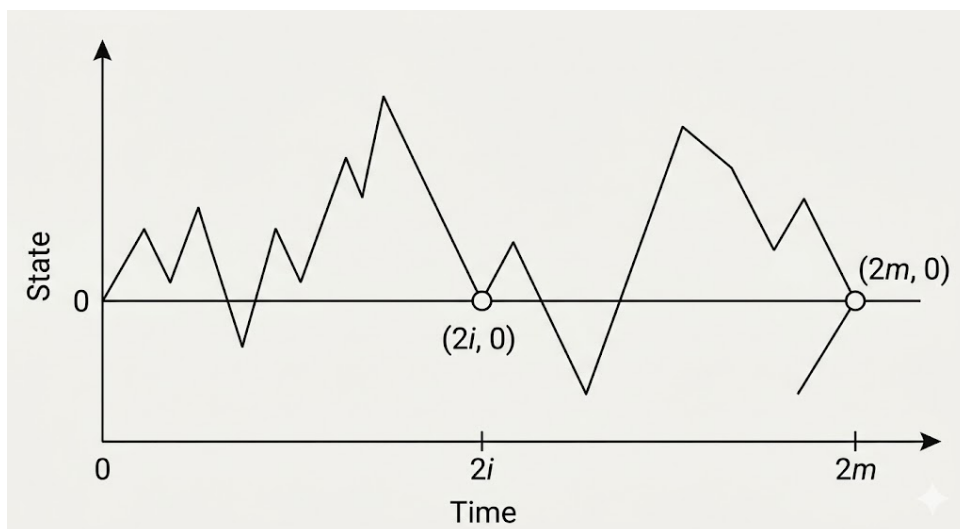
$$P(S_{2m} = 0, S_{2k} \neq 0 \text{ for all } k = 1, \dots, m-1)$$

$$= P(\text{the first return happens at time } 2i, i = 1, \dots, m-1)$$

$$B_{2i} = \{S_1 \neq 0, \dots, S_{2i-1} \neq 0, S_{2i} = 0\},$$

$$B_{2i} \cap B_{2j} = \emptyset \quad (i \neq j).$$

$$= P(S_{2m} = 0, B_{2i} \text{ happens for some } i = 1, \dots, m-1) = P\left(\{S_{2m} = 0\} \cap \left(\bigcup_{i=1}^{m-1} B_{2i}\right)\right).$$



Random Walk Return Probability Decomposition

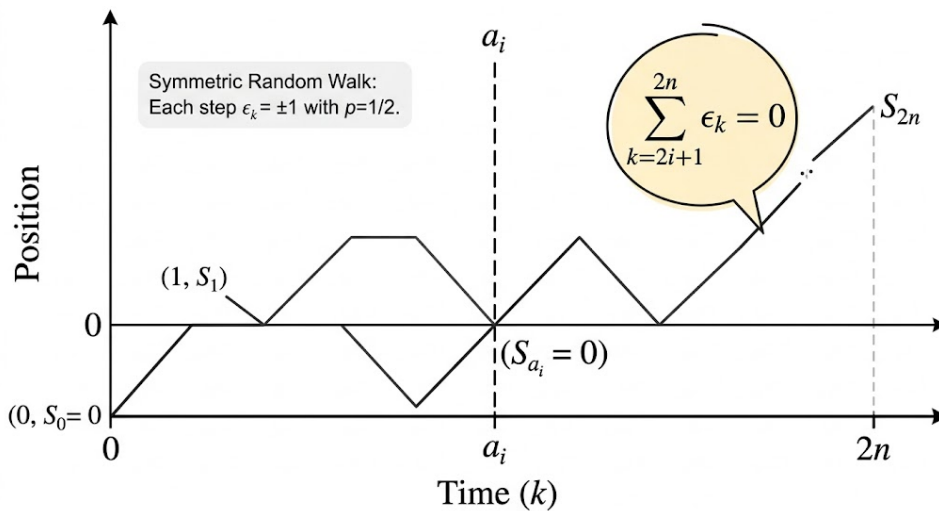
$$\begin{aligned}
 &= P\left(\bigcup_{i=1}^{m-1} (\{S_{2n} = 0\} \cap B_{2i})\right) \\
 &= \sum_{i=1}^{m-1} P(B_{2i} \cap \{S_{2n} = 0\})
 \end{aligned}$$

$$P(B_{2i} \cap \{S_{2n} = 0\}) = P(S_1 \neq 0, \dots, S_{2i-1} \neq 0, S_{2i} = 0, S_{2n} = 0)$$

$$= P(S_1 \neq 0, \dots, S_{2i-1} \neq 0, S_{2i} = 0)$$

The second factor depends on $(\varepsilon_{2i+1}, \dots, \varepsilon_{2n})$ with

$$\sum_{k=2i+1}^{2n} \varepsilon_k = 0 \quad (S_{2n} - S_{2i} = 0).$$



Recurrence for Return Probabilities

$$\begin{aligned}
 &= P(S_1 \neq 0, \dots, S_{2i-1} \neq 0, S_{2i} = 0) \\
 P(S_{2n} - S_{2i} = 0) &= f_{2i} P(S_{2n-2i} = 0) \\
 &= f_{2i} U_{2n-2i}
 \end{aligned}$$

$$U_{2n} = f_{2n} + \sum_{i=1}^{n-1} f_{2i} U_{2n-2i}$$

$$\sum_{i=0}^n f_{2i} U_{2n-2i}$$

Right column:

$$\begin{aligned} E_1 + E_2 + E_3 + \dots \\ \frac{d}{dx} = E_{2n-3} + E_{2n-1} + 6E_{2n-1} + E_{2n} \\ \sum_{k=2i+1}^{2n} E_k = E_1 + \dots + E_{2n-2i} \end{aligned}$$

Convolution of Sequences

$\{U_{2m}\}$ is a convolution of $\{f_{2m}\}$ and $\{U_{2m}\}$, except for $m = 0$.

$$F(s) = \sum_{m \geq 0} f_m s^m$$

demo

$$C_m = \sum f_k u_{m-k}$$

$$k \rightarrow m$$

$$C_0 = 0$$

For $m = 0$, $U_0 \neq f_0 U_0 = 0$ (relation does not hold).

$$\begin{aligned} U(s) &= \sum_{m=0}^{\infty} U_{2m} s^{2m} = 1 + \sum_{m=1}^{\infty} U_{2m} s^{2m} \\ &= 1 + U(s)F(s) = 1 + \sum_{n=1}^{\infty} C_n s^{2n} \\ &= 1 + \sum_{m=1}^{\infty} C_{2m} s^{2m} \\ &= 1 + \sum_{m=1}^{\infty} C_m s^m \quad (m \geq 0) \end{aligned}$$

Abel's Lemma and Generating Functions

$$S_0, \quad U(s) = 1 + U(s)F(s)$$

$$F(s) = 1 - \frac{1}{U(s)}$$

$$\sum_{n=0}^{\infty} f_n = 1 - \frac{1}{\sum_{n=0}^{\infty} u_n}$$

By Abel's lemma,

$$\sum_{n=0}^{\infty} f_n = \lim_{s \rightarrow 1} F(s) = 1 - \frac{1}{\lim_{s \rightarrow 1} U(s)}$$

$$\sum_{n=0}^{\infty} f_n = 1 \quad \text{and} \quad \sum_{n=0}^{\infty} u_n = \infty$$

Probability of No Return Until Time $2n$

Theorem. Let $a_n = u_{2n}$. Define

$$C_k = \{S_1 \neq 0, \dots, S_k \neq 0\}.$$

Then

$$P(S_1 \neq 0, S_2 \neq 0, \dots, S_{2n} \neq 0) = u_{2n}.$$

Moreover,

$$P(S_1 \neq 0, \dots, S_{2n-2} \neq 0, S_{2n} = 0) = f_{2n},$$

and

$$\{S_1 \neq 0, \dots, S_{2n-2} \neq 0\} = C_{2n-2} \cap \{S_{2n-1} \neq 0\}.$$

Recurrence Relations

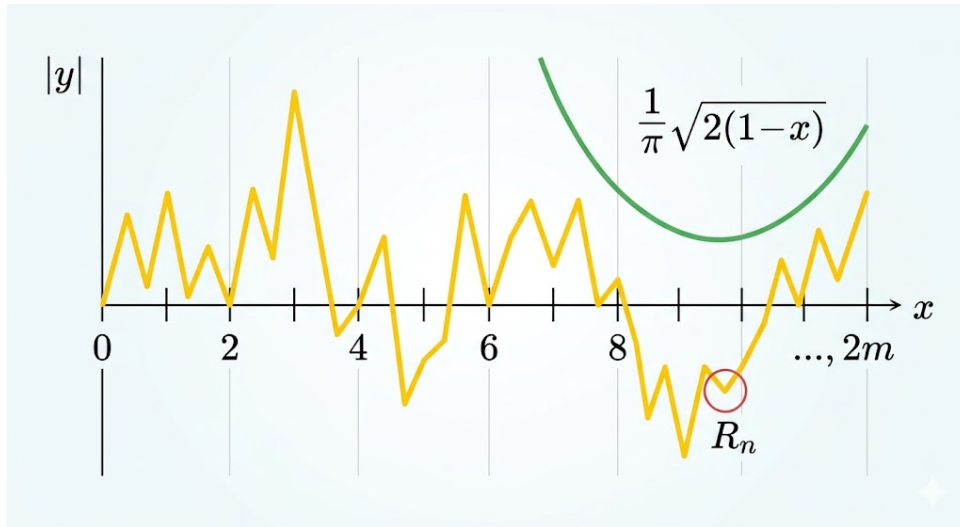
$$\begin{aligned} C_{2n-2} &= C_{2n-2} \{S_{2n-1} \neq 0\} B_{2n} \\ &= C_{2n-2} \{S_{2n-1} \neq 0, S_{2n} = 0\} \end{aligned}$$

$$\bigcup (C_{2n-2} \{S_{2n-1} \neq 0, S_{2n} \neq 0\})$$

$$P(C_{2n-2}) = P(B_{2n} \cup C_{2n}) = P(B_{2n}) + P(C_{2n})$$

$$u_{2n-2} = f_{2n} + u_{2n}$$

Maximum of Weighted Random Walk



Equation:

$$W_{2n} = \max\{k \in (0, 1 - 2)W_i; S_k^0\}$$

Arcsine Distribution Limit

$$P(W_n = \alpha^k) = U_{\alpha^k} U_{\alpha^{M-2k}}$$

$$\boxed{\frac{W_{2n}}{2^M} \longrightarrow W}$$

where W has the arcsine distribution.

$$\sim \frac{1}{\sqrt{\pi k}} \cdot \frac{1}{\sqrt{\pi(M-k)}} = \frac{1}{\pi \sqrt{k(M-k)}}$$

$W \sim$ arcsine distribution.

Random Walk Probabilities

$$P(W_{2n} = 2k) = \begin{cases} P(S_1 \neq 0, \dots, S_{2n} \neq 0) & k = 0 \\ & = u_{2n} \cdot 1 = u_{2n} \cdot u_0 \\ P(S_{2n} = 0) & k = m \\ & = u_{2n} \cdot u_0 \end{cases}$$

For $1 \leq k \leq n-1$,

$$\begin{aligned}
 P(W_{2n} = 2k) &= P(S_{2k} = 0, S_i \neq 0 \text{ for } i = 2k+1, \dots, 2n) \\
 &= P(S_{2k} = 0, S_i - S_{2k} \neq 0 \text{ for } i = 2k+1, \dots, 2n) \\
 &= P(S_{2k} = 0) P(E_{2k+1} \dots E_{2n}) \\
 &= u_{2k} u_{2n-2k}.
 \end{aligned}$$

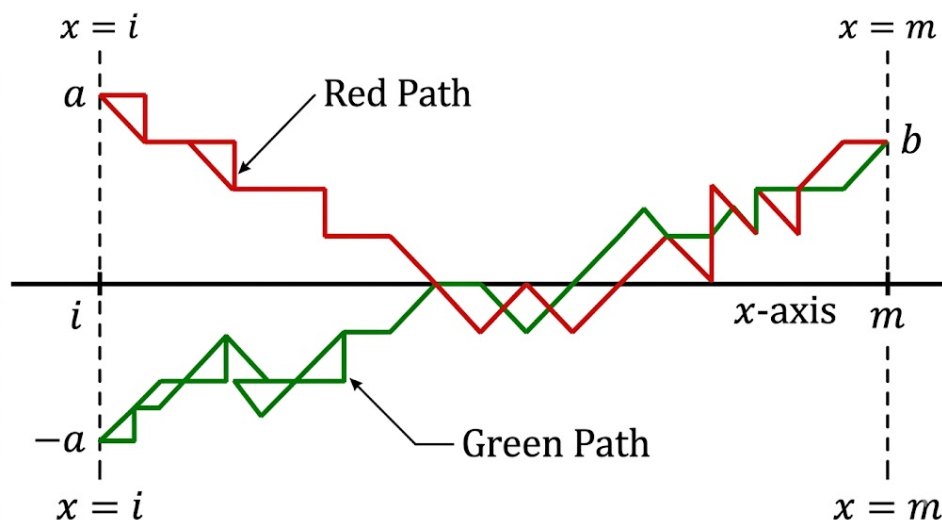
Random Walk

$f_{2n} :=$ Probability of no return until time $2n$ is the same as the probability of return at time $2n$.

$$P(S_1 \neq 0, \dots, S_{2n} \neq 0) = u_{2n} = \binom{2n}{n} 2^{-2n} = P(S_{2n} = 0)$$

Reflection Principle

$a, b > 0$



The number of paths from (i, a) to (m, b) which touches or crosses the x -axis is the same as the number of paths from $(i, -a)$ to (m, b) .

Path Sets and Bijection

$A = \{ \text{paths from } (i, a) \text{ to } (m, b) \text{ which touch or cross } x\text{-axis} \}.$

$B = \{ \text{paths from } (i, -a) \text{ to } (m, b) \}.$

$$|A| = |B|.$$

$$f : A \rightarrow B$$

$$f \text{ is 1-1 \& onto.}$$

Reflection of Paths

$\pi = \text{Path in } A$,

$$\pi = (\delta_i, \delta_{i+1}, \dots, \delta_n)$$

with $\delta_i = a, \delta_n = b$.

Since $\pi \in A$, there exists some $i < k < n$ such that

$$\delta_k = 0 \quad \text{and} \quad |\delta_i - \delta_{i+1}| = 1.$$

Choose the smallest such k .

Now define the reflected path

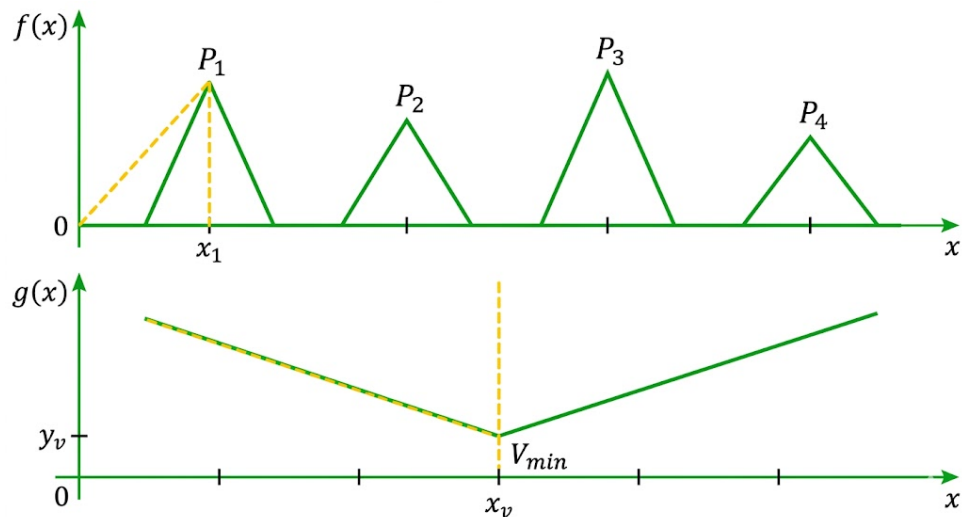
$$\pi' = (-\delta_i, -\delta_{i+1}, \dots, -\delta_{k-1}, 0, \delta_k, \dots, \delta_n).$$

Define $f(\pi) = \pi'$. Then $\pi' \in B$.

Check that π' is a path starting from $-a$.

Show that f is 1-1 and Onto

Show that f is one-to-one and onto.



Ballot Theorem

Two candidates, A and B, poll p and q votes respectively, with $p > q$. Assuming all orderings of votes are equally likely, the probability that candidate A is always strictly ahead of candidate B is

$$\frac{p - q}{p + q}.$$

Proof of the Ballot Theorem

Pf: Let

$$E_i = \begin{cases} +1 & \text{if } i\text{th voter votes for } A \\ -1 & \text{otherwise} \end{cases}$$

$$S_0 = 0, S_i = \sum_{k=1}^i E_k.$$

S_i tracks the lead of Candidate A over B after i votes.

Need to show

$$P(S_1 > 0, S_2 > 0, \dots, S_{p+q} > 0 \mid S_{p+q} = p - q).$$

Conditional Probability via Reflection

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$$\begin{aligned} &= \frac{P(S_1 > 0, \dots, S_{p+q} > 0, S_{p+q} = p - q)}{P(S_{p+q} = p - q)} \\ &= \frac{P(S_1 = 1, \dots, S_{p+q} = p - q)}{P(S_{p+q} = p - q)} = \left\{ S_1 = 1, \dots, S_{p+q} = p - q \right\} \cdot 2^{-(p+q)}. \end{aligned}$$

$$P(S_{p+q} = p - q) = \binom{p+q}{p} 2^{-(p+q)}.$$

Ballot Theorem via Reflection Principle

$$|\{S_1 = b, S_i > 0, \dots, S_{p+q} = b - q\}| = N_{(p+q, b-q)}^{(1,1)} - N_{(p+q, b-q)}^{(1,-1)}.$$

Finally,

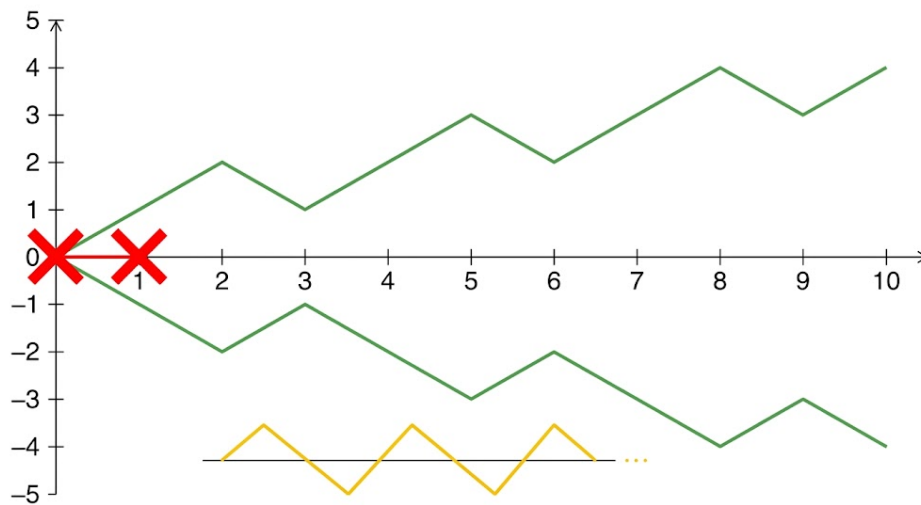
$$\begin{aligned} P(S_1 > 0, \dots, S_{p+q} > 0 \mid S_{p+q} = b - q) &= \frac{N_{(p+q, b-q)}^{(1,1)} - N_{(p+q, b-q)}^{(1,-1)}}{N_{(p+q, b-q)}^{(0,0)}} \\ &= \frac{b - q}{p + q}. \end{aligned}$$

Symmetry of Nonzero Random Walk Paths

$$\begin{aligned} P(S_1 \neq 0, \dots, S_{2n} \neq 0) &= P(S_1 = 1, S_2 > 0, \dots, S_{2n} > 0) \\ &\quad + P(S_1 = -1, S_2 < 0, \dots, S_{2n} < 0) \\ &= 2P(S_1 = 1, S_2 > 0, \dots, S_{2n} > 0). \end{aligned}$$

For increments ε_i , the change of sign gives the corresponding reflected event:

$$\begin{aligned} P(\varepsilon_1 = 1, \varepsilon_1 + \varepsilon_2 > 0, \dots, \varepsilon_1 + \dots + \varepsilon_{2n} > 0) &= P(-\varepsilon_1 = 1, -\varepsilon_1 - \varepsilon_2 > 0, \dots, -\varepsilon_1 - \dots - \varepsilon_{2n} > 0) \\ &= P(\varepsilon_1 = -1, \varepsilon_1 + \varepsilon_2 < 0, \dots, \varepsilon_1 + \dots + \varepsilon_{2n} < 0). \end{aligned}$$



Decomposition into Positive Endpoint Events

$$P(S_1 = 1, S_2 > 0, \dots, S_{2n} > 0) = 2^{-2n}|A|,$$

where

$$A = \{S_1 = 1, S_2 > 0, \dots, S_{2n} > 0\}.$$

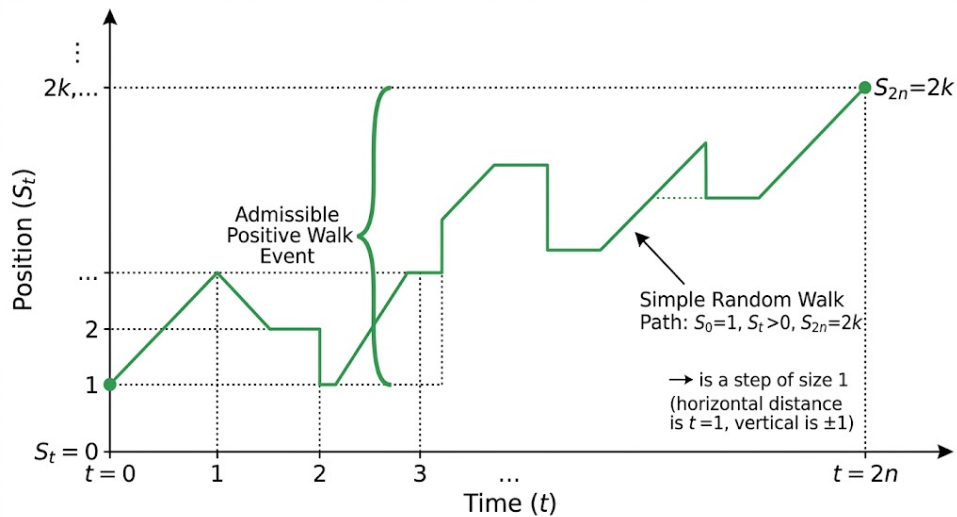
Since the endpoint after $2n$ steps is even and positive, decompose

$$A = \bigcup_{k=1}^{\infty} A_k,$$

where

$$A_k = \{S_1 = 1, S_2 > 0, \dots, S_{2n} > 0, S_{2n} = 2k\}.$$

For $k > n$, $A_k = \emptyset$.



Counting the Sets A_k

For each $k \geq 1$,

$$|A_k| = N_{(2n, 2k)}^{(1, 1)} - N_{(2n, 2k)}^{(1, -1)}.$$

We use the convention

$$\binom{m}{k} = 0 \quad \text{if } k > m.$$

Therefore,

$$\begin{aligned}
 |A| &= \sum_{k=1}^{\infty} |A_k| \\
 &= \sum_{k=1}^{\infty} \left[N_{(2n,2k)}^{(1,1)} - N_{(2n,2k)}^{(1,-1)} \right] \\
 &= \sum_{k=1}^{\infty} \left[\binom{2n-1}{n+k-1} - \binom{2n-1}{n+k} \right].
 \end{aligned}$$

Evaluation of the Positive Path Probability

Using the count from the previous page,

$$\begin{aligned}
 P(S_1 = 1, S_2 > 0, \dots, S_{2n} > 0) &= \frac{|A|}{2^{2n}} \\
 &= \frac{1}{2^{2n}} \binom{2n-1}{n} \\
 &= \frac{1}{2^{2n}} \frac{(2n-1)!}{n!(n-1)!} \\
 &= \frac{1}{2} \frac{(2n)!}{2^{2n} n! n!}.
 \end{aligned}$$

Thus, if

$$u_{2n} = \binom{2n}{n} 2^{-2n},$$

then

$$P(S_1 = 1, S_2 > 0, \dots, S_{2n} > 0) = \frac{1}{2} u_{2n}.$$

Maximum of a Simple Random Walk

Define the maximum of the random walk by

$$M_n = \max(S_0, S_1, \dots, S_n).$$

For $k \geq 0$, the reflection principle gives

$$P(M_n \geq k) = 2P(S_n > k) + P(S_n = k), \quad k = 0, 1, \dots$$

For $k = 0$, this is consistent with

$$\text{LHS} = P(M_n \geq 0) = 1,$$

and

$$\begin{aligned}
 \text{RHS} &= 2P(S_n > 0) + P(S_n = 0) \\
 &= P(S_n > 0) + P(S_n = 0) + P(S_n < 0) \\
 &= 1.
 \end{aligned}$$

Maximum Identity for $k \geq 1$

For $k \geq 1$,

$$\begin{aligned} P(M_n \geq k) &= P(M_n \geq k, S_n > k) \\ &\quad + P(M_n \geq k, S_n = k) \\ &\quad + P(M_n \geq k, S_n < k). \end{aligned}$$

Since $S_n > k$ and $S_n = k$ both imply $M_n \geq k$, this becomes

$$P(M_n \geq k) = P(S_n > k) + P(S_n = k) + P(M_n \geq k, S_n < k).$$

By reflecting paths at their first visit to level k ,

$$P(M_n \geq k, S_n < k) = P(S_n > k).$$

Hence,

$$P(M_n \geq k) = 2P(S_n > k) + P(S_n = k), \quad k = 1, 2, \dots$$

Reflection at Level k

For $k \geq 1$, reflect the path after its first visit to level k . This gives a one-to-one correspondence between paths satisfying

$$M_n \geq k, \quad S_n < k$$

and reflected paths satisfying

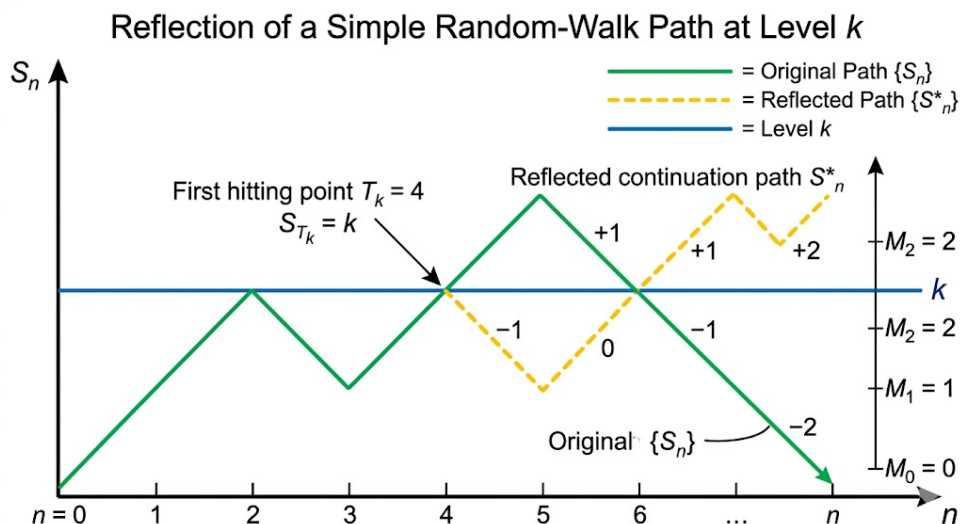
$$M_n \geq k, \quad S_n > k.$$

Hence,

$$P(M_n \geq k, S_n < k) = P(S_n > k).$$

Substituting into the decomposition gives

$$\begin{aligned} P(M_n \geq k) &= P(S_n > k) + P(S_n = k) + P(M_n \geq k, S_n < k) \\ &= 2P(S_n > k) + P(S_n = k). \end{aligned}$$



Markov Chain

Let $\mathcal{S}$ be a countable set, called the state space.

A sequence of random variables

$$\{X_n : n \geq 0\}$$

taking values in $\mathcal{S}$ is called a Markov chain if it satisfies the Markov property:

$$P(X_{n+1} = j \mid X_0 = i_0, X_1 = i_1, \dots, X_n = i) = P(X_{n+1} = j \mid X_n = i),$$

whenever the conditional probabilities are defined.

The chain is discrete time and has discrete state space.

Markov Property

A stochastic process $\{X_n : n \geq 0\}$ with countable state space S is a Markov chain if the conditional distribution of the future state depends only on the present state and not on the past history.

For all $n \geq 0$ and $i, j, i_0, i_1, \dots, i_{n-1} \in S$,

$$P(X_{n+1} = j \mid X_n = i, X_{n-1} = i_{n-1}, \dots, X_0 = i_0) = P(X_{n+1} = j \mid X_n = i),$$

whenever the probabilities are well defined.

Equivalently,

$$P(X_{n+1} = j \mid X_n = i) = p_{ij}.$$

Homogeneous Markov Chains

For a time-homogeneous Markov chain,

$$P(X_{n+1} = j \mid X_n = i) = P(X_1 = j \mid X_0 = i), \quad n \geq 1.$$

Example

Show that for a Markov chain, given the present, the past and the future are independent.

For $i, j, k \in S$,

$$\begin{aligned} P(X_{n+1} = j, X_{n-1} = k \mid X_n = i) \\ = P(X_{n+1} = j \mid X_n = i, X_{n-1} = k) P(X_{n-1} = k \mid X_n = i). \end{aligned}$$

By the Markov property,

$$P(X_{n+1} = j \mid X_n = i, X_{n-1} = k) = P(X_{n+1} = j \mid X_n = i).$$

Therefore,

$$P(X_{n+1} = j, X_{n-1} = k \mid X_n = i) = P(X_{n+1} = j \mid X_n = i) P(X_{n-1} = k \mid X_n = i).$$

Transition Probabilities

Define the one-step transition probability

$$p_{ij} = P(X_{n+1} = j \mid X_n = i) = P(X_1 = j \mid X_0 = i), \quad i, j \in S.$$

The transition probabilities satisfy

$$0 \leq p_{ij} \leq 1, \quad i, j \in S,$$

and, for each fixed $i \in S$,

$$\sum_{j \in S} p_{ij} = 1.$$

The matrix

$$P = \left((p_{ij}) \right)_{i, j \in S}$$

is called the transition probability matrix.

Stochastic Matrix Property

For a transition probability matrix $P = (p_{ij})$, each row sums to one:

$$\sum_{j \in S} p_{ij} = \sum_{j \in S} P(X_1 = j \mid X_0 = i) = 1,$$

since the events $\{X_1 = j\}$, $j \in S$, are disjoint and exhaustive.

Let

$$\mathbf{1} = \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix}.$$

Then

$$P\mathbf{1} = \mathbf{1}.$$

Thus $\mathbf{1}$ is an eigenvector of P corresponding to the eigenvalue 1.

Initial Distribution and Random Recursion

The distribution of X_0 is called the initial distribution.

Problem

Let $\{X_n : n \geq 0\}$ take values in a state space S , with

$$X_0 \sim \mu.$$

Let $\{\xi_n : n \geq 1\}$ be i.i.d. random variables taking values in a set C , independent of X_0 . Assume that

$$X_{n+1} = f(X_n, \xi_{n+1}), \quad n \geq 0,$$

where

$$f : S \times C \rightarrow S.$$

Then $\{X_n : n \geq 0\}$ is a Markov chain. Its transition probabilities are

$$p_{ij} = P(X_{n+1} = j \mid X_n = i) = P(f(i, \xi_{n+1}) = j), \quad i, j \in S.$$

Proof of the Random Recursion Markov Property

Let $f : S \times C \rightarrow S$ and suppose

$$X_{n+1} = f(X_n, \xi_{n+1}),$$

where $\{\xi_n : n \geq 1\}$ is an i.i.d. sequence independent of X_0 .

To prove that $\{X_n : n \geq 0\}$ is a Markov chain, fix $i_0, i_1, \dots, i, j \in S$. Then

$$\begin{aligned} P(X_{n+1} = j \mid X_0 = i_0, \dots, X_{n-1} = i_{n-1}, X_n = i) \\ = P(f(i, \xi_{n+1}) = j \mid X_0 = i_0, \dots, X_n = i). \end{aligned}$$

Since ξ_{n+1} is independent of the past and of X_n , this becomes

$$P(f(i, \xi_{n+1}) = j).$$

Thus the conditional probability depends only on the present state i :

$$P(X_{n+1} = j \mid X_0 = i_0, \dots, X_n = i) = P(f(i, \xi_{n+1}) = j).$$

Therefore $\{X_n\}$ is a Markov chain with transition probabilities

$$p_{ij} = P(f(i, \xi_1) = j).$$

Factorization in the Random Recursion Proof

For states $i_0, i_1, \dots, i_k \in S$, the recursion $X_{n+1} = f(X_n, \xi_{n+1})$ gives

$$\begin{aligned} P(X_0 = i_0, X_1 = i_1, \dots, X_k = i_k) \\ = P(X_0 = i_0) \prod_{r=1}^k P(f(i_{r-1}, \xi_r) = i_r), \end{aligned}$$

because $X_0, \xi_1, \xi_2, \dots$ are independent.

Similarly,

$$\begin{aligned} P(X_0 = i_0, \dots, X_n = i, X_{n+1} = j) \\ = P(X_0 = i_0) \left[\prod_{r=1}^n P(f(i_{r-1}, \xi_r) = i_r) \right] P(f(i, \xi_{n+1}) = j). \end{aligned}$$

Dividing by $P(X_0 = i_0, \dots, X_n = i)$ yields

$$P(X_{n+1} = j \mid X_n = i, X_{n-1} = i_{n-1}, \dots, X_0 = i_0) = P(f(i, \xi_{n+1}) = j).$$

Thus the conditional probability depends only on $X_n = i$.

Transition Probability from the Recursion

From the recursion $X_{n+1} = f(X_n, \xi_{n+1})$,

$$P(X_{n+1} = j \mid X_n = i) = P(f(i, \xi_{n+1}) = j).$$

Equivalently,

$$P(X_{n+1} = j \mid X_n = i) = P(X_{n+1} = j, X_n = i) / P(X_n = i),$$

and the numerator can be decomposed by summing over all possible past states:

$$P(X_{n+1} = j, X_n = i) = \sum_{i_0, i_1, \dots, i_{n-1} \in S} P(X_0 = i_0, X_1 = i_1, \dots, X_n = i, X_{n+1} = j).$$

Using independence of $X_0, \xi_1, \dots, \xi_{n+1}$, this becomes

$$P(X_n = i) P(f(i, \xi_{n+1}) = j),$$

so after dividing by $P(X_n = i)$,

$$P(X_{n+1} = j \mid X_n = i) = P(f(i, \xi_{n+1}) = j).$$

Homogeneity of the Random Recursion Chain

For the process defined by

$$X_{n+1} = f(X_n, \xi_{n+1}),$$

we have

$$P(X_{n+1} = j \mid X_n = i) = P(f(i, \xi_{n+1}) = j).$$

Since the random variables $\xi_1, \xi_2, \dots$ are identically distributed,

$$P(f(i, \xi_{n+1}) = j) = P(f(i, \xi_1) = j).$$

Thus

$$P(X_{n+1} = j \mid X_n = i) = P(f(i, \xi_1) = j),$$

which does not depend on n .

Hence $\{X_n : n \geq 0\}$ is a homogeneous Markov chain with transition probabilities

$$p_{ij} = P(X_1 = j \mid X_0 = i) = P(f(i, \xi_1) = j).$$

Example: I.I.D. Sequence as a Markov Chain

Example

Let $\{X_n : n \geq 0\}$ be an i.i.d. sequence taking values in S . Then it is a Markov chain.

Indeed, for $i_0, i_1, \dots, i_n, j \in S$,

$$P(X_{n+1} = j \mid X_n = i_n, X_{n-1} = i_{n-1}, \dots, X_0 = i_0) = P(X_{n+1} = j),$$

because X_{n+1} is independent of the past.

Since the sequence is identically distributed,

$$P(X_{n+1} = j) = P(X_1 = j).$$

Thus the transition probabilities are

$$p_{ij} = P(X_{n+1} = j \mid X_n = i) = P(X_1 = j), \quad i, j \in S.$$

They do not depend on the present state i .

Transition Matrix for an I.I.D. Chain

For the i.i.d. Markov chain example, suppose the state space is

$$S = \mathbb{Z}_+$$

and

$$P(X_n = j) = p_j, \quad j \in S.$$

Then

$$p_{ij} = P(X_{n+1} = j \mid X_n = i) = P(X_{n+1} = j) = p_j.$$

Thus the transition probabilities do not depend on the current state i .

The transition matrix is

$$P = \begin{pmatrix} p_0 & p_1 & p_2 & p_3 & \cdots \\ p_0 & p_1 & p_2 & p_3 & \cdots \\ p_0 & p_1 & p_2 & p_3 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}.$$

Every row is repeated.

Example: Simple Random Walk

Example

Let $\{\varepsilon_i : i \geq 1\}$ be independent random variables taking values in $\{-1, 1\}$, with

$$P(\varepsilon_i = 1) = p, \quad P(\varepsilon_i = -1) = q, \quad p + q = 1.$$

Let $S_0 = s_0 \in \mathbb{Z}$ and define

$$S_n = S_0 + \sum_{i=1}^n \varepsilon_i, \quad n \geq 0.$$

Then $\{S_n : n \geq 0\}$ is a Markov chain.

Its transition probabilities are

$$p_{ij} = P(S_{n+1} = j \mid S_n = i) = \begin{cases} p, & j = i + 1, \\ q, & j = i - 1, \\ 0, & \text{otherwise.} \end{cases}$$

Simple Symmetric Random Walk Transition Matrix

For the simple random walk,

$$p_{ij} = P(f(i, \varepsilon) = j) = P(i + \varepsilon = j) = P(\varepsilon = j - i).$$

In the symmetric case,

$$P(\varepsilon = 1) = P(\varepsilon = -1) = \frac{1}{2}.$$

Therefore

$$p_{ij} = \begin{cases} \frac{1}{2}, & j = i + 1 \text{ or } j = i - 1, \\ 0, & \text{otherwise.} \end{cases}$$

With state space $\mathbb{Z}$, the transition matrix has one-half on the two off-diagonals:

$$P = \begin{pmatrix} \ddots & \ddots & \ddots & & & \\ & \frac{1}{2} & 0 & \frac{1}{2} & & \\ & & \frac{1}{2} & 0 & \frac{1}{2} & \\ & & & \frac{1}{2} & 0 & \ddots \\ & & & & \ddots & \ddots \end{pmatrix}.$$

General Random Walk as a Markov Chain

If

$$P(\varepsilon_i = +1) = p, \quad P(\varepsilon_i = -1) = 1 - p,$$

then $\{S_n : n \geq 0\}$ is a Markov chain with

$$p_{ij} = \begin{cases} p, & j = i + 1, \\ 1 - p, & j = i - 1, \\ 0, & \text{otherwise.} \end{cases}$$

More generally, let $\{\varepsilon_i : i \geq 1\}$ be i.i.d. random variables taking values in $\mathbb{Z}$ and define

$$S_{n+1} = S_n + \varepsilon_{n+1}.$$

Then $\{S_n : n \geq 0\}$ is a Markov chain on $\mathbb{Z}$.

The transition probabilities are

$$p_{ij} = P(S_{n+1} = j \mid S_n = i) = P(i + \varepsilon_{n+1} = j) = P(\varepsilon_{n+1} = j - i).$$

Higher-Dimensional Random Walk

Example

Let $\{\varepsilon_n : n \geq 1\}$ be i.i.d. random variables taking values in $\mathbb{Z}^d$. Define the higher-dimensional random walk

$$S_n = S_0 + \sum_{k=1}^n \varepsilon_k.$$

Then $\{S_n : n \geq 0\}$ is a Markov chain.

If

$$P(\varepsilon_1 = x) = \alpha_x, \quad x \in \mathbb{Z}^d,$$

then

$$\sum_{x \in \mathbb{Z}^d} \alpha_x = 1.$$

The transition probabilities are

$$p_{ij} = P(S_{n+1} = j \mid S_n = i) = P(i + \varepsilon_{n+1} = j) = P(\varepsilon_{n+1} = j - i) = \alpha_{j-i}.$$

Nearest-Neighbor Random Walk in Higher Dimensions

For a random walk on $\mathbb{Z}^d$,

$$S_{n+1} = S_n + \varepsilon_{n+1},$$

the transition probabilities are

$$p_{ij} = P(\varepsilon_1 = j - i) = \alpha_{j-i}.$$

Special Case

Let $e_1, \dots, e_d$ be the standard unit vectors in $\mathbb{Z}^d$. Suppose the increments take values in

$$\{-e_1, e_1, -e_2, e_2, \dots, -e_d, e_d\},$$

with

$$P(\varepsilon_1 = e_k) = \alpha_k, \quad P(\varepsilon_1 = -e_k) = \alpha_{-k}, \quad k = 1, \dots, d.$$

Then

$$\sum_{k=1}^d (\alpha_k + \alpha_{-k}) = 1.$$

The transition probabilities are

$$p_{ij} = \begin{cases} \alpha_k, & j = i + e_k, \\ \alpha_{-k}, & j = i - e_k, \\ 0, & \text{otherwise.} \end{cases}$$

Markov Chain Property

A process $\{X_n : n \geq 0\}$ is a Markov chain if, for all admissible states,

$$P(X_{n+1} = j \mid X_n = i, X_{n-1} = i_{n-1}, \dots, X_0 = i_0) = P(X_{n+1} = j \mid X_n = i).$$

In words, once the present state $X_n = i$ is known, the conditional distribution of the next state does not depend on the earlier history $X_{n-1}, \dots, X_0$.

Random Walks on Graphs

Examples of Markov chains include iid sequences and random walks. We now describe random walks on graphs.

Let

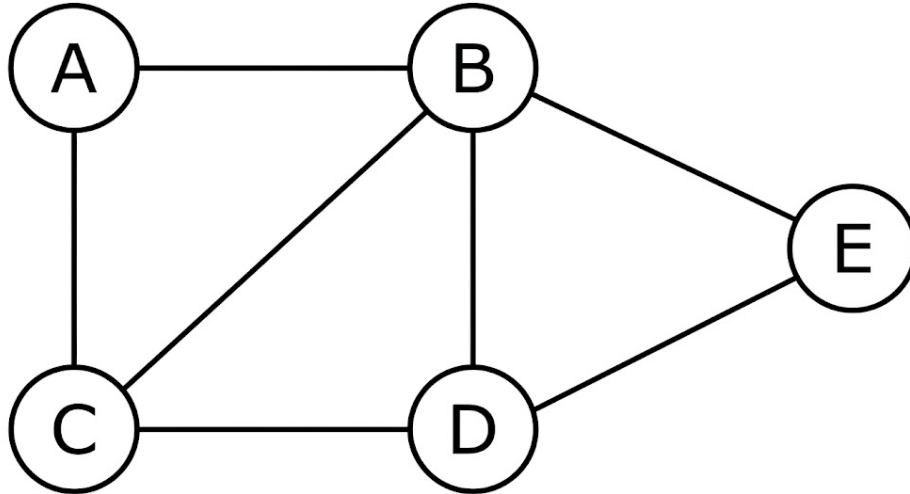
$$G = (V, E)$$

be a graph, where V is the set of vertices and E is the set of edges between the respective vertices. In the example below,

$$V = \{A, B, C, D, E\},$$

and

$$E = \{(A, B), (A, C), (B, C), (B, D), (B, E), (C, D), (D, E)\}.$$



Neighbourhoods and Degrees

For a vertex $v \in V$, define its neighbourhood by

$$N(v) = \{\text{neighbours of } v\}.$$

The degree of v is the number of vertices in its neighbourhood:

$$\deg(v) = |N(v)|.$$

For the graph from the previous page,

$$N(A) = \{B, C\}, \quad N(B) = \{A, C, D, E\}.$$

The simple random walk on the graph has transition probabilities

$$p_{uv} = \begin{cases} \frac{1}{\deg(u)}, & v \in N(u), \\ 0, & \text{otherwise.} \end{cases}$$

Transition Matrix for the Graph

For the graph with vertices ordered as A, B, C, D, E , the transition matrix of the simple random walk is

$$P = \begin{bmatrix} & A & B & C & D & E \\ A & 0 & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ B & \frac{1}{4} & 0 & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ C & \frac{1}{3} & \frac{1}{3} & 0 & \frac{1}{3} & 0 \\ D & 0 & \frac{1}{3} & \frac{1}{3} & 0 & \frac{1}{3} \\ E & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \end{bmatrix}.$$

Weighted Version

An extended version assigns symmetric weights

$$w_{uv} = w_{vu}$$

to every pair of vertices $u, v \in V$.

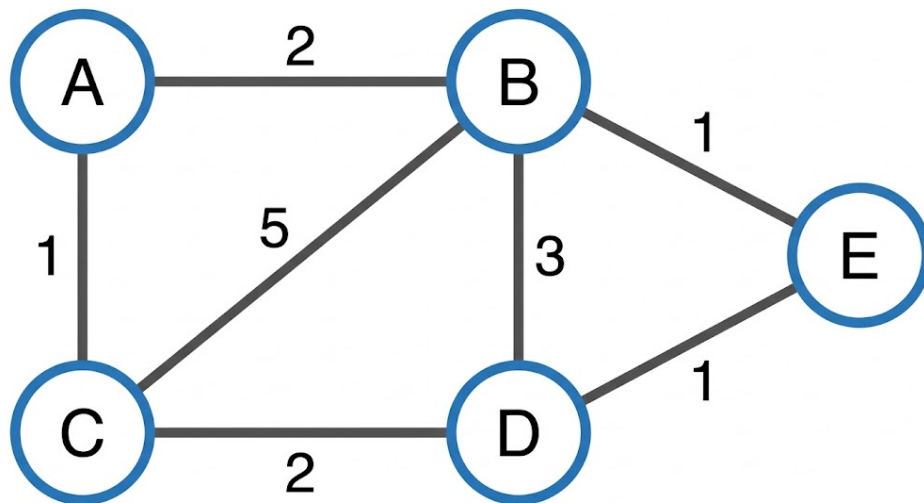
Weighted Random Walks on Graphs

Weights are non-negative, and some of them may be zero. For any $u \in V$, define

$$w_u = \sum_{v \in V} w_{uv} < \infty.$$

Then the transition probability from u to v is

$$p_{uv} = \frac{w_{uv}}{w_u}.$$



The degree model is a special case of the weighted model, obtained by setting

$$w_{uv} = \begin{cases} 1, & (u, v) \in E, \\ 0, & (u, v) \notin E. \end{cases}$$

PageRank Modification

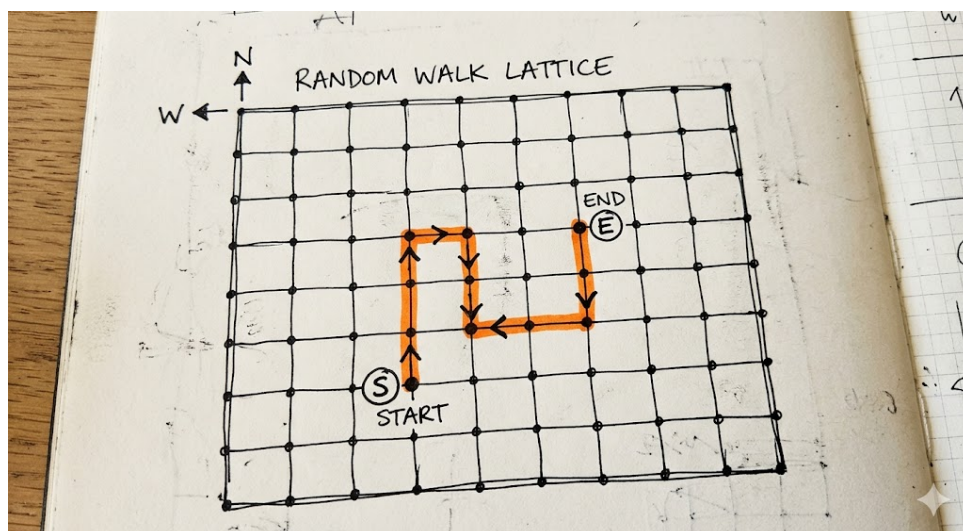
For the integer line graph,

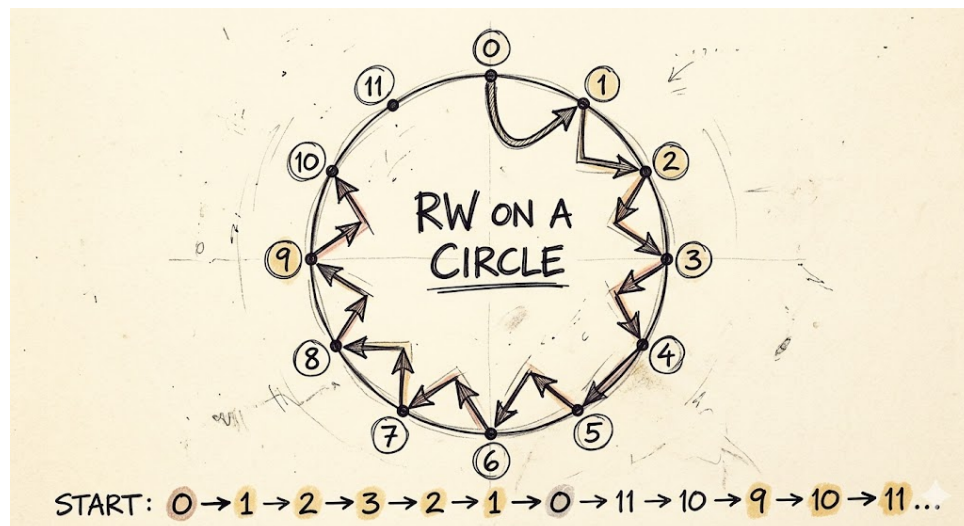
$$V = \mathbb{Z}, \quad E = \{(i, i+1) : i \in \mathbb{Z}\},$$

we can modify the transition probabilities using a finite set $\Gamma \subseteq V$.

$$p'_{uv} = (1 - \alpha)p_{uv} + \alpha q_{uv},$$
$$q_{uv} = \begin{cases} \frac{1}{|\Gamma|}, & v \in \Gamma, \\ 0, & \text{otherwise.} \end{cases}$$

Random Walks on a Lattice and a Circle





For a random walk considered modulo 4,

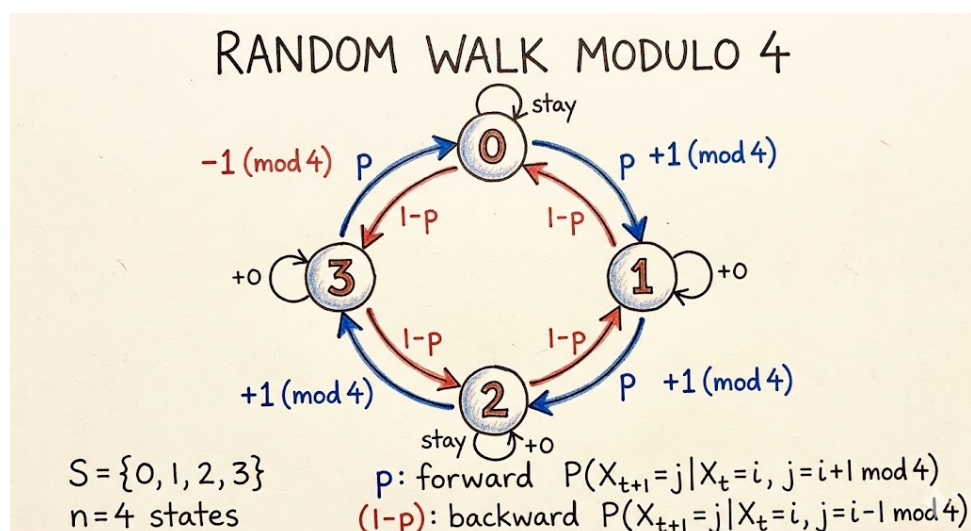
$$\lim_{n \rightarrow \infty} P((S_n \bmod 4) = 0) = \frac{1}{4}.$$

Think of rolling a die. If ξ_i denotes the result at time i , then

$$S_n = \sum_{i=1}^n \xi_i.$$

The event of interest is that S_n is divisible by 4.

Random Walk Modulo 4



Let

$$X_n \equiv S_n \pmod{4}, \quad X_0 = 0.$$

Then $\{X_n : n \geq 0\}$ is a Markov chain on the state space

$$S = \{0, 1, 2, 3\}.$$

A stochastic matrix is called doubly stochastic if every column sum is 1. For a nearest-neighbour walk on the four-state cycle, with clockwise probability p and counter-clockwise probability $q = 1 - p$, the transition matrix is

$$P = \begin{array}{c|cccc} & 0 & 1 & 2 & 3 \\ \hline 0 & 0 & p & 0 & q \\ 1 & q & 0 & p & 0 \\ 2 & 0 & q & 0 & p \\ 3 & p & 0 & q & 0 \end{array}.$$

Two-State Chain and Gambler's Ruin

Two-State Markov Chain

For a two-state Markov chain with state space $S = \{0, 1\}$,

$$P = \begin{array}{c|cc} & 0 & 1 \\ \hline 0 & p_{00} & p_{01} \\ 1 & p_{10} & p_{11} \end{array}, \quad p_{00} + p_{01} = 1, \quad p_{10} + p_{11} = 1.$$

Gambler's Ruin

Let X_n denote the fortune of player A after n trials. If initially

$$X_0 = a,$$

and the total fortune is $N = a + b$, then the state space is

$$S = \{0, 1, \dots, N\}.$$

Finite Gambler's Ruin

For the finite gambler's ruin chain, 0 and N are absorbing states. For $i = 1, \dots, N - 1$,

$$p(i, i + 1) = p, \quad p(i, i - 1) = q, \quad q = 1 - p.$$

The transition matrix has the form

$$P = \begin{array}{c|cccccc} & 0 & 1 & 2 & \dots & N-1 & N \\ \hline 0 & 1 & 0 & 0 & \dots & 0 & 0 \\ 1 & q & 0 & p & \dots & 0 & 0 \\ 2 & 0 & q & 0 & \ddots & 0 & 0 \\ \vdots & \vdots & \ddots & \ddots & \ddots & p & \vdots \\ N-1 & 0 & 0 & 0 & \dots & 0 & p \\ N & 0 & 0 & 0 & \dots & 0 & 1 \end{array}.$$

Gambler's Ruin Against an Infinitely Rich Opponent

When the opponent has infinite fortune, the state space is

$$S = \{0, 1, 2, \dots\}.$$

State 0 is absorbing, and for $i \geq 1$,

$$p(i, i+1) = p, \quad p(i, i-1) = q, \quad q = 1 - p.$$

The transition matrix is

$$P = \begin{array}{c|ccccc} & 0 & 1 & 2 & 3 & \cdots \\ \hline 0 & 1 & 0 & 0 & 0 & \cdots \\ 1 & q & 0 & p & 0 & \cdots \\ 2 & 0 & q & 0 & p & \cdots \\ 3 & 0 & 0 & q & 0 & \ddots \\ \vdots & \vdots & \vdots & \ddots & \ddots & \ddots \end{array}.$$

Birth and Death Chains

A Markov chain on the state space $\{0, 1, \dots, N\}$ is called a birth and death chain if

$$p_{ij} = 0 \quad \text{whenever } |i - j| > 1.$$

For a finite state space, the transition matrix has tridiagonal form

$$P = \begin{pmatrix} r_0 & p_0 & 0 & \cdots & 0 \\ q_1 & r_1 & p_1 & \cdots & 0 \\ 0 & q_2 & r_2 & p_2 & \cdots \\ \vdots & \vdots & \ddots & \ddots & \ddots \\ 0 & 0 & \cdots & q_N & r_N \end{pmatrix}.$$

The row sums give

$$r_0 + p_0 = 1, \quad q_N + r_N = 1, \quad q_i + r_i + p_i = 1 \quad (1 \leq i \leq N-1).$$

Finite State Birth and Death Chain

For the finite state space

$$S = \{0, 1, \dots, N\},$$

the transition matrix is

$$P = \begin{array}{c|ccccc} & 0 & 1 & 2 & \cdots & N \\ \hline 0 & r_0 & p_0 & 0 & \cdots & 0 \\ 1 & q_1 & r_1 & p_1 & \cdots & 0 \\ 2 & 0 & q_2 & r_2 & \ddots & 0 \\ \vdots & \vdots & \vdots & \ddots & \ddots & p_{N-1} \\ N & 0 & 0 & \cdots & q_N & r_N \end{array}.$$

The probabilities satisfy

$$r_0 + p_0 = 1, \quad q_N + r_N = 1, \quad q_i + r_i + p_i = 1.$$

Infinite State Birth and Death Chain

For the infinite state space $S = \{0, 1, 2, \dots\}$, the transition matrix is

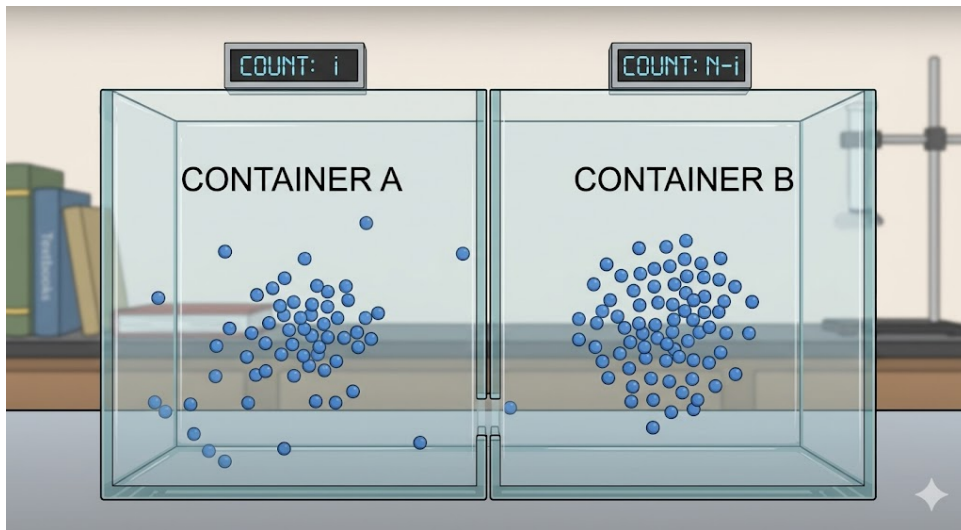
$$P = \begin{array}{c|ccccc} & 0 & 1 & 2 & 3 & \dots \\ \hline 0 & r_0 & p_0 & 0 & 0 & \dots \\ 1 & q_1 & r_1 & p_1 & 0 & \dots \\ 2 & 0 & q_2 & r_2 & p_2 & \dots \\ 3 & 0 & 0 & q_3 & r_3 & \ddots \\ \vdots & \vdots & \vdots & \vdots & \ddots & \ddots \end{array}$$

Here

$$r_0 + p_0 = 1, \quad q_i + r_i + p_i = 1 \quad (i \geq 1).$$

Ehrenfest Model of Diffusion of Gases

A total number N of particles are distributed in two containers A and B .

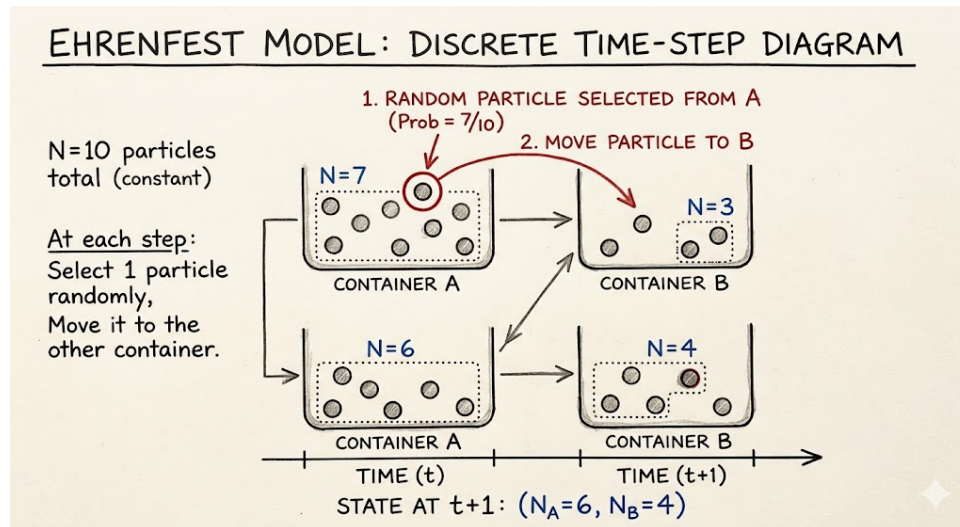


Let X_n be the number of particles in container A . Then

$$N - X_n$$

is the number of particles in container B .

Ehrenfest Model Dynamics



At every time point, choose one particle at random and move it to the other container.

If $X_n = i$, then

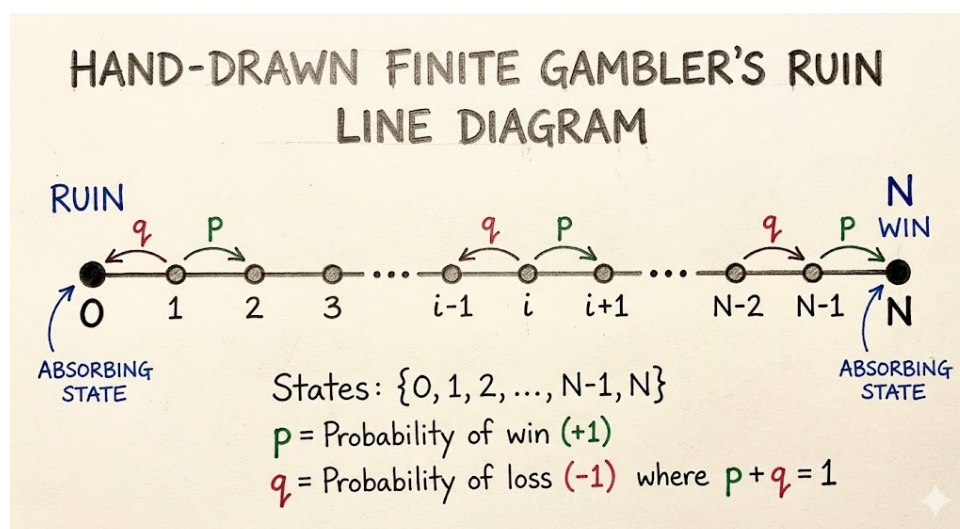
$$P(X_{n+1} = i + 1 \mid X_n = i) = \frac{N - i}{N} = 1 - \frac{i}{N},$$

and

$$P(X_{n+1} = i - 1 \mid X_n = i) = \frac{i}{N}.$$

Comments on Finite Gambler's Ruin and Examples

In the finite-state gambler's ruin problem, the endpoints are absorbing.



Some examples of Markov chains are:

1. Random walks.
2. Birth and death chains, including the Ehrenfest model of diffusion and the Bernoulli–Laplace model.
3. Queueing models.
4. Wright–Fisher models.
5. Branching processes.

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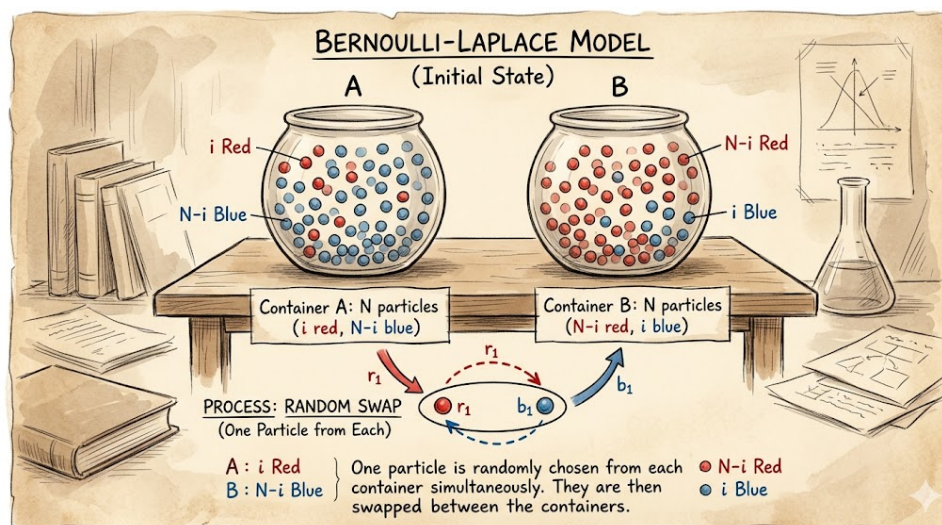
Markov Chain Examples

Examples of Markov chains:

1. Random walks.
2. Birth and death chains:

Ehrenfest model of diffusion, Bernoulli–Laplace model.
3. Queueing models.
4. Wright–Fisher models.
5. Branching processes.

Bernoulli–Laplace Model



There are two types of particles. The total number of particles in each container is N .

Dynamics

Pick one particle from each container and swap them.

Bernoulli–Laplace Transition Matrix

Let X_n be the number of particles of type 1 in container A . Then $\{X_n : n \geq 0\}$ is a Markov chain with state space

$$S = \{0, 1, 2, \dots, N\}.$$

The transition matrix has nonzero entries only on the three diagonals:

$$P = \begin{array}{c|cccccc} & 0 & 1 & 2 & \cdots & N-1 & N \\ \hline 0 & 0 & 1 & 0 & \cdots & 0 & 0 \\ 1 & p_{1,0} & p_{1,1} & p_{1,2} & \cdots & 0 & 0 \\ 2 & 0 & p_{2,1} & p_{2,2} & p_{2,3} & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \ddots & \ddots & \vdots \\ N-1 & 0 & 0 & \cdots & p_{N-1,N-2} & p_{N-1,N-1} & p_{N-1,N} \\ N & 0 & 0 & \cdots & 0 & 1 & 0 \end{array}.$$

For $0 \leq i \leq N$,

$$p_{i,i-1} = \left(\frac{i}{N}\right)^2, \quad p_{i,i} = \frac{2i(N-i)}{N^2}, \quad p_{i,i+1} = \left(\frac{N-i}{N}\right)^2.$$

Bernoulli–Laplace Probabilities and Queueing Model

The transition probabilities b_{ij} of the Bernoulli–Laplace model are

$$b_{ij} = \begin{cases} 0, & |i-j| > 1, \\ \frac{2i(N-i)}{N^2}, & j = i, \\ \left(\frac{N-i}{N}\right)^2, & j = i+1, \\ \left(\frac{i}{N}\right)^2, & j = i-1. \end{cases}$$

Queueing Model

Think of a Poisson-process setting where, at each time point, a bus comes and takes the first customer if there is any customer waiting.

Queueing Model

The bus then leaves. Between two time points, many customers may arrive.

Let ξ_{n+1} be the number of customers arriving between time points n and $n+1$. Assume that

$$\{\xi_n : n \geq 1\}$$

are i.i.d. random variables, and

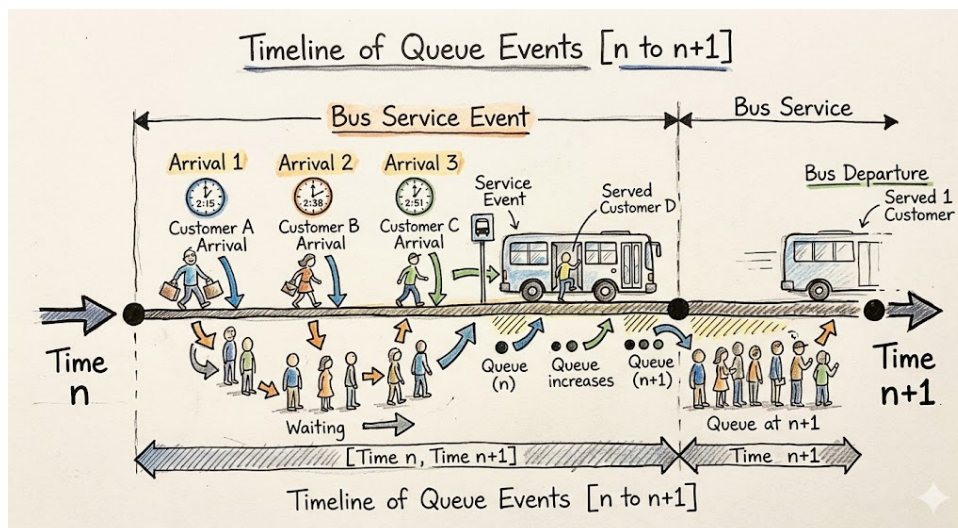
$$P(\xi_n = k) = \alpha_k, \quad k = 0, 1, 2, \dots,$$

with

$$\sum_{k=0}^{\infty} \alpha_k = 1.$$

Let X_n be the number of customers waiting at time n .

Queueing Model Update



If $X_n = 0$, then the handwritten update is

$$X_{n+1} = \max(0, \xi_{n+1} - 1).$$

If $X_n \geq 1$, then

$$X_{n+1} = X_n - 1 + \xi_{n+1}.$$

Thus

$$X_{n+1} = f(X_n, \xi_{n+1}),$$

and $\{X_n : n \geq 0\}$ is a Markov chain with state space

$$S = \{0, 1, 2, \dots\}.$$

Queueing Model Transition Matrix

For the queueing chain, the first row is special:

$$p_{0j} = \begin{cases} \alpha_0 + \alpha_1, & j = 0, \\ \alpha_{j+1}, & j \geq 1. \end{cases}$$

For $i \geq 1$,

$$p_{ij} = \begin{cases} 0, & j < i - 1, \\ \alpha_{j-i+1}, & j \geq i - 1. \end{cases}$$

Hence

$$P = \begin{array}{c|ccccc} & 0 & 1 & 2 & 3 & \cdots \\ \hline 0 & \alpha_0 + \alpha_1 & \alpha_2 & \alpha_3 & \alpha_4 & \cdots \\ 1 & \alpha_0 & \alpha_1 & \alpha_2 & \alpha_3 & \cdots \\ 2 & 0 & \alpha_0 & \alpha_1 & \alpha_2 & \cdots \\ 3 & 0 & 0 & \alpha_0 & \alpha_1 & \cdots \\ \vdots & \vdots & \vdots & \ddots & \ddots & \ddots \end{array}.$$

Wright–Fisher Model

Let X_n be the number of positions occupied by type A at time n . There are N positions, each occupied by one of the two types a or A .

Given X_n , the next generation is binomial:

$$X_{n+1} \sim \text{Bin}\left(N, \frac{X_n}{N}\right).$$

Thus if $X_n = i$, then the probability that an A is placed in any position of the next generation is i/N .

Wright–Fisher Transition Probabilities

The transition probability is

$$p_{ij} = P(X_{n+1} = j \mid X_n = i) = P\left(\text{Bin}\left(N, \frac{i}{N}\right) = j\right).$$

Hence

$$p_{ij} = \binom{N}{j} \left(\frac{i}{N}\right)^j \left(1 - \frac{i}{N}\right)^{N-j}.$$

The boundary states 0 and N are absorbing.

Branching Process

For a branching process, if the population at time n is Z_n , then

$$Z_{n+1} = \sum_{i=1}^{Z_n} \xi_i^{(n)}.$$

For $k \geq 1$, write

$$f(k, \epsilon) = \sum_{i=1}^k \epsilon_i.$$

Then

$$Z_{n+1} = f(Z_n, \epsilon_{n+1}).$$

The random variables ϵ_i are i.i.d. offspring variables taking values in

$$\{0, 1, 2, \dots\}.$$

Branching Process Transition Probabilities

For the branching process,

$$p_{ij} = P(Z_{n+1} = j \mid Z_n = i) = P\left(\sum_{m=1}^i \epsilon_m = j\right).$$

If $G(s)$ is the probability generating function of the offspring distribution, then p_{ij} is the coefficient of s^j in

$$\left(G(s)\right)^i.$$

Generic Finite Example

The notes next introduce a generic finite example with state space

$$S = \{0, 1, 2, 3, 4, 5, 6, 7\}.$$

Generic Finite Example Matrix

For the generic finite example with

$$S = \{0, 1, 2, 3, 4, 5, 6, 7\},$$

the handwritten transition matrix lists several visible rows as follows (blank entries are interpreted as 0):

$$P = \begin{array}{c|cccccccc} & 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ \hline 0 & \frac{1}{4} & \frac{3}{4} & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & \frac{1}{2} & 0 & \frac{1}{3} & 0 & 0 & \frac{1}{6} & 0 & 0 \\ 2 & 0 & 0 & \frac{1}{3} & 0 & \frac{1}{3} & 0 & \frac{1}{3} & 0 \\ 3 & \frac{1}{2} & 0 & 0 & \frac{1}{3} & 0 & \frac{1}{3} & 0 & 0 \\ 4 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 5 & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} \end{array}.$$

Polya's Urn Model

At time 0, an urn contains r red balls and b blue balls. Draw one ball at random, note its colour, replace it, and add c many balls of the same colour. Let X_n denote the colour of the ball drawn at the n -th draw.

Polya's Urn Is Not Markov in the Drawn Colour

The process

$$\{X_n : n \geq 0\}$$

with $X_n \in \{R, B\}$ is not a Markov chain.

For example, the probabilities

$$P(X_2 = R \mid X_1 = R) \quad \text{and} \quad P(X_2 = R \mid X_1 = R, X_0 = B)$$

need not be equal. The future draw depends on the composition of the urn, not only on the most recent colour.

Transition Diagrams

Given a transition matrix

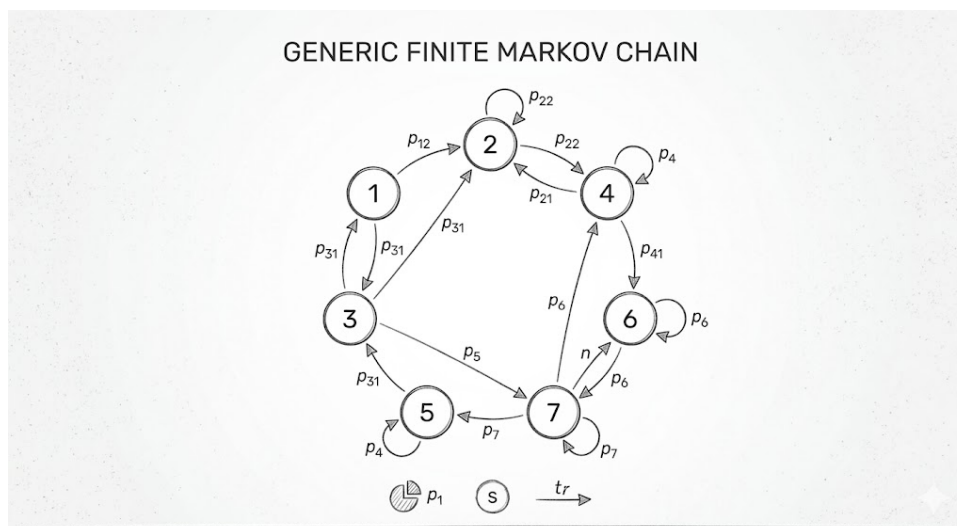
$$P = ((p_{ij})),$$

we form the directed graph whose vertices are the states in the state space S . There is a directed edge from i to j whenever

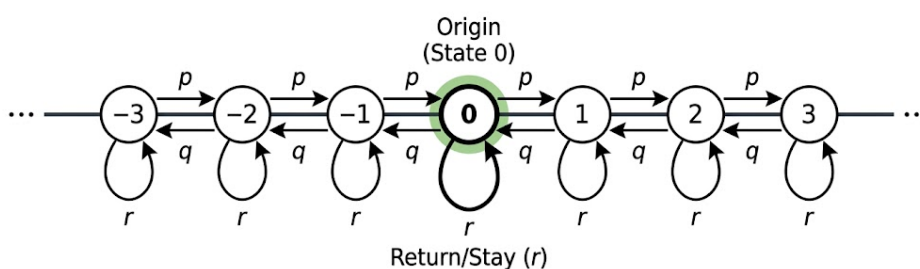
$$p_{ij} > 0.$$

Examples of Transition Diagrams

The page gives several hand-drawn transition diagrams.



One-Dimensional Random Walk Transition Diagram

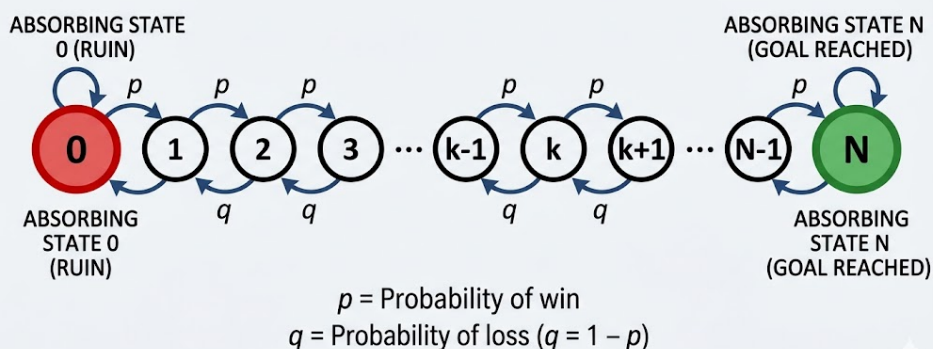


Transition Probabilities: p (Forward), q (Backward), r (Stay), with $p + q + r = 1$.

Transition Probabilities: p (Forward), q (Backward), r (Stay), with $p + q + r = 1$.

Gambler's Ruin Transition Diagram on a Finite Line

A model of random walk on $\{0, 1, \dots, N\}$ with two absorbing states.



The examples include a generic finite transition diagram, a random-walk transition diagram, and a gambler's ruin transition diagram.

Higher Transitions

For events $A_0, A_1, \dots, A_n$, the Markov property has the form

$$P(A_n \mid A_{n-1}, A_{n-2}, \dots, A_0) = P(A_n \mid A_{n-1}).$$

Equivalently, for a Markov chain,

$$P(A_n \mid A_0, A_1, \dots, A_{n-1}) = P(A_n \mid A_{n-1}).$$

Joint Distribution of a Markov Chain

For states $i_0, i_1, \dots, i_m$,

$$\begin{aligned} P(X_m = i_m, X_{m-1} = i_{m-1}, \dots, X_0 = i_0) \\ = P(X_m = i_m \mid X_{m-1} = i_{m-1}, \dots, X_0 = i_0) \\ \times P(X_{m-1} = i_{m-1}, \dots, X_0 = i_0). \end{aligned}$$

By the Markov property,

$$P(X_m = i_m \mid X_{m-1} = i_{m-1}, \dots, X_0 = i_0) = p_{i_{m-1}, i_m}.$$

Product Form of the Joint Distribution

Iterating the Markov property gives

$$P(X_m = i_m, X_{m-1} = i_{m-1}, \dots, X_0 = i_0) = \pi_{i_0} p_{i_0, i_1} p_{i_1, i_2} \cdots p_{i_{m-1}, i_m},$$

where

$$\pi_{i_0} = P(X_0 = i_0).$$

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Markov Chain: Joint Distribution

For events $A_0, A_1, \dots, A_n$,

$$\begin{aligned} P(A_n \cap A_{n-1} \cap \cdots \cap A_0) &= P(A_n \mid A_{n-1} \cap \cdots \cap A_0) \\ &\times P(A_{n-1} \mid A_{n-2} \cap \cdots \cap A_0) \\ &\times \cdots \times P(A_1 \mid A_0) P(A_0). \end{aligned}$$

For a Markov chain this becomes

$$P(X_m = i_m, \dots, X_0 = i_0) = P(X_0 = i_0) \prod_{r=1}^m p_{i_{r-1}, i_r}.$$

m -Step Transition Probabilities

The m -step transition probability is denoted by

$$p_{ij}^{(m)} = P(X_m = j \mid X_0 = i).$$

For $m = 2$,

$$\begin{aligned} p_{ij}^{(2)} &= P(X_2 = j \mid X_0 = i) \\ &= \sum_{k \in S} P(X_2 = j \mid X_1 = k, X_0 = i) P(X_1 = k \mid X_0 = i) \\ &= \sum_{k \in S} p_{ik} p_{kj}. \end{aligned}$$

Markov Chain Joint Distribution and Higher Transitions

Using the chain rule and the Markov property,

$$\begin{aligned} P(X_m = i_m, X_{m-1} = i_{m-1}, \dots, X_0 = i_0) \\ &= \left[P(X_m = i_m \mid X_{m-1} = i_{m-1}, \dots, X_0 = i_0) \right] \\ &\quad \times P(X_{m-1} = i_{m-1}, \dots, X_0 = i_0) \\ &= p_{i_{m-1}, i_m} P(X_{m-1} = i_{m-1}, \dots, X_0 = i_0). \end{aligned}$$

Continuing recursively gives

$$P(X_m = i_m, \dots, X_0 = i_0) = \pi_{i_0} p_{i_0, i_1} p_{i_1, i_2} \cdots p_{i_{m-1}, i_m}.$$

Powers of the Transition Matrix

For the two-step transition probabilities,

$$P^2 = \left((p_{ij}^{(2)}) \right), \quad p_{ij}^{(2)} = P_i(X_2 = j).$$

Here

$$p_{ij}^{(2)} \geq 0, \quad \sum_{j \in S} p_{ij}^{(2)} = 1.$$

Thus the entries are non-negative and the row sums are one, so P^2 is again a stochastic matrix.

$$\sum_{j \in S} p_{ij}^{(2)} = \sum_{j \in S} P_i(X_2 = j) = 1.$$

Exercise noted in class: show that not every stochastic matrix can arise as P^2 for another stochastic matrix P .

Induction for n -Step Transitions

Claim:

$$P^n = \left(\left(p_{ij}^{(n)} \right) \right) \quad \text{for all } n \geq 1.$$

Assume the claim is true for n . Then

$$p_{ij}^{(n+1)} = P(X_{n+1} = j \mid X_0 = i).$$

Split according to the state at time n :

$$\begin{aligned} p_{ij}^{(n+1)} &= \sum_{k \in S} P(X_{n+1} = j, X_n = k \mid X_0 = i) \\ &= \sum_{k \in S} P(X_{n+1} = j \mid X_n = k, X_0 = i) P(X_n = k \mid X_0 = i). \end{aligned}$$

The conditional-probability identity being used is

$$P(A \cap B \mid C) = P(A \mid B \cap C) P(B \mid C).$$

Induction Step Continued

By the Markov property,

$$P(X_{n+1} = j \mid X_n = k, X_0 = i) = P(X_{n+1} = j \mid X_n = k) = p_{kj}.$$

Therefore

$$\begin{aligned} p_{ij}^{(n+1)} &= \sum_{k \in S} p_{kj} P(X_n = k \mid X_0 = i) \\ &= \sum_{k \in S} p_{kj} p_{ik}^{(n)}. \end{aligned}$$

This is exactly the (i, j) -entry of $P^n P$. Hence

$$P^{n+1} = P^n P,$$

and the induction is complete.

Consequently, each P^n is also a stochastic matrix.

Chapman–Kolmogorov Equations

For every $m \geq 1$,

$$P^m = \left(\left(p_{ij}^{(m)} \right) \right).$$

Thus, for $m, n \geq 1$,

$$P^{m+n} = P^m P^n = \left(\left(p_{ij}^{(m)} \right) \right) \left(\left(p_{ij}^{(n)} \right) \right).$$

Equivalently, for $i, j \in S$,

$$p_{ij}^{(m+n)} = \sum_{k \in S} p_{ik}^{(m)} p_{kj}^{(n)}.$$

These are the Chapman–Kolmogorov equations.

Including Zero Steps

Define

$$p_{ij}^{(0)} = \begin{cases} 1, & i = j, \\ 0, & i \neq j, \end{cases} = \delta_{ij}.$$

Thus

$$P^0 = I.$$

With this convention, for all $m, n \geq 0$,

$$P^{m+n} = P^m P^n,$$

or entrywise,

$$p_{ij}^{(m+n)} = \sum_{k \in S} p_{ik}^{(m)} p_{kj}^{(n)}.$$

Examples of Transition Matrix Powers

IID Sequence

Let $\{X_n : n \geq 0\}$ be iid with

$$P(X_n = j) = \alpha_j, \quad j \in S.$$

Then, for every $m \geq 1$,

$$P(X_m = j \mid X_0 = i) = P(X_m = j) = \alpha_j.$$

Hence all rows of the transition matrix are identical:

$$P = P^m = \begin{pmatrix} \alpha_0 & \alpha_1 & \alpha_2 & \cdots \\ \alpha_0 & \alpha_1 & \alpha_2 & \cdots \\ \alpha_0 & \alpha_1 & \alpha_2 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}.$$

Two-State Markov Chain

Let

$$S = \{0, 1\}, \quad P = \begin{pmatrix} p_{00} & p_{01} \\ p_{10} & p_{11} \end{pmatrix},$$

with $0 < p_{ij} < 1$. Write

$$P^m = \begin{pmatrix} p_{00}^{(m)} & p_{01}^{(m)} \\ p_{10}^{(m)} & p_{11}^{(m)} \end{pmatrix}.$$

By Chapman–Kolmogorov,

$$\begin{aligned} p_{00}^{(m+1)} &= \sum_{k \in S} p_{0k}^{(m)} p_{k0} \\ &= p_{00}^{(m)} p_{00} + p_{01}^{(m)} p_{10}. \end{aligned}$$

Two-State Recursion

Since

$$p_{01}^{(m)} = 1 - p_{00}^{(m)},$$

we get

$$\begin{aligned} p_{00}^{(m+1)} &= p_{00}^{(m)} p_{00} + (1 - p_{00}^{(m)}) p_{10} \\ &= p_{10} + p_{00}^{(m)} (p_{00} - p_{10}). \end{aligned}$$

Let

$$\lambda_2 = p_{00} - p_{10}.$$

Then

$$p_{00}^{(m+1)} = p_{10} + \lambda_2 p_{00}^{(m)}.$$

Iterating,

$$\begin{aligned} p_{00}^{(m+1)} &= p_{10} + \lambda_2 (p_{10} + \lambda_2 p_{00}^{(m-1)}) \\ &= p_{10} + p_{10} \lambda_2 + \lambda_2^2 p_{00}^{(m-1)} \\ &= \dots \end{aligned}$$

Two-State Limit

The iteration gives

$$p_{00}^{(m)} = p_{10} \sum_{j=0}^{m-1} (p_{00} - p_{10})^j (p_{00} - p_{10})^m.$$

Since

$$1 - (p_{00} - p_{10}) = p_{01} + p_{10},$$

we may write

$$p_{00}^{(m)} = p_{10} \frac{1 - \lambda_2^m}{p_{01} + p_{10}} + \lambda_2^m = \pi_0 + \lambda_2^m (1 - \pi_0),$$

where

$$\pi_0 = \frac{p_{10}}{p_{01} + p_{10}}.$$

Because $|\lambda_2| < 1$,

$$p_{00}^{(m)} \longrightarrow \pi_0.$$

Also

$$p_{01}^{(m)} = 1 - p_{00}^{(m)} \longrightarrow \pi_1, \quad \pi_1 = 1 - \pi_0 = \frac{p_{01}}{p_{01} + p_{10}}.$$

Remaining Two-State Entries

Similarly,

$$p_{11}^{(m)} = p_{01} \sum_{j=0}^{m-1} (p_{11} - p_{01})^j (p_{11} - p_{01})^m.$$

For a two-state stochastic matrix,

$$p_{11} - p_{01} = p_{00} - p_{10} = \lambda_2.$$

Therefore

$$p_{11}^{(m)} = \pi_1 + \lambda_2^m(1 - \pi_1) \longrightarrow \pi_1.$$

Consequently,

$$p_{10}^{(m)} = 1 - p_{11}^{(m)} \longrightarrow \pi_0.$$

Thus every row of P^m converges to (π_0, π_1) .

Rate of Convergence in the Two-State Chain

From the preceding formula,

$$|p_{00}^{(m)} - \pi_0| = |\lambda_2^m(1 - \pi_0)| \leq C|\lambda_2|^m.$$

Also,

$$p_{01}^{(m)} = 1 - p_{00}^{(m)} = \pi_1 + \lambda_2^m C_{01},$$

and hence

$$|p_{01}^{(m)} - \pi_1| \leq C'|\lambda_2|^m.$$

Here

$$\pi_0 = \frac{p_{10}}{p_{10} + p_{01}}, \quad \pi_1 = \frac{p_{01}}{p_{10} + p_{01}}.$$

Convergence of the Other Row

The same calculation gives

$$\begin{aligned} p_{11}^{(m)} &= \frac{p_{01}}{p_{01} + p_{10}} + (p_{11} - p_{01})^m \left(1 - \frac{p_{01}}{p_{01} + p_{10}}\right) \\ &= \pi_1 + \lambda_2^m C_{11}. \end{aligned}$$

Thus

$$p_{11}^{(m)} \longrightarrow \pi_1.$$

Since

$$p_{10}^{(m)} = 1 - p_{11}^{(m)},$$

we also have

$$p_{10}^{(m)} \longrightarrow \pi_0.$$

Stationary Distribution for the Two-State Chain

Combining the limits,

$$\begin{pmatrix} p_{00}^{(m)} & p_{01}^{(m)} \\ p_{10}^{(m)} & p_{11}^{(m)} \end{pmatrix} \longrightarrow \begin{pmatrix} \pi_0 & \pi_1 \\ \pi_0 & \pi_1 \end{pmatrix}.$$

Let

$$\bar{\pi} = (\pi_0, \pi_1).$$

Check that

$$\bar{\pi}P = \bar{\pi}, \quad \bar{\pi}\mathbf{1} = 1.$$

The equation

$$\bar{\pi}P = \bar{\pi}$$

is called the balance equation, and $\bar{\pi}$ is called a stationary distribution.

Stationarity Check

Assume

$$P(X_0 = 0) = \pi_0, \quad P(X_0 = 1) = \pi_1.$$

Then

$$\begin{aligned} P(X_1 = 0) &= P(X_1 = 0 \mid X_0 = 0)P(X_0 = 0) \\ &\quad + P(X_1 = 0 \mid X_0 = 1)P(X_0 = 1) \\ &= \pi_0 p_{00} + \pi_1 p_{10} \\ &= \pi_0. \end{aligned}$$

Similarly,

$$P(X_1 = 1) = \pi_1.$$

Thus, if $X_0 \sim \bar{\pi}$, then $X_1 \sim \bar{\pi}$.

Recurrence and Transience

A state i is recurrent if

$$f_{ii}^* = P(X_n = i \text{ for some } n \geq 1 \mid X_0 = i) = 1.$$

Theorem:

$$i \text{ is recurrent} \iff \sum_{n=0}^{\infty} p_{ii}^{(n)} = \infty.$$

Notes:

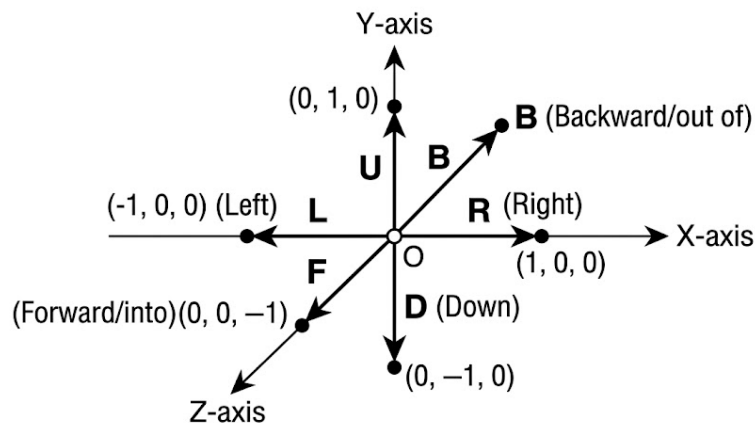
- The one-dimensional simple symmetric random walk is recurrent, but the asymmetric random walk is transient.
- The two-dimensional simple symmetric random walk is recurrent.

Example: Three-Dimensional Simple Symmetric Random Walk

For the three-dimensional simple symmetric random walk, from each point the chain moves to one of the six nearest neighbours with probability $1/6$.

3D Simple Symmetric Random Walk at Origin O

Six possible steps ($p = 1/6$ each) from origin to nearest neighbors: L, R, U, D, F, B.



For the origin $0 = (0, 0, 0)$,

$$p_{00}^{(n)} = 0 \quad \text{if } n \text{ is odd.}$$

For $n = 2m$, a return path must contain equal numbers of positive and negative moves in each coordinate direction. If the three pairs of opposite directions are used

$$l, \quad t, \quad m - l - t$$

times, respectively, then

$$0 \leq l, t, \quad l + t \leq m.$$

Return Probability in Three Dimensions

For $n = 2m$,

$$p_{00}^{(2m)} = \sum_{\substack{0 \leq l, t \\ l+t \leq m}} \frac{(2m)!}{(l! t! (m-l-t)!)} \frac{1}{6^{2m}}.$$

Rewrite this as

$$p_{00}^{(2m)} = \binom{2m}{m} \frac{1}{2^{2m}} \sum_{\substack{0 \leq l, t \\ l+t \leq m}} \left[\frac{m!}{l! t! (m-l-t)!} \frac{1}{3^m} \right]^2.$$

Let

$$\alpha_{l,t} = \frac{m!}{l! t! (m-l-t)!} \frac{1}{3^m}.$$

This is the multinomial probability

$$\alpha_{l,t} = P\{M_m = (l, t, m-l-t)\},$$

where M_m has cell probabilities $(1/3, 1/3, 1/3)$.

Bounding the Multinomial Sum

Since the $\alpha_{l,t}$'s are the probabilities in the multinomial distribution,

$$\sum_{\substack{0 \leq l,t \\ l+t \leq m}} \alpha_{l,t} = 1.$$

Therefore

$$\begin{aligned} p_{\mathbf{00}}^{(2m)} &= \binom{2m}{m} \frac{1}{2^{2m}} \sum_{\substack{0 \leq l,t \\ l+t \leq m}} \alpha_{l,t}^2 \\ &\leq \binom{2m}{m} \frac{1}{2^{2m}} \sum_{\substack{0 \leq l,t \\ l+t \leq m}} \left(\max_{\substack{0 \leq l,t \\ l+t \leq m}} \alpha_{l,t} \right) \alpha_{l,t} \\ &= \binom{2m}{m} \frac{1}{2^{2m}} \max_{\substack{0 \leq l,t \\ l+t \leq m}} \alpha_{l,t}. \end{aligned}$$

Maximal Multinomial Term

Show that

$$\alpha_{l,t} \leq \begin{cases} \alpha_{k,k}, & m = 3k, \\ \alpha_{k+1,k}, & m = 3k+1, \\ \alpha_{k+1,k+1}, & m = 3k+2. \end{cases}$$

For the case $m = 3k$,

$$\begin{aligned} p_{\mathbf{00}}^{(6k)} &\leq \binom{6k}{3k} \frac{1}{2^{6k}} \alpha_{k,k} \\ &= \frac{(6k)!}{(3k)!(3k)!} \frac{1}{2^{6k}} \cdot \frac{(3k)!}{k!k!k!} \frac{1}{3^{3k}}. \end{aligned}$$

Stirling Bound

Using Stirling's formula,

$$n! \sim \sqrt{2\pi n} \left(\frac{n}{e}\right)^n,$$

the preceding expression is of order

$$\frac{\sqrt{6\pi k} \left(\frac{6k}{e}\right)^{6k}}{\left(\sqrt{3\pi k} \left(\frac{3k}{e}\right)^{3k}\right)^2} \cdot \frac{1}{2^{6k}} \cdot \frac{\sqrt{3\pi k} \left(\frac{3k}{e}\right)^{3k}}{\left(\sqrt{\pi k} \left(\frac{k}{e}\right)^k\right)^3} \cdot \frac{1}{3^{3k}}.$$

Hence

$$p_{\mathbf{00}}^{(6k)} \leq \frac{C_1}{k^{3/2}}.$$

The other residue classes are bounded similarly:

$$p_{\mathbf{00}}^{(6k+2)} \leq \frac{C_2}{k^{3/2}}, \quad p_{\mathbf{00}}^{(6k+4)} \leq \frac{C_3}{k^{3/2}}.$$

Three-Dimensional Walk is Transient

Thus, for $r = 0, 2, 4$,

$$p_{\mathbf{00}}^{(6k+r)} \leq \frac{C}{k^{3/2}}.$$

Therefore

$$\sum_{n=0}^{\infty} p_{\mathbf{00}}^{(n)} = \sum_{m=0}^{\infty} p_{\mathbf{00}}^{(2m)} = \sum_{k=0}^{\infty} p_{\mathbf{00}}^{(6k)} + \sum_{k=0}^{\infty} p_{\mathbf{00}}^{(6k+2)} + \sum_{k=0}^{\infty} p_{\mathbf{00}}^{(6k+4)} < \infty.$$

Hence 0 is transient. By irreducibility of the three-dimensional simple symmetric random walk, every state is transient.

Communication Classes

We define a relation between states of a chain.

We say that i leads to j , and write

$$i \rightarrow j,$$

if

$$p_{ij}^{(n)} > 0 \quad \text{for some } n \geq 0.$$

We say that i communicates with j , and write

$$i \leftrightarrow j,$$

if

$$i \rightarrow j \quad \text{and} \quad j \rightarrow i.$$

Communication is an Equivalence Relation

Communication is an equivalence relation.

- Reflexive: by definition,

$$p_{ii}^{(0)} = 1 > 0,$$

so $i \leftrightarrow i$.

- Symmetric: if $i \leftrightarrow j$, then $j \leftrightarrow i$.
- Transitive: if $i \leftrightarrow j$ and $j \leftrightarrow k$, then $i \leftrightarrow k$.

Proof of Transitivity

Suppose

$$i \leftrightarrow j \quad \text{and} \quad j \leftrightarrow k.$$

Then there exist $m_1, n_1 \geq 0$ such that

$$p_{ij}^{(m_1)} > 0, \quad p_{ji}^{(n_1)} > 0,$$

and there exist $m_2, n_2 \geq 0$ such that

$$p_{jk}^{(m_2)} > 0, \quad p_{kj}^{(n_2)} > 0.$$

By Chapman–Kolmogorov,

$$p_{ik}^{(m_1+m_2)} \geq p_{ij}^{(m_1)} p_{jk}^{(m_2)} > 0,$$

so $i \rightarrow k$. Similarly,

$$p_{ki}^{(n_2+n_1)} \geq p_{kj}^{(n_2)} p_{ji}^{(n_1)} > 0,$$

so $k \rightarrow i$. Thus $i \leftrightarrow k$.

Equivalence Classes

Similarly, if $k \rightarrow i$, then for suitable $m_1, m_2 \geq 0$,

$$p_{ki}^{(m_2+m_1)} \geq p_{kj}^{(m_2)} p_{ji}^{(m_1)} > 0.$$

Thus communication is an equivalence relation. The equivalence relation partitions the state space into disjoint sets called equivalence classes, or communicating classes.

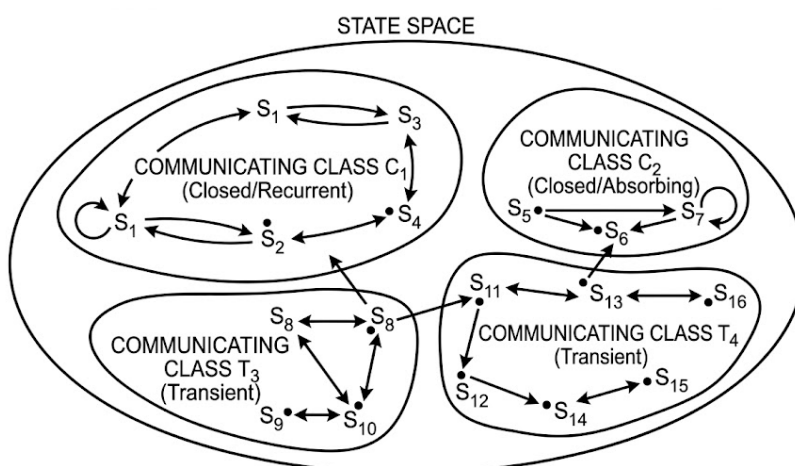


FIGURE 2.1: State Space Partitioned into Disjoint Communicating Classes (C_1, C_2, T_3, T_4)

Class Properties

For any two states $i, j \in C$, communication holds:

$$i \leftrightarrow j.$$

If $i \in C_1$ and $j \in C_2$ with $C_1 \neq C_2$, then i and j do not communicate; either $i \nleftrightarrow j$, or $j \nleftrightarrow i$, or both.

Theorem: recurrence/transience is a class property.

Theorem: positive recurrence/null recurrence is also a class property.

Two-State Communication Examples

Let

$$S = \{1, 2\}, \quad P = \begin{pmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{pmatrix}.$$

For example,

$$P = \begin{pmatrix} 1 & 0 \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}.$$

If $p_{12} = 0$, then $1 \nleftrightarrow 2$. If $p_{21} > 0$, then $2 \rightarrow 1$. Hence the communicating classes are

$$C_1 = \{1\}, \quad C_2 = \{2\}.$$

If instead

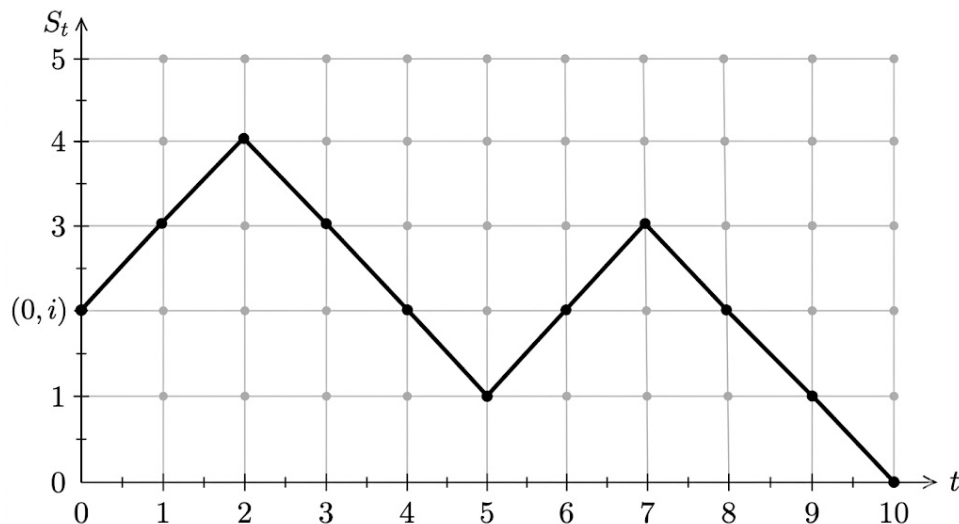
$$p_{12} > 0, \quad p_{21} > 0,$$

then

$$C_1 = \{1, 2\}.$$

Random Walk on the Integers

For a random walk on $\mathbb{Z}$, all states communicate with each other.



If $i > 0$, then one path from 0 to i consists of i upward/rightward moves, so

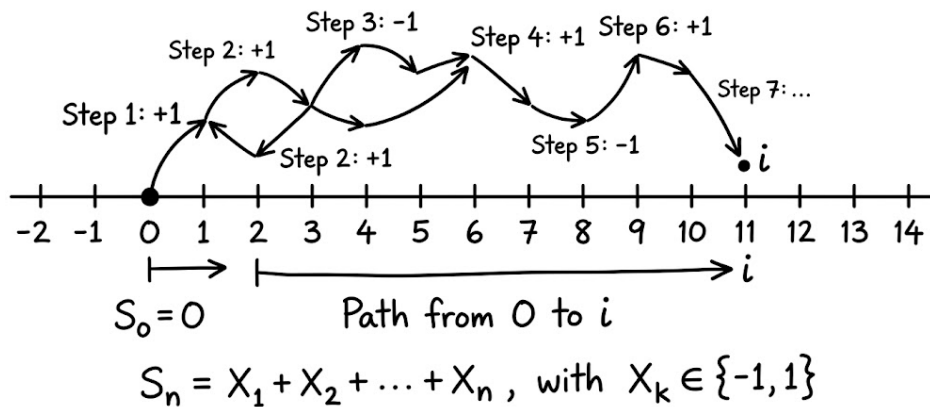
$$p_{0i}^{(i)} \geq p^i > 0.$$

If $i < 0$, then one path from 0 to i consists of $-i$ downward/leftward moves, so

$$p_{0i}^{(-i)} \geq q^{-i} > 0.$$

Similarly $i \rightarrow 0$, so $i \leftrightarrow 0$.

ONE-DIMENSIONAL RANDOM WALK ON INTEGERS



Hence all states communicate with each other and the single communicating class is

$$C = S.$$

Definition: if a chain has only one communicating class, it is called irreducible.

Example of Communicating Classes

Let

$$S = \{0, 1, 2, 3, 4, 5, 6, 7\}.$$

In the transition matrix below, an asterisk denotes a positive transition probability and 0 denotes a zero transition probability:

	0	1	2	3	4	5	6	7
0	*	0	0	*	0	0	0	0
1	*	0	*	0	0	0	*	0
2	0	0	0	0	*	0	0	*
3	*	0	0	*	0	0	0	0
4	0	0	*	0	*	0	0	*
5	*	*	*	*	*	*	*	*
6	*	0	0	0	*	0	0	*
7	0	0	*	0	*	0	0	*

The communicating classes indicated are

$$C_1 = \{0, 3\}, \quad C_2 = \{2, 4, 7\}.$$

Both C_1 and C_2 are recurrent classes.

The singleton classes

$$C_3 = \{1\}, \quad C_4 = \{6\}, \quad C_5 = \{5\}$$

are transient.

Gambler's Ruin

For gambler's ruin on

$$S = \{0, 1, 2, \dots, N\},$$

the transition matrix has absorbing endpoints and nearest-neighbour transitions in the interior:

$$P = \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \\ q & 0 & p & \cdots & 0 \\ 0 & q & 0 & p & \cdots \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & q & 0 & p \\ 0 & \cdots & 0 & 0 & 1 \end{pmatrix}.$$

The communicating classes are

$$C_1 = \{0\}, \quad C_2 = \{N\},$$

and both are recurrent. The remaining class is

$$C_3 = \{1, 2, \dots, N-1\},$$

which is transient.

For instance,

$$1 \rightarrow 0 \quad \text{but} \quad 0 \not\rightarrow 1.$$

Recurrence and Transience Criterion

The scan repeats the recurrence/transience criterion. A state i is recurrent if

$$f_{ii}^* = P(X_n = i \text{ for some } n \geq 1 \mid X_0 = i) = 1.$$

Equivalently,

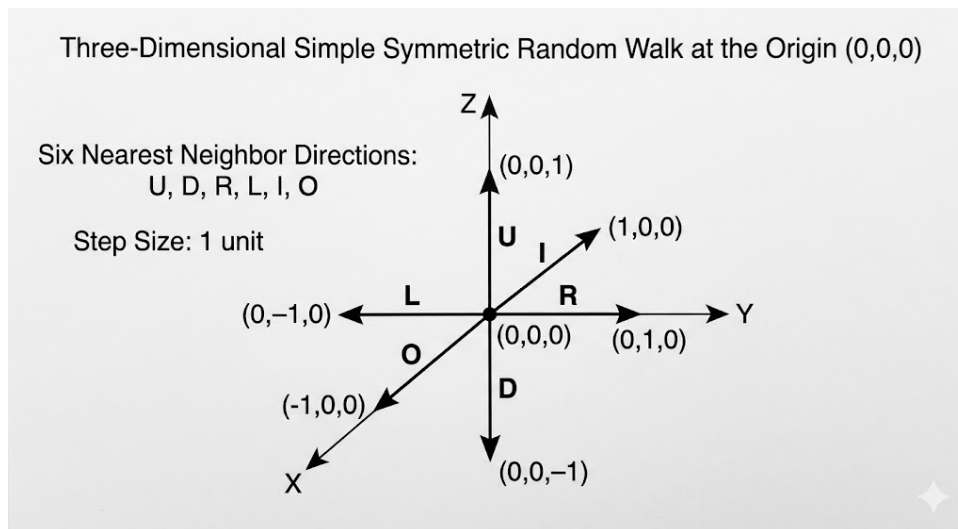
$$i \text{ is recurrent} \iff \sum_{n=0}^{\infty} p_{ii}^{(n)} = \infty.$$

The one-dimensional simple symmetric random walk is recurrent, whereas the asymmetric random walk is transient. The two-dimensional simple symmetric random walk is recurrent.

Three-Dimensional Simple Symmetric Random Walk

For the three-dimensional simple symmetric random walk,

$$p_{00}^{(n)} = 0 \quad \text{if } n \text{ is odd.}$$



For $n = 2m$, returning to 0 requires equal numbers of moves in each pair of opposite directions. If the numbers of such pairs are l , t , and $m - l - t$, then

$$0 \leq l, t, \quad l + t \leq m.$$

For fixed l, t , the number of admissible words is

$$\frac{(2m)!}{l! l! t! t! (m - l - t)! (m - l - t)!},$$

and each word has probability $(1/6)^{2m}$.

Return Probability Formula Repeated

For $n = 2m$,

$$p_{00}^{(2m)} = \sum_{\substack{0 \leq l, t \\ l+t \leq m}} \frac{(2m)!}{(l! l! t! t! (m - l - t)! (m - l - t)!)} \frac{1}{6^{2m}}.$$

Equivalently,

$$p_{00}^{(2m)} = \binom{2m}{m} \frac{1}{2^{2m}} \sum_{\substack{0 \leq l, t \\ l+t \leq m}} (\alpha_{l,t})^2,$$

where

$$\alpha_{l,t} = \frac{m!}{l! l! t! t! (m - l - t)! (m - l - t)!} \frac{1}{3^m}.$$

This is the probability of the multinomial cell $(l, t, m - l - t)$ with probabilities $(1/3, 1/3, 1/3)$.

Bounding the Repeated Multinomial Sum

Again,

$$\sum_{\substack{0 \leq l, t \\ l+t \leq m}} \alpha_{l,t} = 1.$$

Thus

$$\begin{aligned} p_{00}^{(2m)} &= \binom{2m}{m} \frac{1}{2^{2m}} \sum_{\substack{0 \leq l, t \\ l+t \leq m}} \alpha_{l,t}^2 \\ &\leq \binom{2m}{m} \frac{1}{2^{2m}} \max_{\substack{0 \leq l, t \\ l+t \leq m}} \alpha_{l,t}. \end{aligned}$$

The maximum occurs at the most balanced multinomial cell.

Maximal Term Cases Repeated

The maximal multinomial term satisfies

$$\alpha_{l,t} \leq \begin{cases} \alpha_{k,k}, & m = 3k, \\ \alpha_{k+1,k}, & m = 3k+1, \\ \alpha_{k+1,k+1}, & m = 3k+2. \end{cases}$$

In the case $m = 3k$,

$$\begin{aligned} p_{00}^{(6k)} &\leq \binom{6k}{3k} \frac{1}{2^{6k}} \alpha_{k,k} \\ &= \frac{(6k)!}{(3k)!(3k)!} \frac{1}{2^{6k}} \cdot \frac{(3k)!}{k!k!k!} \frac{1}{3^{3k}}. \end{aligned}$$

Transience Estimate (continued)

Using Stirling's approximation in the return-probability estimate gives a term of the form

$$\frac{\sqrt{\pi 6k} \left(\frac{6k}{e}\right)^{6k}}{\left(\sqrt{\pi 3k} \left(\frac{3k}{e}\right)^{3k}\right)^2} \cdot \frac{1}{2^{6k}}.$$

The same Stirling estimate is applied to the neighbouring even values of the time. Hence there is a constant $C > 0$ such that

$$p_{0,0}^{(6k)} \leq \frac{C}{k^{3/2}}, \quad p_{0,0}^{(6k+2)} \leq \frac{C}{k^{3/2}}, \quad p_{0,0}^{(6k+4)} \leq \frac{C}{k^{3/2}}.$$

Transience Estimate (conclusion)

Thus, for $n = 0, 2, 4$,

$$p_{0,0}^{(6k+n)} \leq \frac{C}{k^{3/2}}.$$

Therefore

$$\begin{aligned}\sum_{n=0}^{\infty} p_{0,0}^{(n)} &= \sum_{m=0}^{\infty} p_{0,0}^{(2m)} \\ &= \sum_{k=0}^{\infty} p_{0,0}^{(6k)} + \sum_{k=0}^{\infty} p_{0,0}^{(6k+2)} + \sum_{k=0}^{\infty} p_{0,0}^{(6k+4)} \\ &< \infty.\end{aligned}$$

Hence 0 is transient.

Communication Classes

We define a relation between states of a Markov chain.

We say that i leads to j , written

$$i \rightsquigarrow j,$$

if and only if

$$p_{ij}^{(n)} > 0 \quad \text{for some } n > 0.$$

We say that i communicates with j , written $i \leftrightarrow j$, if

$$i \rightsquigarrow j \quad \text{and} \quad j \rightsquigarrow i.$$

Communication is an Equivalence Relation

Communication is an equivalence relation.

Reflexive. By definition,

$$p_{ii}^{(0)} = 1 > 0.$$

Hence i communicates with i .

Symmetric. If i communicates with j , then by definition

$$i \rightsquigarrow j \quad \text{and} \quad j \rightsquigarrow i.$$

Thus j communicates with i .

Transitivity

Suppose

$$i \leftrightarrow j \quad \text{and} \quad j \leftrightarrow k.$$

Then there exist $n_1, m_1 \geq 0$ such that

$$p_{ij}^{(n_1)} > 0, \quad p_{jk}^{(m_1)} > 0,$$

and there exist $n_2, m_2 \geq 0$ such that

$$p_{ji}^{(n_2)} > 0, \quad p_{kj}^{(m_2)} > 0.$$

By Chapman–Kolmogorov,

$$p_{ik}^{(n_1+m_1)} \geq p_{ij}^{(n_1)} p_{jk}^{(m_1)} > 0.$$

Hence $i \rightsquigarrow k$.

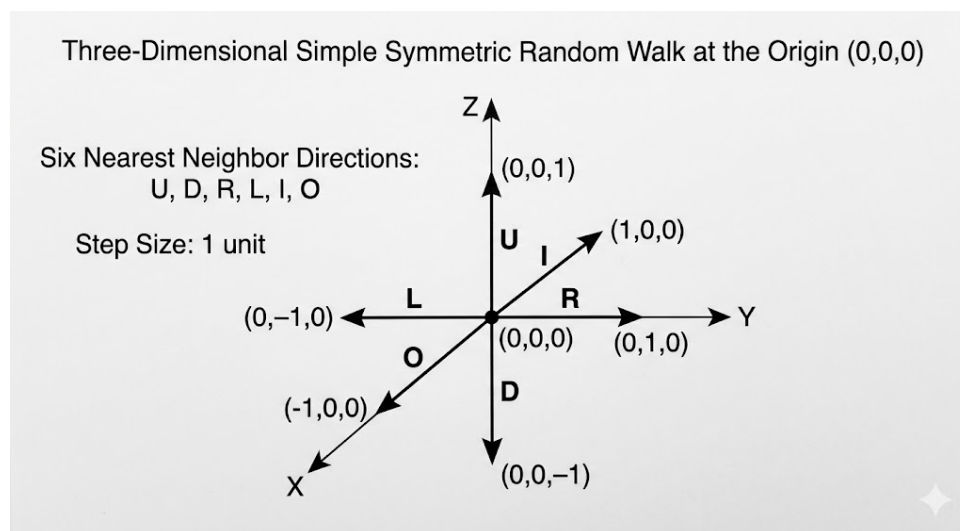
Transitivity (continued)

Similarly,

$$p_{ki}^{(m_2+n_2)} \geq p_{kj}^{(m_2)} p_{ji}^{(n_2)} > 0.$$

Hence $k \rightsquigarrow i$. Therefore $i \leftrightarrow k$, and communication is transitive.

The equivalence relation partitions the state space into disjoint sets called communicating classes.



Communicating Classes

For any two states $i, j \in C$, where C is a communicating class,

$$i \leftrightarrow j.$$

If $i \in C_1$, $j \in C_2$, and $C_1 \neq C_2$, then either

$$i \nrightarrow j, \quad j \nrightarrow i,$$

or both.

Class Property

Recurrence and transience are class properties.

Example: Two-State Chain

Let

$$S = \{1, 2\}, \quad P = \begin{pmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{pmatrix}.$$

If $p_{12} = 0$, then $1 \not\leftrightarrow 2$. If $p_{21} > 0$, then

$$2 \rightsquigarrow 1 \quad (n = 1).$$

If $p_{12} > 0$ and $p_{21} > 0$, then both states communicate and

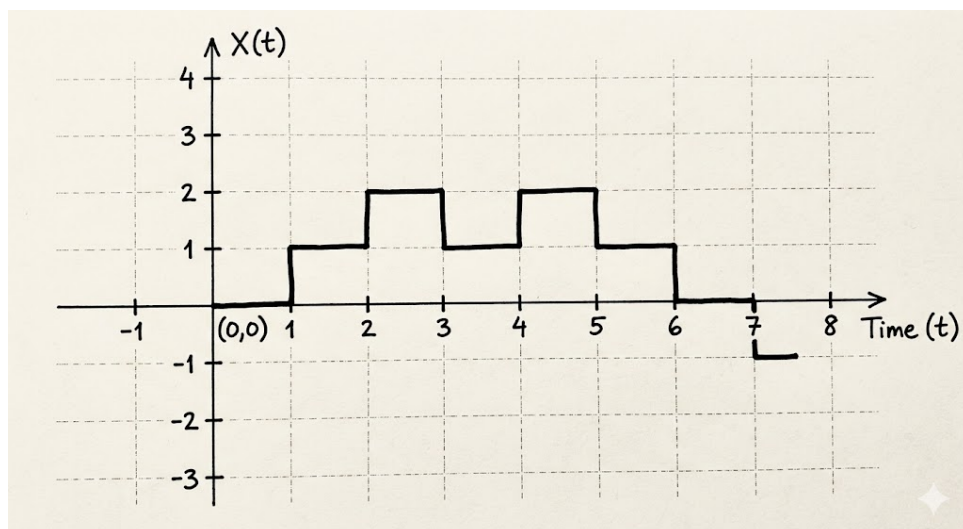
$$C_1 = \{1, 2\}.$$

Otherwise the communicating classes are

$$C_1 = \{1\}, \quad C_2 = \{2\}.$$

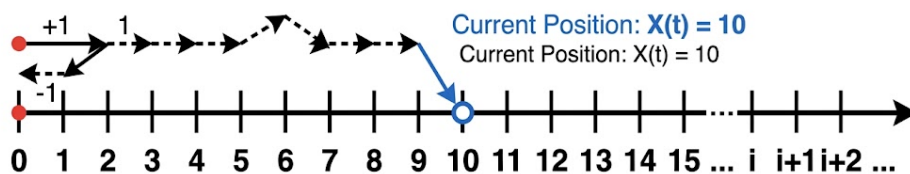
Example: Simple Random Walk

For a one-dimensional random walk, the state space is the set of integer points.



One-Dimensional Random Walk: Horizontal Integer Number Line

State Positions ($X(t)$)



Integer States (Positions)

For $i > 0$,

$$p_{0i}^{(i)} \geq p^i > 0,$$

and for $i < 0$,

$$p_{0i}^{(-i)} \geq q^{-i} > 0.$$

Thus

$$0 \leftrightarrow i \quad (i \in \mathbb{Z}).$$

All states communicate with one another, so $C = S$.

If a chain has only one communicating class, it is called irreducible.

Example: Communicating Classes from a Matrix

Let

$$S = \{0, 1, 2, 3, 4, 5, 6, 7\}.$$

The transition matrix on the page is written schematically with * for positive entries and 0 for impossible transitions. Reading the positive-entry pattern gives the communicating classes

$$C_1 = \{0, 3\}, \quad C_2 = \{2, 4, 7\}, \quad C_3 = \{1\}, \quad C_4 = \{6\}, \quad C_5 = \{5\}.$$

The closed classes C_1 and C_2 are recurrent classes, while the singleton classes listed above are not closed in the displayed example.

Example: Gambler's Ruin

For Gambler's ruin,

$$S = \{0, 1, \dots, N\}.$$

The transition matrix has absorbing endpoints:

$$P = \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \\ q & 0 & p & & 0 \\ 0 & q & 0 & \ddots & \vdots \\ \vdots & & \ddots & \ddots & p \\ 0 & \cdots & 0 & 0 & 1 \end{pmatrix}.$$

The communicating classes are

$$C_1 = \{0\}, \quad C_2 = \{N\}, \quad C_3 = \{1, 2, \dots, N-1\}.$$

The endpoint classes C_1 and C_2 are recurrent, while C_3 is transient.

Recurrence and Transience as Class Properties

Recall that

$$i \rightsquigarrow j \iff p_{ij}^{(n)} > 0 \quad \text{for some } n > 0,$$

and

$$i \leftrightarrow j \iff i \rightsquigarrow j \text{ and } j \rightsquigarrow i.$$

Communication is an equivalence relation.

Theorem

Recurrence and transience are class properties.

Suppose $i \leftrightarrow j$ and i is recurrent. We show that j is recurrent.

Proof of the Class Property

Since i is recurrent,

$$\sum_{n=0}^{\infty} p_{ii}^{(n)} = \infty.$$

We want to show that

$$\sum_{n=0}^{\infty} p_{jj}^{(n)} = \infty.$$

Since $i \leftrightarrow j$, there exist $n_1, m_1 \geq 0$ such that

$$p_{ij}^{(n_1)} > 0, \quad p_{ji}^{(m_1)} > 0.$$

Proof of the Class Property (continued)

For any $n \geq 0$,

$$p_{jj}^{(m_1+n+n_1)} \geq p_{ji}^{(m_1)} p_{ii}^{(n)} p_{ij}^{(n_1)}.$$

Let

$$C(i, j) = p_{ji}^{(m_1)} p_{ij}^{(n_1)} > 0.$$

Then

$$\begin{aligned} \sum_{r=0}^{\infty} p_{jj}^{(r)} &\geq \sum_{n=0}^{\infty} p_{jj}^{(m_1+n+n_1)} \\ &\geq C(i, j) \sum_{n=0}^{\infty} p_{ii}^{(n)} = \infty. \end{aligned}$$

Thus j is recurrent. Hence recurrence, and therefore transience, is a class property.

Periodicity of a State

For a state i , define

$$A(i) = \{n \geq 1 : p_{ii}^{(n)} > 0\}.$$

The period of i is the greatest common divisor of the elements of $A(i)$:

$$\lambda(i) = \gcd A(i).$$

If $\lambda(i) = 1$, then the state i is called aperiodic.

A Basic Property of $A(i)$

Let d be a divisor of every element of $A(i)$. Equivalently,

$$d \mid m \quad \text{for every } m \in A(i).$$

The period may be written as

$$\lambda(i) = \sup\{d : d \text{ divides every element of } A(i)\}.$$

If $n, m \in A(i)$, then

$$p_{ii}^{(n+m)} \geq p_{ii}^{(n)} p_{ii}^{(m)} > 0.$$

Hence

$$n + m \in A(i).$$

Example: One-Dimensional Simple Random Walk

For the simple one-dimensional random walk,

$$p_{00}^{(2)} \geq p_{01} p_{10} > 0.$$

Thus $2 \in A(0)$, so $\lambda(0) \leq 2$.

Also,

$$p_{00}^{(2n+1)} = 0,$$

because the walk cannot return to its starting point in an odd number of steps. Therefore

$$n \in A(0) \iff n = 2m \quad \text{for some } m \geq 1.$$

Hence

$$\lambda(0) = 2.$$

The same argument applies to every state of the simple random walk.

Examples of Periodicity

Gambler's ruin. For the absorbing states,

$$p_{00} = 1, \quad p_{NN} = 1.$$

Hence

$$1 \in A(0), \quad 1 \in A(N),$$

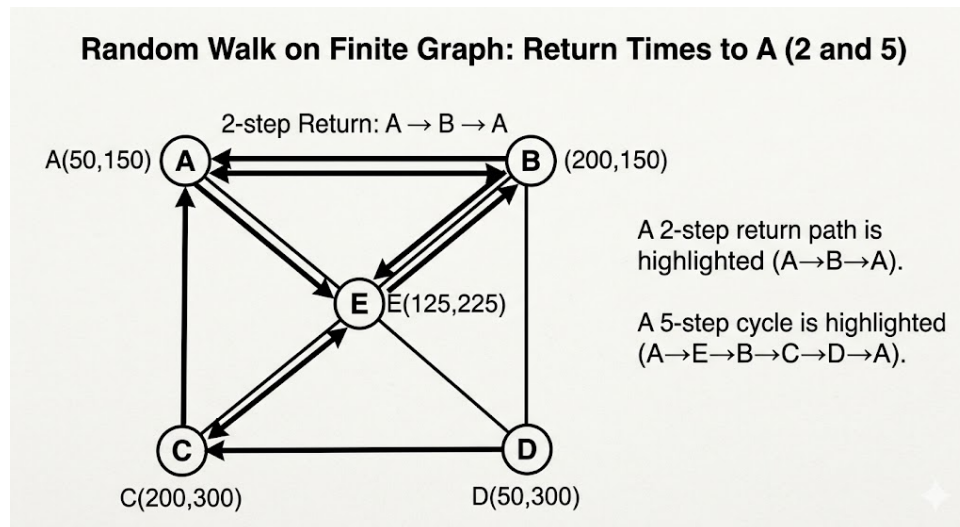
and

$$\lambda(0) = 1, \quad \lambda(N) = 1.$$

For $1 \leq i \leq N-1$, returns occur only after an even number of steps, so

$$\lambda(i) = 2.$$

Random walk on a graph.



The graph has return paths of lengths 2 and 5 to the state A . Thus

$$2, 5 \in A(A) \quad \Rightarrow \quad \lambda(A) = 1.$$

Example: Reflecting Random Walk

For a reflecting random walk on $\{0, 1, 2, \dots\}$, the transition matrix has the form

$$P = \begin{array}{c|ccccc} & 0 & 1 & 2 & 3 & \dots \\ \hline 0 & \alpha & 1 - \alpha & 0 & 0 & \dots \\ 1 & q & 0 & p & 0 & \dots \\ 2 & 0 & q & 0 & p & \dots \\ \vdots & \vdots & \vdots & \vdots & \ddots & \ddots \end{array}.$$

If $\alpha > 0$, then

$$1 \in A(0), \quad \lambda(0) = 1.$$

If $\alpha = 0$, then the possible return times to 0 are even, and

$$\lambda(0) = 2.$$

Periodicity as a Class Property

Theorem

Periodicity is also a class property.

Let

$$A(i) = \{n \geq 1 : p_{ii}^{(n)} > 0\}.$$

Suppose $i \leftrightarrow j$. Then there exist $n_1, n_2 \geq 0$ such that

$$p_{ij}^{(n_1)} > 0, \quad p_{ji}^{(n_2)} > 0.$$

Let d be a divisor of $A(i)$. Since

$$p_{ii}^{(n_1+n_2)} \geq p_{ij}^{(n_1)} p_{ji}^{(n_2)} > 0,$$

we have $n_1 + n_2 \in A(i)$. Therefore

$$d \mid (n_1 + n_2).$$

Periodicity as a Class Property (continued)

If $n \in A(j)$, then $p_{jj}^{(n)} > 0$. Hence

$$p_{ii}^{(n_1+n+n_2)} \geq p_{ij}^{(n_1)} p_{jj}^{(n)} p_{ji}^{(n_2)} > 0.$$

Thus

$$n_1 + n + n_2 \in A(i).$$

Since d divides every element of $A(i)$,

$$d \mid (n_1 + n + n_2).$$

But also $d \mid (n_1 + n_2)$. Therefore

$$d \mid n.$$

Thus every divisor of $A(i)$ is also a divisor of $A(j)$.

Periodicity as a Class Property (conclusion)

Since $\lambda(i)$ is the largest divisor of all elements of $A(i)$, and $\lambda(i)$ is also a divisor of $A(j)$, we get

$$\lambda(i) \leq \lambda(j).$$

Reversing the roles of i and j ,

$$\lambda(j) \leq \lambda(i).$$

Therefore

$$\lambda(i) = \lambda(j).$$

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Behaviour of $p_{ij}^{(n)}$

Theorem

For states i and j , the behaviour of $p_{ij}^{(n)}$ is summarized as follows.

1. If i is transient, then

$$\sum_{n=0}^{\infty} p_{ij}^{(n)} < \infty.$$

2. If i is recurrent and $i \leftrightarrow j$, then

$$\sum_{n=0}^{\infty} p_{ij}^{(n)} = \infty.$$

3. If i is null recurrent, then

$$p_{ij}^{(n)} \rightarrow 0.$$

Null Recurrence and Positive Recurrence

Class Property

Null recurrence and positive recurrence are also class properties.

Suppose $i \leftrightarrow j$. Then there exist $m_1, m_2 \geq 0$ such that

$$p_{ij}^{(m_1)} > 0, \quad p_{ji}^{(m_2)} > 0.$$

For every $n \geq 0$,

$$p_{ii}^{(m_1+n+m_2)} \geq p_{ij}^{(m_1)} p_{jj}^{(n)} p_{ji}^{(m_2)}.$$

Example: A Periodic Two-State Chain

Consider

$$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Then

$$p_{11}^{(2n)} = 1, \quad p_{11}^{(2n+1)} = 0.$$

Hence

$$p_{11}^{(n)} \not\rightarrow 0.$$

Also,

$$\lambda(1) = 2.$$

This example shows the effect of periodicity on the sequence $p_{ii}^{(n)}$.

Null Recurrence as a Class Property

From the inequality

$$p_{ii}^{(m_1+n+m_2)} \geq p_{ij}^{(m_1)} p_{jj}^{(n)} p_{ji}^{(m_2)},$$

we get

$$p_{jj}^{(n)} \leq \frac{p_{ii}^{(m_1+n+m_2)}}{p_{ij}^{(m_1)} p_{ji}^{(m_2)}}.$$

If i is null recurrent, then i is recurrent and

$$p_{ii}^{(n)} \rightarrow 0.$$

Thus

$$p_{jj}^{(n)} \rightarrow 0.$$

Since recurrence is already a class property, j is recurrent; hence j is null recurrent.

Positive Recurrence as a Class Property

Let $i \leftrightarrow j$. Since recurrence is a class property, if i is recurrent then j is recurrent.

If i is positive recurrent and j were null recurrent, then by the null recurrence class property proved above, i would also be null recurrent. This is a contradiction.

Therefore, if i is positive recurrent and $i \leftrightarrow j$, then j is positive recurrent.

Example: Transient Singleton Classes

For the example with

$$S = \{0, 1, \dots, 7\},$$

the communicating classes are

$$C_1 = \{0, 3\}, \quad C_2 = \{2, 4, 7\}, \quad C_3 = \{1\}, \quad C_4 = \{6\}, \quad C_5 = \{5\}.$$

Here C_1 and C_2 are closed classes. Also,

$$p_{10} = q > 0,$$

so $1 \rightsquigarrow 0$. If 1 were recurrent, then recurrence would force $0 \rightsquigarrow 1$, and hence $0 \leftrightarrow 1$, contradicting $0 \in C_1$ and $1 \in C_3$. Therefore 1 is transient.

Similarly,

$$6 \rightsquigarrow 0 \quad \Rightarrow \quad 6 \text{ is transient,}$$

and

$$5 \rightsquigarrow 0 \quad \Rightarrow \quad 5 \text{ is transient.}$$

A Recurrence Proposition

Proposition

If i is recurrent and $i \rightsquigarrow j$, then $j \rightsquigarrow i$.

Let $m_1 \geq 1$ be such that

$$p_{ij}^{(m_1)} > 0.$$

We prove the result by contradiction. Suppose that $j \not\leftrightarrow i$. Then

$$p_{ji}^{(n)} = 0 \quad \text{for all } n \geq 0.$$

But since i is recurrent, starting from i the chain must return to i with probability 1.

Proof of the Recurrence Proposition

Since i is recurrent,

$$\mathbb{P}_i(X_n = i \text{ for some } n \geq m_1) = 1.$$

Conditioning on the state at time m_1 ,

$$1 = \sum_{k \in S} \mathbb{P}_i(X_n = i \text{ for some } n \geq m_1 \mid X_{m_1} = k) p_{ik}^{(m_1)}.$$

By the Markov property, the conditional probability only depends on the state k .

Proof of the Recurrence Proposition (conclusion)

If $j \not\leftrightarrow i$, then

$$\mathbb{P}(X_n = i \text{ for some } n \geq 1 \mid X_0 = j) = 0.$$

Thus the term corresponding to $k = j$ in the conditional sum is zero. Since $p_{ij}^{(m_1)} > 0$,

$$\mathbb{P}_i(X_n = i \text{ for some } n \geq m_1) \leq 1 - p_{ij}^{(m_1)} < 1,$$

which contradicts recurrence of i . Hence $j \rightsquigarrow i$.

Recurrence and Transience

Proposition

If i is recurrent and $i \rightsquigarrow j$, then $i \leftrightarrow j$. Equivalently, a recurrent state cannot lead to a state outside its communicating class.

In particular, if i is recurrent and $i \rightsquigarrow j$, then j must be recurrent.

Proof

Since i is recurrent,

$$\mathbb{P}(X_n = i \text{ for some } n \geq m_0 \mid X_0 = i) = 1$$

for every fixed $m_0 \geq 1$.

Conditioning on X_{m_0} ,

$$\begin{aligned} 1 &= \sum_{k \in S} \mathbb{P}(X_n = i \text{ for some } n \geq m_0 \mid X_{m_0} = k, X_0 = i) \mathbb{P}(X_{m_0} = k \mid X_0 = i) \\ &= \sum_{k \in S} \mathbb{P}(X_n = i \text{ for some } n \geq 1 \mid X_0 = k) p_{ik}^{(m_0)}. \end{aligned}$$

Proof and Example

If $j \not\rightsquigarrow i$, then

$$\mathbb{P}(X_n = i \text{ for some } n \geq 1 \mid X_0 = j) = 0.$$

Since $p_{ij}^{(m_0)} > 0$ for some m_0 , the conditional sum is strictly less than 1, contradicting recurrence. Hence $j \rightsquigarrow i$.

Example. Consider the reflecting random walk with

$$P = \begin{array}{c|cccc} & 0 & 1 & 2 & \cdots \\ \hline 0 & 0 & 1 & 0 & \cdots \\ 1 & q & 0 & p & \cdots \\ 2 & 0 & q & 0 & p \\ \vdots & \vdots & \vdots & \ddots & \ddots \end{array}, \quad 0 < p < 1, \quad p + q = 1.$$

Here

$$0 \leftrightarrow 1 \leftrightarrow 2 \leftrightarrow \cdots,$$

so the chain is irreducible. For this chain,

$$p > q \Rightarrow 0 \text{ is transient}, \quad p \leq q \Rightarrow 0 \text{ is recurrent}.$$

Recurrence and Communication (continued)

Let

$$A_n = \{X_n = i\}.$$

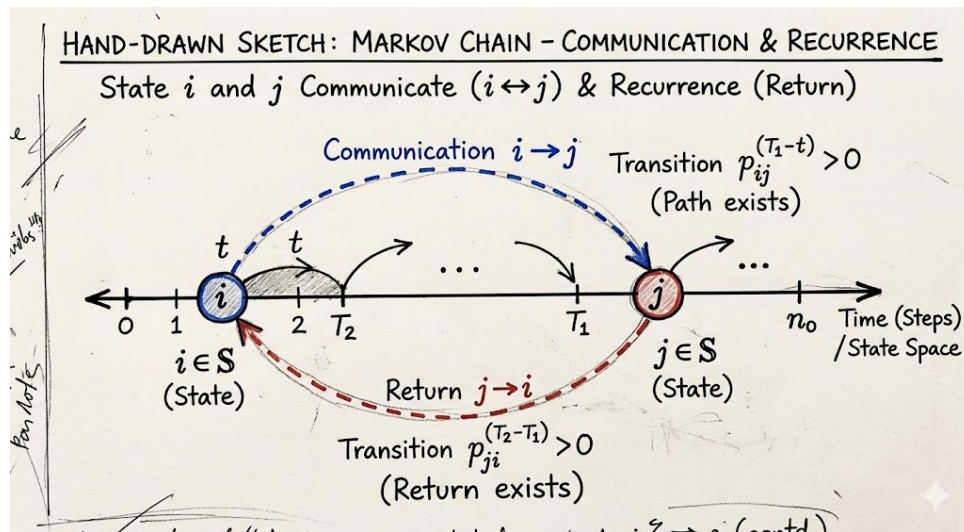
To write the event of ever hitting i as a disjoint union, define

$$B_1 = A_1, \quad B_n = A_n \setminus \bigcup_{k=1}^{n-1} A_k, \quad n \geq 2.$$

Then the events B_n are disjoint and represent first hitting i at time n .

$$\begin{aligned} f_{ji}^* &= P(X_n = i \text{ for some } n \geq 1 \mid X_0 = j) \\ &= P\left(\bigcup_{n=1}^{\infty} \{X_n = i\} \mid X_0 = j\right) \\ &\leq \sum_{n=1}^{\infty} P(X_n = i \mid X_0 = j) \\ &= \sum_{n=1}^{\infty} p_{ji}^{(n)}. \end{aligned}$$

Thus, if $\sum_{n \geq 1} p_{ji}^{(n)} = 0$, then $f_{ji}^* = 0$. This is used in the communication argument between states i and j .



Closed Sets

A subset $C \subseteq S$ is called closed if

$$p_{ij} = 0 \quad \text{for all } i \in C, j \notin C.$$

Equivalently, once the chain enters C , it cannot leave C .

$$C \text{ closed} \iff P(X_{n+1} \in C \mid X_n = i) = 1 \text{ for all } i \in C.$$

Example. For the chain with

$$S = \{0, 1, \dots, 7\},$$

the communication classes listed on the page are

$$\{0, 3\}, \quad \{2, 4, 7\}, \quad \{1\}, \quad \{6\}, \quad \{5\}.$$

Each class is checked for closedness by verifying whether any transition from a state in the class goes outside the class.

Closed Communicating Classes

In probability theory, a recurrent state can lead only to recurrent states. Therefore a recurrent communicating class must be closed.

$$\text{recurrent communicating class} \implies \text{closed communicating class.}$$

Consequently, communicating classes which are not closed are transient.

Theorem. A Markov chain having a finite number of states must have at least one recurrent state.

Existence of a Recurrent State

Proof. Suppose S is finite and that every state is transient. For any $i \in S$ and any $n \geq 0$,

$$\sum_{j \in S} p_{ij}^{(n)} = 1.$$

If j is transient, then

$$\lim_{n \rightarrow \infty} p_{ij}^{(n)} = 0.$$

Since S is finite, we may pass the limit through the finite sum:

$$1 = \lim_{n \rightarrow \infty} \sum_{j \in S} p_{ij}^{(n)} = \sum_{j \in S} \lim_{n \rightarrow \infty} p_{ij}^{(n)} = 0,$$

which is impossible. Hence at least one state must be recurrent.

The page also records the renewal decomposition

$$p_{ij}^{(n)} = \sum_{k=1}^n f_{ij}^{(k)} p_{jj}^{(n-k)},$$

where the chain first reaches j at time k and then returns from j to j in the remaining $n - k$ steps.

Restricting the Chain to a Closed Class

Let C be a closed class. Define

$$\tilde{P} = (\tilde{p}_{ij})_{i,j \in C}, \quad \tilde{p}_{ij} = p_{ij}, \quad i, j \in C.$$

Then $\tilde{P}$ is a transition matrix on C , because for every $i \in C$,

$$\sum_{j \in C} \tilde{p}_{ij} = \sum_{j \in C} p_{ij} = 1.$$

The last equality follows from closedness:

$$p_{ij} = 0 \quad (i \in C, j \notin C).$$

Similarly,

$$p_{ij}^{(m)} = 0 \quad \text{for all } m \geq 1, i \in C, j \notin C.$$

Chapman–Kolmogorov on a Closed Class

For the restricted transition matrix $\tilde{P}$,

$$\tilde{p}_{ij}^{(m+n)} = \sum_{k \in C} \tilde{p}_{ik}^{(m)} \tilde{p}_{kj}^{(n)}, \quad i, j \in C.$$

Since $\tilde{p}_{ij} = p_{ij}$ on C ,

$$\begin{aligned}\tilde{p}_{ij}^{(m+n)} &= \sum_{k \in C} p_{ik}^{(m)} p_{kj}^{(n)} \\ &= \sum_{k \in S} p_{ik}^{(m)} p_{kj}^{(n)} \\ &= p_{ij}^{(m+n)}.\end{aligned}$$

The terms with $k \notin C$ vanish because C is closed.

Therefore, for $i, j \in C$,

$$\tilde{p}_{ij}^{(m)} = p_{ij}^{(m)}, \quad m \geq 0.$$

Finite Closed Classes

Consider the Markov chain on the state space C with transition matrix $\tilde{P}$, where C is a closed communicating class.

Theorem. A finite closed communicating class is recurrent.

Proof. Since C is finite, the restricted Markov chain on C has at least one recurrent state. But C is communicating, so every state in C communicates with that recurrent state.

Finite Closed Classes (continued)

All states in a communicating class have the same type: they are either all recurrent or all transient. Hence, if one state in the finite closed class C is recurrent, then every state in C is recurrent.

$$C \text{ finite, closed, communicating} \implies C \text{ recurrent.}$$

Theorem. In a finite state-space chain, all recurrent states are positive recurrent.

Proof. Let i be recurrent and let C be the communicating class containing i . Since i is recurrent, the class C is closed. Also C is finite because S is finite.

Positive Recurrence in Finite Chains

Restrict the chain to the finite closed communicating class C . Suppose, for contradiction, that $i \in C$ is null recurrent. Then every state in C is null recurrent.

For each $n \geq 0$,

$$\sum_{j \in C} p_{ij}^{(n)} = 1.$$

But for a transient or null recurrent state j ,

$$\lim_{n \rightarrow \infty} p_{ij}^{(n)} = 0.$$

Since C is finite,

$$1 = \lim_{n \rightarrow \infty} \sum_{j \in C} p_{ij}^{(n)} = \sum_{j \in C} \lim_{n \rightarrow \infty} p_{ij}^{(n)} = 0,$$

a contradiction. Therefore every recurrent state in a finite state-space chain is positive recurrent.

First Step Analysis

Gambler's Ruin

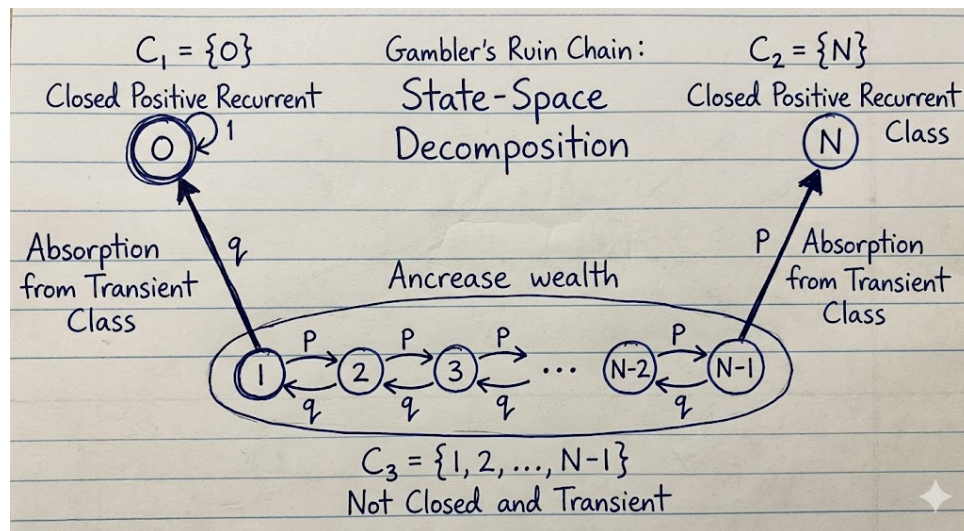
For the gambler's ruin chain with state space

$$S = \{0, 1, \dots, N\},$$

the communicating classes are

$$C_1 = \{0\}, \quad C_2 = \{N\}, \quad C_3 = \{1, 2, \dots, N-1\}.$$

The classes C_1 and C_2 are closed and positive recurrent. The class C_3 is not closed, hence it is transient.



Thus, starting from $i \in C_3$, the chain is eventually absorbed in one of the closed classes:

$$P(X_n \in \{0, N\} \text{ for some } n \geq 0 \mid X_0 = i) = 1.$$

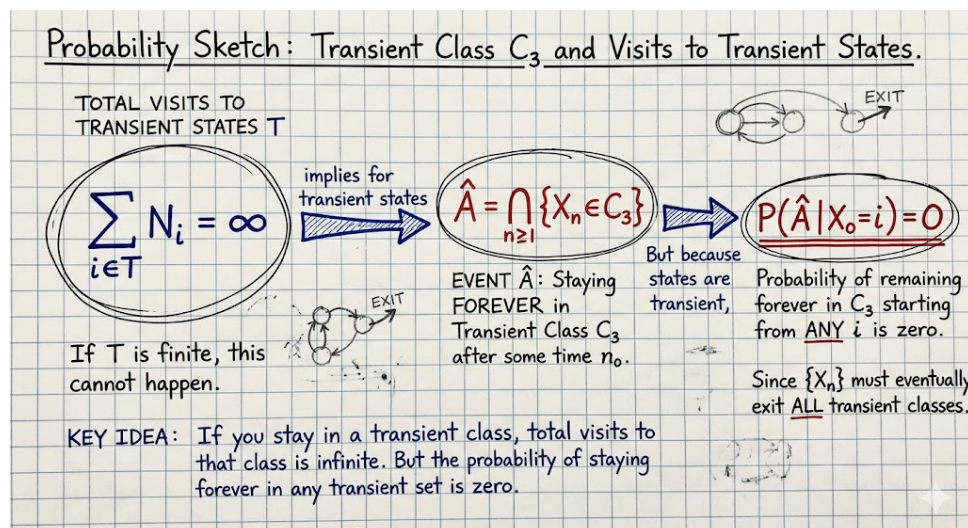
Leaving the Transient Class

Let

$$\hat{A} = \bigcap_{n=1}^{\infty} \{X_n \in C_3\}$$

be the event that the chain stays forever in the transient class C_3 . Then

$$P(\hat{A} \mid X_0 = i) = 0.$$



Let T be a finite set of transient states, and let N_i denote the number of visits to state i . For each transient state i ,

$$N_i < \infty \quad \text{with probability 1.}$$

Hence

$$\sum_{i \in T} N_i < \infty \quad \text{with probability 1.}$$

The visit-count estimate recorded on the page is

$$P(N_i \geq k \mid X_0 = i) = (f_{ii}^*)^{k-1}, \quad k \geq 1.$$

Eventual Absorption

Since every N_i is finite with probability one for $i \in T$,

$$P\left(\sum_{i \in T} N_i = \infty\right) = 1 - P\left(\sum_{i \in T} N_i < \infty\right) = 1 - 1 = 0.$$

Definition. Eventual absorption in a recurrent class C_j is defined by the probability

$$P(X_n \in C_j \text{ for some } n \geq 0 \mid X_0 = i).$$

In the gambler's ruin problem this is interpreted as the probability of eventual absorption, given initial capital i .

First Step Analysis for Absorption

Write the state space as

$$S = C_1 \cup C_2 \cup \dots \cup C_t \cup T,$$

where $C_1, \dots, C_t$ are recurrent classes and T is the set of transient states.

If T is finite and $i \in T$, then

$$P(X_n \in T \text{ for all } n \geq 1 \mid X_0 = i) = 0.$$

Therefore

$$P\left(X_n \in \bigcup_{\ell=1}^t C_\ell \text{ for some } n \geq 0 \mid X_0 = i\right) = 1.$$

Absorption Probabilities

For $i \in T$, the total probability of eventual absorption in one of the recurrent classes is

$$\sum_{j=1}^t P(X_n \in C_j \text{ for some } n \geq 0 \mid X_0 = i) = 1.$$

The eventual absorption probability in recurrent class C_j is

$$P(X_n \in C_j \text{ for some } n \geq 0 \mid X_0 = i).$$

For gambler's ruin,

$$S = \{0, 1, \dots, N\}, \quad T = \{1, 2, \dots, N-1\},$$

and the recurrent classes are

$$C_1 = \{0\}, \quad C_2 = \{N\}.$$

Absorption into a Closed Class

In gambler's ruin, for $i \in T$, absorption in $C_1 = \{0\}$ is the probability of ruin when the initial capital is i .

Let C be a closed class. Define

$$\alpha_i^{(C)} = P(X_n \in C \text{ for some } n \geq 0 \mid X_0 = i).$$

If $i \in C$, then

$$\alpha_i^{(C)} = 1.$$

If i belongs to another closed class $D \neq C$, then

$$\alpha_i^{(C)} = 0.$$

Approximation by Finite Time

Define the finite-time absorption probability

$$\alpha_i^{(C)}(n) = P(X_m \in C \text{ for some } 0 \leq m \leq n \mid X_0 = i).$$

Let

$$A_n = \{X_m \in C \text{ for some } 0 \leq m \leq n\}.$$

Then

$$A_n \subseteq A_{n+1}, \quad \bigcup_{n=0}^{\infty} A_n = \{X_m \in C \text{ for some } m \geq 0\}.$$

Therefore, by continuity from below,

$$\alpha_i^{(C)} = \lim_{n \rightarrow \infty} \alpha_i^{(C)}(n).$$

Finite-Time Recursion

The absorption probability satisfies

$$\boxed{\alpha_i^{(C)} = \lim_{n \rightarrow \infty} \alpha_i^{(C)}(n)}.$$

For $i \notin C$,

$$\alpha_i^{(C)}(0) = 0, \quad \alpha_i^{(C)}(1) = \sum_{j \in C} p_{ij}.$$

Write

$$p_{iC} = \sum_{j \in C} p_{ij}.$$

Then, for $m \geq 0$,

$$\alpha_i^{(C)}(m+1) = p_{iC} + \sum_{j \in T} p_{ij} \alpha_j^{(C)}(m), \quad i \in T.$$

The boundary values are

$$\alpha_i^{(C)} = 1 \quad (i \in C), \quad \alpha_i^{(C)} = 0 \quad (i \in D, \ D \neq C \text{ closed}).$$

Deriving the First-Step Equation

For $i \in T$,

$$\begin{aligned} \alpha_i^{(C)}(m+1) &= P(X_n \in C \text{ for some } 0 \leq n \leq m+1 \mid X_0 = i) \\ &= \sum_{j \in C} p_{ij} + \sum_{j \in T} p_{ij} P(X_n \in C \text{ for some } 0 \leq n \leq m \mid X_0 = j) \\ &= p_{iC} + \sum_{j \in T} p_{ij} \alpha_j^{(C)}(m). \end{aligned}$$

Thus

$$\boxed{\alpha_i^{(C)}(m+1) = p_{iC} + \sum_{j \in T} p_{ij} \alpha_j^{(C)}(m)}.$$

Taking $m \rightarrow \infty$,

$$\alpha_i^{(C)} = p_{iC} + \sum_{j \in T} p_{ij} \alpha_j^{(C)}, \quad i \in T.$$

Smallest Nonnegative Solution

Let $\{y_i : i \in T\}$ be a solution of the system

$$y_i = p_{iC} + \sum_{j \in T} p_{ij} y_j, \quad i \in T,$$

with

$$0 \leq y_i \leq 1.$$

For any $i \in T$,

$$y_i \geq 0 = \alpha_i^{(C)}(0).$$

We want to show, by induction, that

$$y_i \geq \alpha_i^{(C)}(n) \quad \text{for all } n \geq 0, \quad i \in T.$$

Smallest Solution Theorem

Assume that, for some $n \geq 0$,

$$y_i \geq \alpha_i^{(C)}(n) \quad \text{for all } i \in T.$$

Then

$$\begin{aligned} y_i &= p_{iC} + \sum_{j \in T} p_{ij} y_j \\ &\geq p_{iC} + \sum_{j \in T} p_{ij} \alpha_j^{(C)}(n) \\ &= \alpha_i^{(C)}(n+1). \end{aligned}$$

Thus $y_i \geq \alpha_i^{(C)}(n)$ for all n . Letting $n \rightarrow \infty$, every solution dominates $\alpha_i^{(C)}$.

Theorem. The eventual absorption probability into a closed class C is the smallest solution of

$$y_i = p_{iC} + \sum_{j \in T} p_{ij} y_j, \quad 0 \leq y_i \leq 1, \quad i \in T.$$

Example of Absorption Equations

The eventual absorption probabilities into a closed class are given by the smallest solution of the linear system

$$y_i = p_{iC} + \sum_{j \in T} p_{ij} y_j, \quad i \in T.$$

In the example on the page, the closed classes and transient set are recorded as

$$C_1 = \{0, 3\}, \quad C_2 = \{2, 4, 7\}, \quad T = \{1, 5, 6\}.$$

To find the probability of eventual absorption in C_1 , set

$$y_i = \alpha_i^{(C_1)}, \quad i \in T.$$

The unknowns are

$$\alpha_1^{(C_1)}, \quad \alpha_5^{(C_1)}, \quad \alpha_6^{(C_1)}.$$

Example Equations

For the example with transient states $T = \{1, 5, 6\}$, the equations for absorption into C_1 are

$$\begin{aligned} y_1 &= \frac{1}{2} + p_{11}y_1 + p_{15}y_5 + p_{16}y_6, \\ y_5 &= \frac{1}{4} + p_{51}y_1 + p_{55}y_5 + p_{56}y_6, \\ y_6 &= \frac{1}{3} + p_{61}y_1 + p_{65}y_5 + p_{66}y_6. \end{aligned}$$

Here

$$p_{1C_1} = \frac{1}{2}, \quad p_{5C_1} = \frac{1}{4}, \quad p_{6C_1} = \frac{1}{3}.$$

The next example is a birth and death chain.

Birth and Death Chain

A birth and death chain on $\{0, 1, \dots, N\}$ has transition matrix

$$P = \begin{array}{c|ccccccc} & 0 & 1 & 2 & \cdots & N-2 & N-1 & N \\ \hline 0 & r_0 & p_0 & 0 & \cdots & 0 & 0 & 0 \\ 1 & q_1 & r_1 & p_1 & \cdots & 0 & 0 & 0 \\ 2 & 0 & q_2 & r_2 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ N-1 & 0 & 0 & 0 & \cdots & q_{N-1} & r_{N-1} & p_{N-1} \\ N & 0 & 0 & 0 & \cdots & 0 & q_N & r_N \end{array}.$$

Here

$$p_i, q_i \geq 0, \quad i = 1, \dots, N-1.$$

For absorbing endpoints,

$$r_0 = r_N = 1, \quad p_0 = 0, \quad q_N = 0.$$

The closed classes are

$$C_1 = \{0\}, \quad C_2 = \{N\}.$$

For $C_1 = \{0\}$,

$$\alpha_i^{(C_1)} = P(X_m \in C_1 \text{ for some } m \geq 0 \mid X_0 = i).$$

First-Step Equations for Birth and Death Chains

Let

$$y_i = \alpha_i^{(C_1)}, \quad C_1 = \{0\}.$$

Then

$$y_0 = 1, \quad y_N = 0.$$

For $1 \leq i \leq N-1$, first-step analysis gives

$$y_i = q_i y_{i-1} + r_i y_i + p_i y_{i+1}.$$

In particular,

$$\begin{aligned} y_1 &= q_1 + r_1 y_1 + p_1 y_2, \\ y_i &= q_i y_{i-1} + r_i y_i + p_i y_{i+1}, \quad i = 2, \dots, N-2, \\ y_{N-1} &= q_{N-1} y_{N-2} + r_{N-1} y_{N-1}. \end{aligned}$$

Solving the Difference Equation

Since

$$q_i + r_i + p_i = 1,$$

the equation

$$y_i = q_i y_{i-1} + r_i y_i + p_i y_{i+1}$$

becomes

$$(q_i + p_i)y_i = q_i y_{i-1} + p_i y_{i+1}.$$

Therefore

$$q_i(y_i - y_{i-1}) = p_i(y_{i+1} - y_i),$$

and hence

$$y_{i+1} - y_i = \frac{q_i}{p_i}(y_i - y_{i-1}).$$

Iterating,

$$y_{i+1} - y_i = \left(\prod_{j=1}^i \frac{q_j}{p_j} \right) (y_1 - y_0).$$

Difference Equation Coefficients

For $i = 1, \dots, N-1$,

$$y_{i+1} - y_i = \left(\prod_{j=1}^i \frac{q_j}{p_j} \right) (y_1 - y_0).$$

Define

$$a_0 = 1, \quad a_i = \prod_{j=1}^i \frac{q_j}{p_j}, \quad i > 0.$$

Then

$$y_{i+1} - y_i = a_i(y_1 - y_0).$$

Summing from $i = 0$ to $k-1$,

$$\begin{aligned} y_k - y_0 &= \sum_{i=0}^{k-1} (y_{i+1} - y_i) \\ &= \sum_{i=0}^{k-1} a_i (y_1 - y_0). \end{aligned}$$

Boundary Conditions

Using $y_0 = 1$ and $y_N = 0$,

$$y_N - y_0 = \sum_{i=0}^{N-1} a_i (y_1 - y_0) = -1.$$

Thus

$$y_1 - y_0 = -\frac{1}{\sum_{i=0}^{N-1} a_i}.$$

For $1 \leq k \leq N-1$,

$$\begin{aligned} y_k - y_0 &= \sum_{i=0}^{k-1} a_i (y_1 - y_0) \\ &= -\frac{\sum_{i=0}^{k-1} a_i}{\sum_{i=0}^{N-1} a_i}. \end{aligned}$$

Hence

$$y_k = 1 - \frac{\sum_{i=0}^{k-1} a_i}{\sum_{i=0}^{N-1} a_i}.$$

Gambler's Ruin Formula

The absorption probability at 0 is

$$y_k = 1 - \frac{\sum_{i=0}^{k-1} a_i}{\sum_{i=0}^{N-1} a_i}, \quad k = 1, \dots, N-1.$$

For gambler's ruin,

$$p_i = p, \quad q_i = q = 1 - p.$$

Therefore

$$a_i = \prod_{j=1}^i \frac{q}{p} = \left(\frac{q}{p}\right)^i.$$

If $p \neq q$, then

$$y_k = 1 - \frac{\sum_{i=0}^{k-1} \left(\frac{q}{p}\right)^i}{\sum_{i=0}^{N-1} \left(\frac{q}{p}\right)^i}.$$

Evaluating the Gambler's Ruin Formula

For $p \neq q$,

$$\begin{aligned}
 y_k &= 1 - \frac{\frac{1 - \left(\frac{q}{p}\right)^k}{1 - \frac{q}{p}}}{\frac{1 - \left(\frac{q}{p}\right)^N}{1 - \frac{q}{p}}} \\
 &= 1 - \frac{1 - \left(\frac{q}{p}\right)^k}{1 - \left(\frac{q}{p}\right)^N} \\
 &= \frac{\left(\frac{q}{p}\right)^k - \left(\frac{q}{p}\right)^N}{1 - \left(\frac{q}{p}\right)^N}.
 \end{aligned}$$

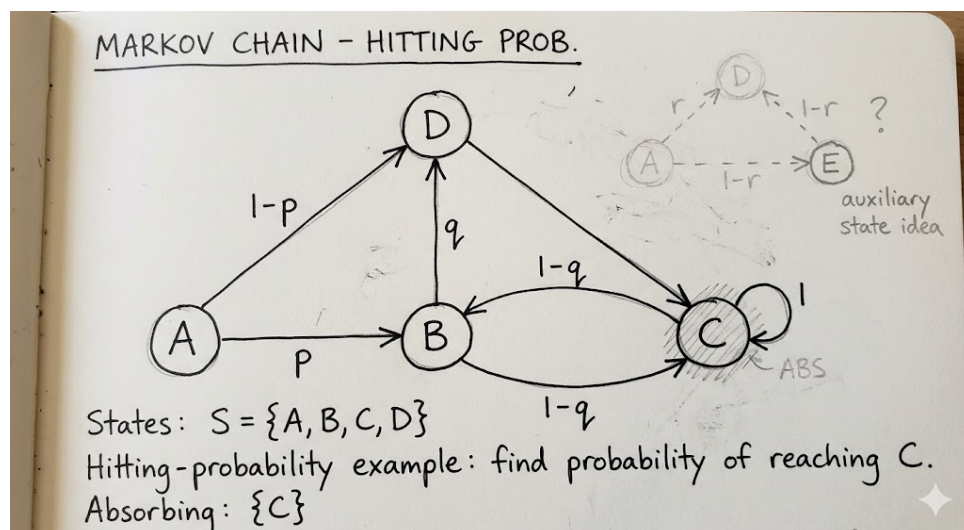
If $p = q = \frac{1}{2}$, then $a_i = 1$, so

$$y_k = 1 - \frac{k}{N} = \frac{N - k}{N}.$$

Infinite Gambler's Ruin and a Hitting Example

In the infinite-state gambler's ruin problem, if $q \geq p$, then eventual ruin occurs with probability 1. If $q < p$, then the eventual ruin probability is less than 1.

The page then begins a finite hitting-probability example using a state diagram.



Let C be the target closed class and write

$$y_i = P(X_n \in C \text{ for some } n \geq 0 \mid X_0 = i).$$

The following page records the first-step equations for the transient states in this diagram.

Hitting Example Equations

For the hitting-probability example, let

$$y_i = P_i(\text{eventual hit of the target closed class}).$$

The first-step equations recorded on the page are

$$\begin{aligned} y_A &= \frac{1}{3}y_B + \frac{1}{3}y_E, \\ y_B &= \frac{1}{3} + \frac{1}{3}y_A, \\ y_E &= \frac{1}{2}y_A. \end{aligned}$$

These equations are solved with the boundary value $y_i = 1$ on the target closed class and $y_i = 0$ on closed classes that cannot reach the target.

Hitting Probabilities Between States

Let

$$f_{ij}^* = P(X_n = j \text{ for some } n \geq 1 \mid X_0 = i).$$

The page records the following cases.

$$f_{ij}^* = \begin{cases} 0, & i \in C, j \in D, C \neq D, \text{ where } C, D \text{ are closed classes,} \\ 1, & i, j \text{ are recurrent and belong to the same communicating class,} \\ \alpha_i^{(C)}, & i \in T, j \in C, \text{ where } C \text{ is recurrent.} \end{cases}$$

For the example with states A, B, C, D, E , the transient set is recorded as

$$T = \{A, B, E\}.$$

The states C and D are absorbing recurrent states in the displayed transition matrix.

First-Step Equations for f_{ij}^*

For transient states, the hitting probabilities f_{ij}^* are found by a first-step decomposition:

$$\begin{aligned} f_{ij}^* &= \sum_{n=1}^{\infty} f_{ij}^{(n)} \\ &= f_{ij}^{(1)} + \sum_{n=2}^{\infty} f_{ij}^{(n)} \\ &= p_{ij} + \sum_{n=2}^{\infty} \sum_{k \in T} p_{ik} f_{kj}^{(n-1)} \\ &= p_{ij} + \sum_{k \in T} p_{ik} \sum_{m=1}^{\infty} f_{kj}^{(m)} \\ &= p_{ij} + \sum_{k \in T} p_{ik} f_{kj}^*. \end{aligned}$$

Thus, for $i, j \in T$,

$$f_{ij}^* = p_{ij} + \sum_{k \in T} p_{ik} f_{kj}^*.$$

Together with the recurrent-class boundary cases, this gives a linear system for all hitting probabilities.

Stationary Distribution

Suppose $\gamma(a)$ is a nonnegative weight assigned to state a , and let

$$R = \sum_{i \in S} \gamma(i).$$

Define

$$\pi(a) = \frac{\gamma(a)}{R}.$$

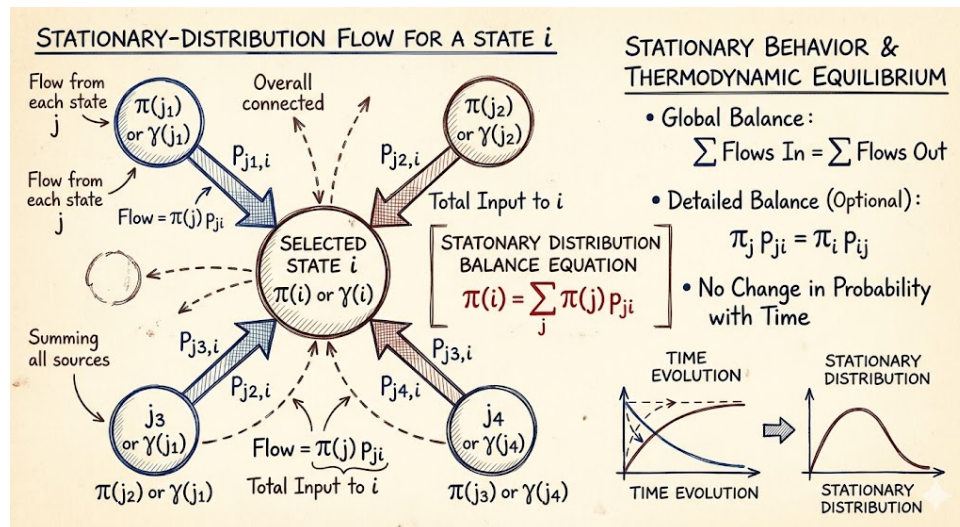
If the weights satisfy the balance relation

$$\gamma(a) = \sum_{j \in S} p_{ja} \gamma(j),$$

then

$$\pi(a) = \frac{\gamma(a)}{R} = \sum_{j \in S} p_{ja} \frac{\gamma(j)}{R} = \sum_{j \in S} \pi(j) p_{ja}.$$

This is the equilibrium, or stationary, relation.



Definition of Stationary Distribution

We call a probability distribution π on the state space a stationary distribution if

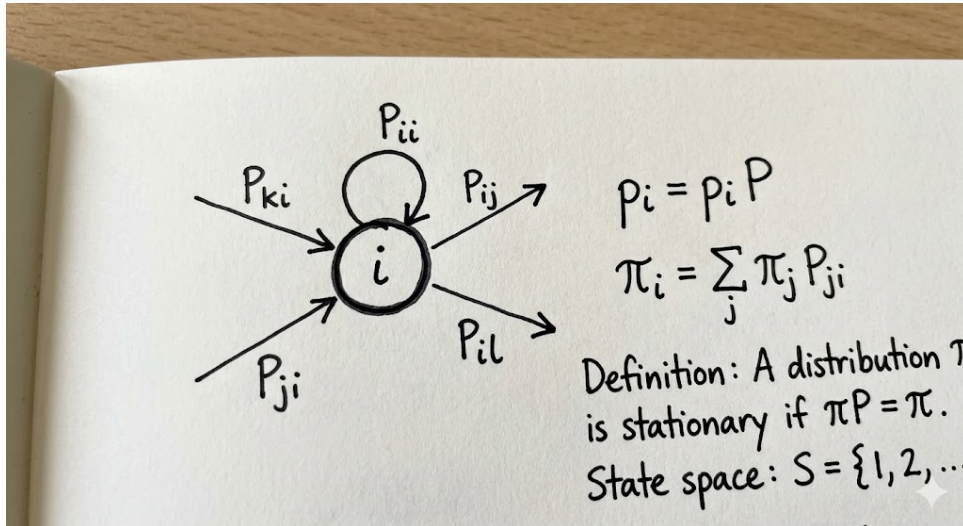
$$\pi = \pi P.$$

Equivalently,

$$\pi(j) = \sum_{i \in S} \pi(i) p_{ij}, \quad j \in S,$$

with

$$\pi(i) \geq 0, \quad \sum_{i \in S} \pi(i) = 1.$$



In some cases there is a nonnegative invariant measure

$$m = (m(i) : i \in S), \quad m(i) \geq 0,$$

satisfying

$$m = mP,$$

but

$$\sum_{i \in S} m(i) = \infty.$$

Such a measure cannot be normalized to a stationary probability distribution.

Balance Equations and Reversibility

The equation

$$\pi(i) = \sum_{j \in S} \pi(j) p_{ji}$$

is called the balance equation.

If, for a probability distribution π , we have $\pi = \pi P$ and

$$\pi(i) p_{ij} = \pi(j) p_{ji}, \quad \forall i, j \in S,$$

then we say that the system is reversible in π .

Detailed Balance Implies Stationarity

In physics notation, the reversibility condition is called detailed balance.

Proposition

If the chain is reversible in π , then π is a stationary distribution.

Proof

For $i \in S$,

$$\pi(i) = \sum_{j \in S} \pi(i)p_{ij} = \sum_{j \in S} \pi(j)p_{ji}.$$

Thus $\pi = \pi P$, so π is stationary.

Stationary Distributions for a Reducible Chain

Let $S = \{0, 1, 2\}$ and

$$P = \begin{pmatrix} 1 & 0 & 0 \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{6} \\ 0 & 0 & 1 \end{pmatrix}.$$

For

$$\pi = (\lambda, 0, 1 - \lambda),$$

we have $\pi P = \pi$.

In particular,

$$\pi_1 = (1, 0, 0), \quad \pi_1 P = \pi_1,$$

and

$$\pi_2 = (0, 0, 1), \quad \pi_2 P = \pi_2.$$

Hence

$$\lambda\pi_1 + (1 - \lambda)\pi_2, \quad 0 \leq \lambda \leq 1,$$

is a stationary distribution.

Two-State Stationary Distribution

Let $S = \{1, 2\}$ and

$$P = \begin{pmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{pmatrix}, \quad 0 < p_{12}, p_{21} < 1.$$

Then a stationary distribution is

$$\pi = \left(\frac{p_{21}}{p_{12} + p_{21}}, \frac{p_{12}}{p_{12} + p_{21}} \right),$$

and it satisfies

$$\pi P = \pi.$$

Initial Distribution and Classes

In the generic example, for the classes

$$C_1 = \{0, 1\}, \quad C_2 = \{2, 4, 7\},$$

one obtains corresponding stationary distributions π_1 and π_2 .

Let μ be any probability distribution on S . Write

$$\mu = (\mu(i) : i \in S),$$

where

$$\sum_{i \in S} \mu(i) = 1, \quad \mu(i) \geq 0.$$

If $X_0 \sim \mu$, then

$$P(X_0 = i) = \mu(i).$$

Distribution After n Steps

For $i \in S$,

$$\begin{aligned} P(X_n = i) &= \sum_{j \in S} P(X_n = i \mid X_0 = j) P(X_0 = j) \\ &= \sum_{j \in S} \mu(j) p_{ji}^{(n)} \\ &= (\mu P^n)_i. \end{aligned}$$

Similarly, for $m \geq 0$,

$$\begin{aligned} P(X_m = i) &= \sum_{v \in S} P(X_m = i \mid X_0 = v) P(X_0 = v) \\ &= \sum_{v \in S} (P^m)_{vi} \mu(v). \end{aligned}$$

By induction,

$$X_n \sim \mu P^n.$$

Stationary Initial Law

If $\mu = \pi$, where π is a stationary distribution, then

$$X_n \sim \pi, \quad \forall n \geq 0.$$

Theorem

For an irreducible positive recurrent Markov chain, there exists a unique stationary distribution π .

Positive Recurrence and Uniform Stationarity

The stationary distribution in a positive recurrent class gives mass only to elements of that class.

Theorem

For an irreducible positive recurrent chain,

$$\pi(i) = \frac{1}{\mu_i},$$

where

$$\mu_i = E(T_i \mid X_0 = i)$$

is the mean return time to state i .

Corollary

If a finite transition matrix is doubly stochastic, then the stationary distribution is uniform:

$$\pi_i = \frac{1}{N}, \quad i = 1, \dots, N.$$

Doubly Stochastic Chains

Let

$$\mathbf{1} = (1, \dots, 1).$$

Since P is doubly stochastic,

$$\mathbf{1}P = \mathbf{1}.$$

Therefore

$$\frac{1}{N}\mathbf{1}P = \frac{1}{N}\mathbf{1}.$$

The vector $\frac{1}{N}\mathbf{1}$ is a probability vector, hence it is a stationary distribution. Since a finite irreducible positive recurrent chain has a unique stationary distribution, this is the only stationary distribution.

Dice Walk Modulo N

Suppose a die is rolled repeatedly, and let E_i be the score of the i -th roll. Define

$$X_n = \left(\sum_{i=1}^n E_i \right) \bmod N.$$

Then

$$P(X_n = i) \longrightarrow \frac{1}{N}.$$

Here the transition matrix P is doubly stochastic, and the chain X_n is irreducible.

Return Times for the Dice Walk

Hence $\frac{1}{N}\mathbf{1}$ is the only stationary distribution. Also,

$$\frac{1}{\mu_i} = \pi(i) = \frac{1}{N},$$

so

$$\mu_i = N, \quad \forall i.$$

For the dice walk

$$X_n = \left(\sum_{i=1}^n E_i \right) \bmod N,$$

we have

$$P(X_n = i) \longrightarrow \frac{1}{N}.$$

The transition matrix is doubly stochastic and X_n is irreducible.

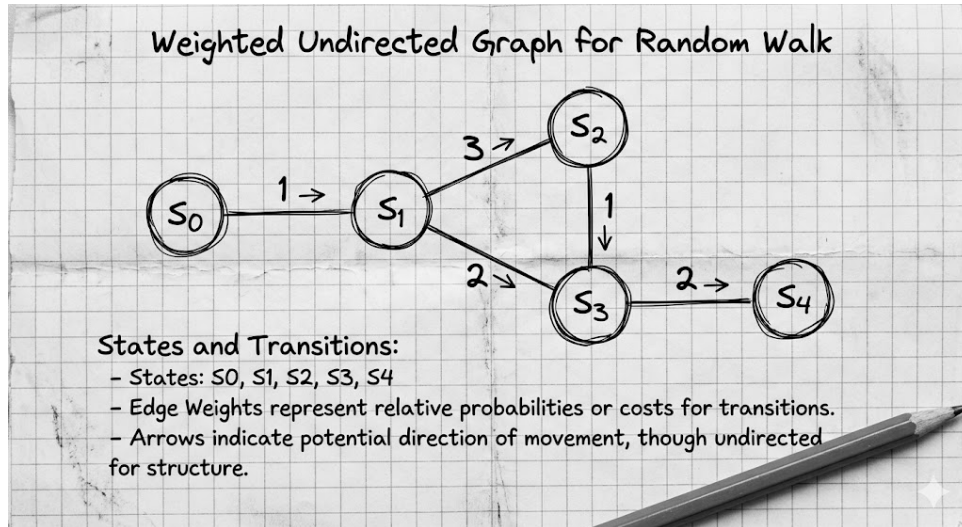
Limiting Distribution

For a positive recurrent irreducible chain,

$$\begin{aligned} \lim_{n \rightarrow \infty} P(X_n = i) &= \lim_{n \rightarrow \infty} \sum_{j \in S} P(X_n = i \mid X_0 = j) P(X_0 = j) \\ &= \sum_{j \in S} \left[\lim_{n \rightarrow \infty} p_{ji}^{(n)} \right] P(X_0 = j) \\ &= \sum_{j \in S} \frac{1}{\mu_i} P(X_0 = j) \\ &= \frac{1}{\mu_i}. \end{aligned}$$

Random Walks on Graphs

Assume the random walk is irreducible. Since the state space is finite, it has a unique stationary distribution.



Let the edge weights satisfy

$$w_{ij} = w_{ji}.$$

Define

$$w_i = \sum_{j \in S} w_{ij} > 0,$$

and also

$$w_i = \sum_j w_{ji}.$$

The unique stationary distribution is

$$\pi(i) = \frac{w_i}{w_{..}}, \quad w_{..} = \sum_{k \in S} w_k.$$

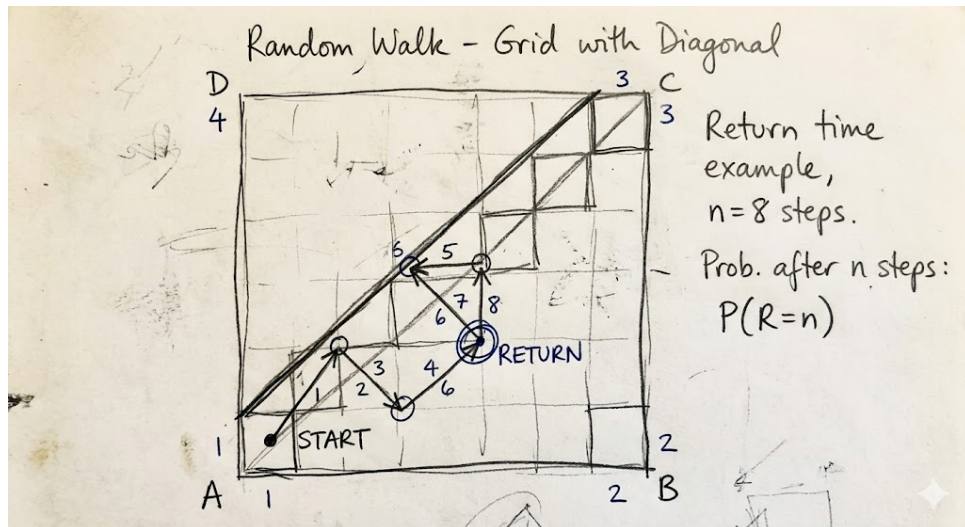
Detailed Balance on Graphs

For the weighted graph random walk,

$$\begin{aligned} \pi(i)p_{ij} &= \frac{w_i}{w_{..}} \frac{w_{ij}}{w_i} \\ &= \frac{w_{ij}}{w_{..}} \\ &= \frac{w_{ji}}{w_{..}} \\ &= \pi(j)p_{ji}. \end{aligned}$$

Thus the detailed balance equations hold, and the unique stationary distribution is

$$\pi(i) = \frac{w_i}{w_{..}}.$$



For the illustrated left-bottom square or corner, the return-time question is determined from the stationary probability.

Stationary Distribution and Detailed Balance

Let π be a probability distribution on S such that

$$\pi = \pi P.$$

The balance equation is

$$\pi(i) = \sum_{j \in S} \pi(j) p_{ji}.$$

If

$$\pi(i) p_{ij} = \pi(j) p_{ji}, \quad \forall i, j \in S,$$

then π is stationary. This is the detailed balance equation.

Birth and Death Chains

For a random walk on a graph with weights $w_{ij} = w_{ji}$,

$$\pi(i) = \frac{w_i}{w_{..}}, \quad w_i = \sum_j w_{ij}, \quad w_{..} = \sum_i w_i,$$

and this satisfies detailed balance.

Birth and Death Chain

Let the state space be

$$\{0, 1, 2, \dots, N\}.$$

The transition matrix has the form

$$P = \begin{pmatrix} r_0 & p_0 & 0 & \cdots & 0 \\ q_1 & r_1 & p_1 & \cdots & 0 \\ 0 & q_2 & r_2 & p_2 & \cdots \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & q_N & r_N \end{pmatrix}.$$

For interior states, $q_i, p_i > 0$. At the boundary one may use the convention

$$q_0 = 0, \quad p_N = 0.$$

Detailed Balance for Birth and Death Chains

Suppose that the detailed balance equations have the form

$$\pi(i)p_i = \pi(i+1)q_{i+1}, \quad i = 0, 1, \dots, N-1.$$

For $i = 0$,

$$\pi(0)p_0 = \pi(1)q_1,$$

and hence

$$\pi(1) = \pi(0) \frac{p_0}{q_1}.$$

For $i = N$, the boundary equation is

$$\pi(N)q_N = \pi(N-1)p_{N-1}.$$

For $i = 1, \dots, N-1$, the same detailed balance relation applies:

$$\pi(i)p_i = \pi(i+1)q_{i+1}.$$

Solving the Balance Equations

The equations are

$$\pi(i)p_i = \pi(i+1)q_{i+1}, \quad i = 0, 1, \dots, N-1.$$

Therefore

$$\pi(i+1) = \pi(i) \frac{p_i}{q_{i+1}}.$$

Define

$$b_0 = 1,$$

and, for $i \geq 0$,

$$b_{i+1} = \prod_{j=0}^i \frac{p_j}{q_{j+1}}.$$

Normalizing the Birth-Death Distribution

From the recursion,

$$\pi(i) = \pi(0)b_i, \quad 0 \leq i \leq N.$$

Now

$$\sum_{i=0}^N \pi(i) = 1$$

implies

$$\sum_{i=0}^N \pi(0)b_i = 1.$$

Thus

$$\pi(0) = \frac{1}{\sum_{i=0}^N b_i}.$$

Hence

$$\pi(i) = \frac{b_i}{\sum_{j=0}^N b_j}.$$

Infinite State Space

For an infinite state space,

$$\sum_{i=0}^{\infty} \pi(i) = 1$$

gives

$$\pi(0) \sum_{i=0}^{\infty} b_i = 1.$$

Case 1

If

$$\sum_{i=0}^{\infty} b_i < \infty,$$

then

$$\pi(0) = \frac{1}{\sum_{i=0}^{\infty} b_i}$$

and

$$\pi(i) = \frac{b_i}{\sum_{j=0}^{\infty} b_j}.$$

Thus there exists a stationary distribution.

Case 2

If

$$\sum_{i=0}^{\infty} b_i = \infty,$$

then there is no stationary distribution.

Infinite Birth-Death Example

If

$$\sum_{j=0}^{\infty} b_j = \infty,$$

then there is no stationary distribution. Hence the chain is not positive recurrent.

Consider the birth-death chain on the nonnegative integers with transition matrix

$$P = \begin{pmatrix} q_0 & p_0 & 0 & \cdots & \\ q & 0 & p & \cdots & \\ 0 & q & 0 & p & \cdots \\ \vdots & \vdots & \ddots & \ddots & \end{pmatrix}.$$

Boundary choices include $q_0 = q$, corresponding to an elastic boundary, or $q_0 = 0$, corresponding to a reflecting boundary.

Criterion for the Infinite Chain

For $j \geq 1$,

$$b_j = \prod_{i=0}^{j-1} \frac{p_i}{q_{i+1}} = \frac{p_0}{q} \left(\frac{p}{q} \right)^{j-1}.$$

Therefore

$$\sum_j b_j < \infty$$

if and only if

$$\frac{p}{q} < 1,$$

that is,

$$p < q.$$

Bernoulli-Laplace Model

For the Bernoulli-Laplace model,

$$\pi(i) = \frac{\binom{N}{i}^2}{\binom{2N}{N}}.$$

Markov Chain Monte Carlo and Ising Model

Markov Chain Monte Carlo

The goal in Markov chain Monte Carlo is to construct a Markov chain whose stationary distribution is a desired target distribution.

Ising Model

Let the spin configurations be

$$s = (s_i : i \in V), \quad s_i \in \{-1, 1\}.$$

The target probability has the form

$$P(s) \propto \exp(H(s)).$$

Equivalently,

$$P(s) = \frac{1}{Z} \exp(H(s)),$$

where Z is the normalizing constant. A typical Hamiltonian written in the notes is of the form

$$H(s) = \sum_{\langle i, j \rangle} h_{ij} s_i s_j.$$

Bayesian Model

Let

$$\theta \in \Theta,$$

and suppose

$$X_1, \dots, X_n \sim f(x \mid \theta).$$

Recall

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

The posterior distribution is

$$\pi(\theta \mid x) = \frac{f(x \mid \theta) \pi(\theta)}{\int_{\Theta} f(x \mid \theta) \pi(\theta) d\theta}.$$

MCMC can be used to sample from this posterior distribution.

Motivation for MCMC

We want a sample, or even the average value, from some distribution. Suppose the target distribution is

$$\pi(i) = \frac{f(i)}{C},$$

where C is an unknown or very large normalizing constant, while $f(i)$ is known and can be computed.

The goal is to construct an irreducible Markov chain such that π is its stationary distribution.

Hastings-Metropolis Algorithm

Start with

$$X_0 \sim \mu_0,$$

where μ_0 is known. Run the Markov chain sufficiently long, and return X_n at that time as the sample.

Hastings-Metropolis Algorithm

Let

$$H = ((h_{ij}))$$

be a stochastic proposal matrix.

Proposal and Acceptance

From the current state i , draw a proposal Y_n with distribution

$$H_i = (h(i, j) : j \in S),$$

so that

$$P(Y_n = j) = h_{ij}.$$

Define an acceptance matrix

$$A = ((a_{ij})), \quad 0 \leq a_{ij} \leq 1.$$

If $X_n = i$ and the proposed state is $Y_n = j$, define

$$X_{n+1} = \begin{cases} j, & \text{with probability } a_{ij}, \\ i, & \text{otherwise.} \end{cases}$$

Metropolis-Hastings Transition Matrix

For $j \neq i$, the transition probabilities are

$$p_{ij} = h_{ij}a_{ij}.$$

Also,

$$p_{ii} = 1 - \sum_{j \neq i} h_{ij}a_{ij}.$$

Choose the acceptance probability as

$$a_{ij} = \min \left\{ 1, \frac{\pi(j)h_{ji}}{\pi(i)h_{ij}} \right\}.$$

If $\pi(i) = f(i)/C$, this becomes

$$a_{ij} = \min \left\{ 1, \frac{f(j)h_{ji}}{f(i)h_{ij}} \right\}.$$

Detailed Balance for Metropolis-Hastings

For $i \neq j$,

$$\begin{aligned}\pi(i)p_{ij} &= \frac{f(i)}{C} h_{ij} a_{ij} \\ &= \frac{f(i)}{C} h_{ij} \min \left\{ 1, \frac{f(j)h_{ji}}{f(i)h_{ij}} \right\}.\end{aligned}$$

Case 1

If

$$f(i)h_{ij} > f(j)h_{ji},$$

then

$$a_{ij} = \frac{f(j)h_{ji}}{f(i)h_{ij}}.$$

Case 2

If

$$f(i)h_{ij} \leq f(j)h_{ji},$$

then

$$a_{ij} = 1.$$

Metropolis-Hastings Case Check

For $i \neq j$, in Case 1,

$$\begin{aligned}\pi(i)p_{ij} &= \frac{f(i)}{C} h_{ij} a_{ij} \\ &= \frac{f(i)h_{ij}}{C} \frac{f(j)h_{ji}}{f(i)h_{ij}} \\ &= \frac{f(j)h_{ji}}{C}.\end{aligned}$$

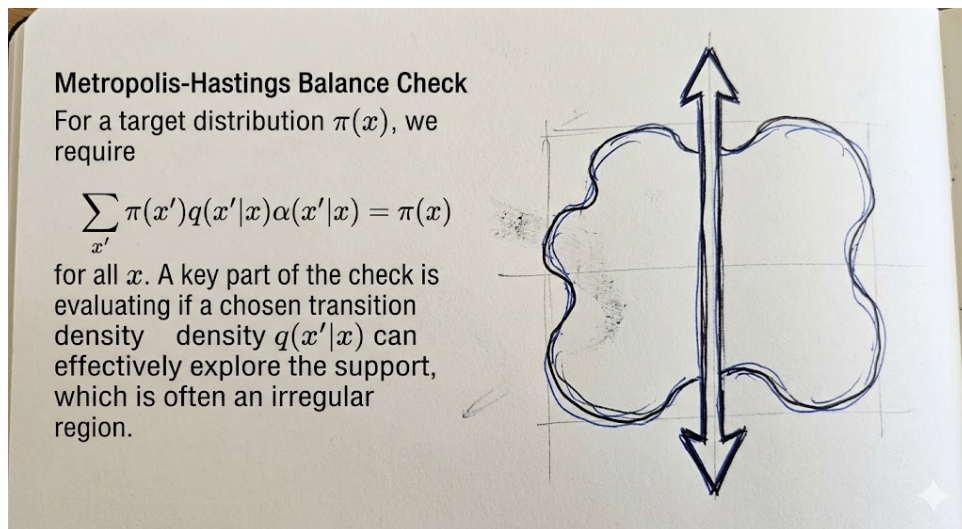
On the other hand,

$$\begin{aligned}\pi(j)p_{ji} &= \frac{f(j)}{C} h_{ji} a_{ji} \\ &= \frac{f(j)}{C} h_{ji} \cdot 1 \\ &= \frac{f(j)h_{ji}}{C}.\end{aligned}$$

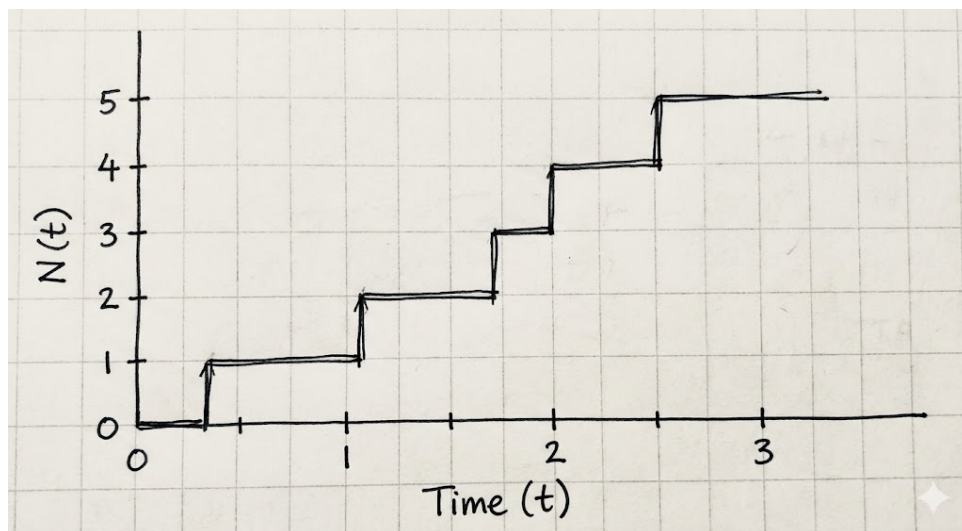
Thus

$$\pi(i)p_{ij} = \pi(j)p_{ji}.$$

The other case is similar.



Counting and Poisson Processes



A process $N(t)$, indexed by $t \in [0, \infty)$, is called a counting process if

$$N(t) \geq 0, \quad N(t) \in \{0, 1, 2, \dots\},$$

and

$$N(t) \geq N(s) \quad \text{if } t > s.$$

Stationary and Independent Increments

The quantity

$$N(t+s) - N(s)$$

is called the increment over the period $[s, t+s]$.

The increments are stationary if the distribution of

$$N(t+s) - N(s)$$

does not depend on s , and depends only on t .

The increments are independent if, for any $n \geq 1$ and

$$0 \leq t_1 < s_1 \leq t_2 < s_2 \leq \cdots \leq t_n < s_n,$$

the random variables

$$N(s_k) - N(t_k), \quad k = 1, \dots, n,$$

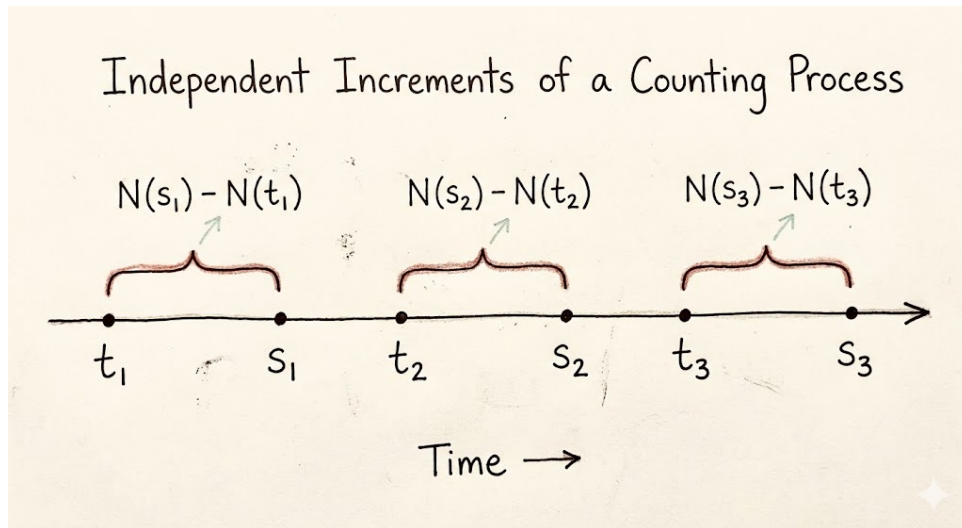
are independent.

Homogeneous Poisson Process

For disjoint intervals, the family

$$\{N(s_1) - N(t_1), \dots, N(s_n) - N(t_n)\}$$

is independent.



A counting process is called a homogeneous Poisson process with intensity $\lambda > 0$ if:

$$N(0) = 0 \quad \text{with probability 1,}$$

the process has independent increments, the increments are stationary, and for $s \geq 0$, as $t \downarrow 0$,

$$P(N(t+s) - N(s) = 1) = \lambda t + o(t),$$

while

$$P(N(t+s) - N(s) \geq 2) = o(t).$$

Poisson Increment Law and Little- o Notation

For a homogeneous Poisson process,

$$P(N(t+s) - N(s) = k) = e^{-\lambda t} \frac{(\lambda t)^k}{k!}, \quad k = 0, 1, 2, \dots, \quad s, t > 0.$$

A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is called $o(h)$ if

$$\lim_{h \rightarrow 0} \frac{f(h)}{h} = 0.$$

For example, if $f(x) = x^2$, then

$$\frac{f(h)}{h} = h \rightarrow 0.$$

Small-Time Poisson Probabilities

If

$$f_1 = o(h), \quad f_2 = o(h),$$

then

$$f_1 + f_2 = o(h).$$

For a homogeneous Poisson process,

$$P(N(t+h) - N(t) = 1) = e^{-\lambda h} \lambda h = \lambda h + o(h).$$

Since

$$e^{-\lambda h} = 1 - \lambda h + o(h),$$

we get

$$\begin{aligned} P(N(t+h) - N(t) \geq 2) &= 1 - e^{-\lambda h} - e^{-\lambda h} \lambda h \\ &= o(h). \end{aligned}$$

Small-Increment Conditions

For a Poisson process with rate λ ,

$$P(N(t+h) - N(t) \geq 2) = 1 - e^{-\lambda h} - e^{-\lambda h} \lambda h.$$

Hence

$$\frac{P(N(t+h) - N(t) \geq 2)}{h} = \frac{1 - e^{-\lambda h} - \lambda h e^{-\lambda h}}{h} \rightarrow 0,$$

so

$$P(N(t+h) - N(t) \geq 2) = o(h).$$

Also,

$$28 + 29 +$$

$$\begin{aligned} P(N(t+h) - N(t) = 0) &= 1 - P(N(t+h) - N(t) \geq 1) \\ &= 1 - \lambda h + o(h). \end{aligned}$$

Converse Characterization of the Poisson Process

Let $\{N(t) : t \geq 0\}$ be a counting process satisfying:

$$P(N(0) = 0) = 1,$$

with independent and stationary increments, and for $h \downarrow 0$,

$$P(N(t+h) - N(t) = 1) = \lambda h + o(h),$$

$$P(N(t+h) - N(t) \geq 2) = o(h).$$

We show that $N(t+s) - N(s)$ has the Poisson distribution with parameter λt . By stationary increments it is enough to take $s = 0$. Write

$$P_k(t) = P(N(t) = k), \quad k = 0, 1, 2, \dots$$

The Case $k = 0$

For $k = 0$,

$$\begin{aligned} P_0(t+h) &= P(N(t+h) = 0) \\ &= P(N(t) = 0, N(t+h) - N(t) = 0). \end{aligned}$$

Using independent increments,

$$\begin{aligned} P_0(t+h) &= P(N(t) = 0)P(N(t+h) - N(t) = 0) \\ &= P_0(t)\{1 - \lambda h + o(h)\}. \end{aligned}$$

Solving for $P_0(t)$

From

$$P_0(t+h) = P_0(t)\{1 - \lambda h + o(h)\},$$

we get

$$\frac{P_0(t+h) - P_0(t)}{h} = \frac{P_0(t)\{1 - \lambda h + o(h)\} - P_0(t)}{h}.$$

Letting $h \downarrow 0$,

$$P'_0(t) = -\lambda P_0(t).$$

Thus

$$\frac{d}{dt} (e^{\lambda t} P_0(t)) = 0,$$

so

$$e^{\lambda t} P_0(t) = C.$$

Since $P_0(0) = 1$, we have $C = 1$, and therefore

$$P_0(t) = e^{-\lambda t}.$$

Now suppose the result is true for a fixed k .

Induction Step

Assume

$$P_k(t) = e^{-\lambda t} \frac{(\lambda t)^k}{k!}.$$

We consider

$$P_{k+1}(t+h) = P(N(t+h) = k+1).$$

Decomposing according to the number of arrivals by time t ,

$$\begin{aligned} P_{k+1}(t+h) &= P(N(t) = k+1, N(t+h) - N(t) = 0) \\ &\quad + P(N(t) = k, N(t+h) - N(t) = 1) \\ &\quad + \sum_{i=0}^{k-1} P(N(t) = i, N(t+h) - N(t) = k+1-i). \end{aligned}$$

In the last summation, $k+1-i \geq 2$.

Using the Small-Increment Estimates

By independent increments,

$$P(N(t) = k, N(t+h) - N(t) = 1) = P_k(t)\{\lambda h + o(h)\}.$$

Also, for $0 \leq i \leq k-1$,

$$\begin{aligned} P(N(t) = i, N(t+h) - N(t) = k+1-i) \\ \leq P(N(t+h) - N(t) \geq 2) = o(h). \end{aligned}$$

Therefore

$$P_{k+1}(t+h) = P_{k+1}(t)\{1 - \lambda h + o(h)\} + P_k(t)\{\lambda h + o(h)\} + o(h).$$

Differential Equation for P_{k+1}

Subtracting $P_{k+1}(t)$ from both sides, dividing by h , and letting $h \downarrow 0$ gives

$$P'_{k+1}(t) = -\lambda P_{k+1}(t) + \lambda P_k(t).$$

Using the induction hypothesis,

$$P'_{k+1}(t) = -\lambda P_{k+1}(t) + \lambda e^{-\lambda t} \frac{(\lambda t)^k}{k!}.$$

Equivalently,

$$P'_{k+1}(t) + \lambda P_{k+1}(t) = \lambda e^{-\lambda t} \frac{(\lambda t)^k}{k!}.$$

Multiplying by $e^{\lambda t}$,

$$\frac{d}{dt} \left(e^{\lambda t} P_{k+1}(t) \right) = \frac{\lambda^{k+1} t^k}{k!}.$$

Continuing the Induction

The induction step has reduced the problem to

$$P'_{k+1}(t) + \lambda P_{k+1}(t) = \lambda P_k(t).$$

Multiplying by the integrating factor $e^{\lambda t}$,

$$\begin{aligned} \frac{d}{dt} \left(e^{\lambda t} P_{k+1}(t) \right) &= e^{\lambda t} \lambda P_k(t) \\ &= e^{\lambda t} \lambda e^{-\lambda t} \frac{(\lambda t)^k}{k!} \\ &= \frac{\lambda^{k+1} t^k}{k!}. \end{aligned}$$

Completion of the Induction

Integrating from 0 to t ,

$$e^{\lambda t} P_{k+1}(t) - P_{k+1}(0) = \int_0^t \frac{\lambda^{k+1} s^k}{k!} ds = \frac{\lambda^{k+1} t^{k+1}}{(k+1)!}.$$

Since $P_{k+1}(0) = 0$,

$$P_{k+1}(t) = e^{-\lambda t} \frac{(\lambda t)^{k+1}}{(k+1)!}.$$

Thus, by induction,

$$P(N(t) = k) = e^{-\lambda t} \frac{(\lambda t)^k}{k!}, \quad k = 0, 1, 2, \dots$$

Arrival and Interarrival Times

The arrival times are

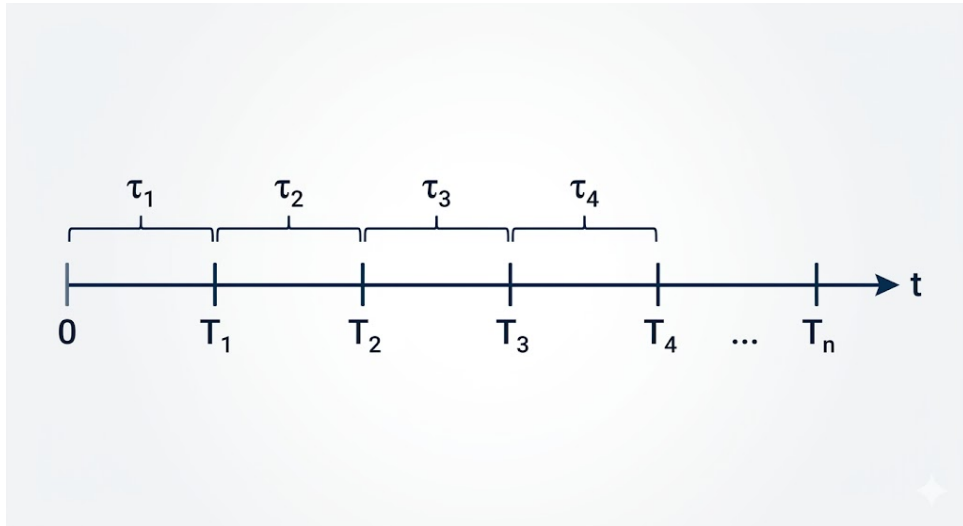
$$T_1 = \inf\{t : N(t) = 1\}, \quad T_2 = \inf\{t : N(t) = 2\}, \quad T_0 = 0.$$

More generally,

$$T_k = \inf\{t : N(t) = k\}.$$

The interarrival times are

$$\tau_1 = T_1, \quad \tau_k = T_k - T_{k-1}, \quad k \geq 2.$$



Distribution of the First Interarrival Time

The arrival epochs are

$$T_i = \inf\{t : N(t) = i\}, \quad i \geq 1.$$

The interarrival times are

$$\tau_k = T_k - T_{k-1}, \quad k \geq 1,$$

where $T_0 = 0$.

For the first interarrival time,

$$\begin{aligned} P(\tau_1 > t) &= P(T_1 > t) \\ &= P(N(t) = 0) \\ &= e^{-\lambda t}. \end{aligned}$$

Hence

$$\tau_1 \sim \text{Exp}(\lambda).$$

The same argument, using independent and stationary increments after each arrival time, gives that the interarrival times are independent and identically distributed exponential random variables with rate λ .

Poisson Process from Exponential Interarrival Times

Let

$$E_1, E_2, \dots$$

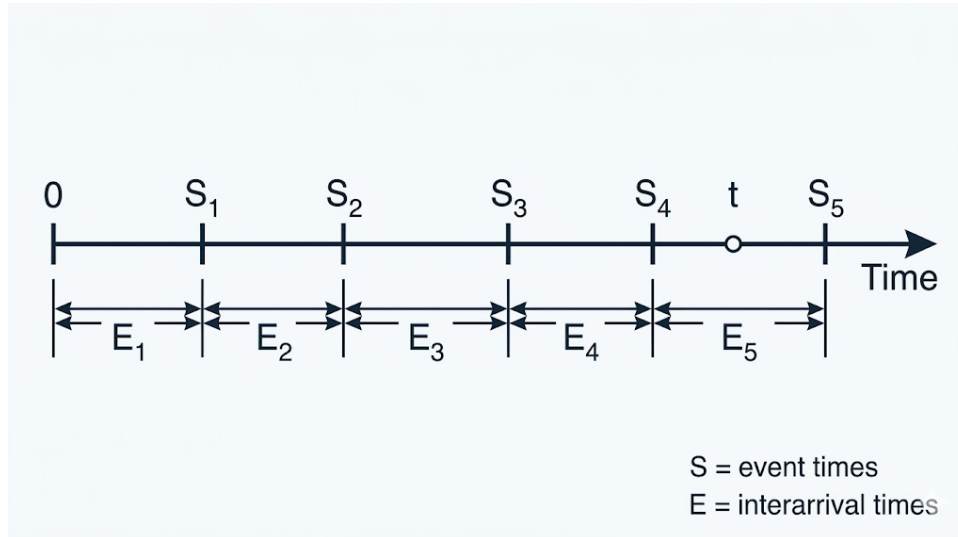
be independent and identically distributed exponential random variables with rate λ . Define

$$S_n = \sum_{i=1}^n E_i, \quad n \geq 1,$$

and

$$N(t) = \max\{n : S_n \leq t\}.$$

Then S_n is the time of the n th arrival, and $E_n = S_n - S_{n-1}$ is the n th interarrival time.



This construction gives a counting process whose arrivals occur after successive exponential waiting times.

Definition of a Poisson Process

A counting process $\{N(t) : t \geq 0\}$ is called a **Poisson process with rate $\lambda > 0$** if:

$$N(0) = 0,$$

it has independent increments, it has stationary increments, and

$$P(N(t) = k) = e^{-\lambda t} \frac{(\lambda t)^k}{k!}, \quad k = 0, 1, 2, \dots$$

Equivalently, the small-increment probabilities are

$$P(N(t+h) - N(t) = 1) = \lambda h + o(h),$$

and

$$P(N(t+h) - N(t) \geq 2) = o(h),$$

as $h \downarrow 0$.

Interarrival Times

Let

$$\tau_0 = 0,$$

and for $k \geq 1$,

$$\tau_k = \inf\{t : N(t) = k\}.$$

The interarrival times are

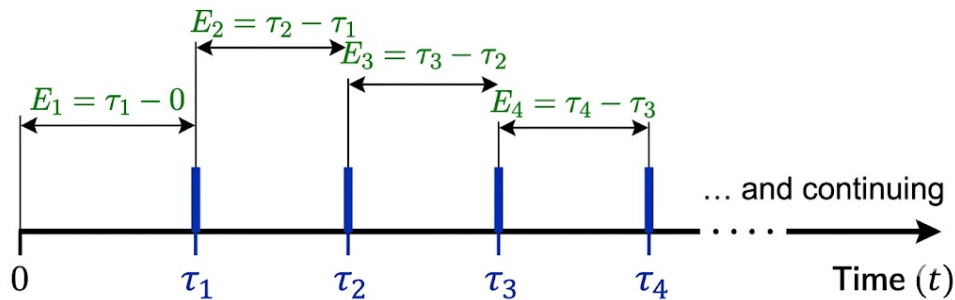
$$E_k = \tau_k - \tau_{k-1}, \quad k \geq 1.$$

For a Poisson process with rate λ ,

$$P(N(t) = k) = e^{-\lambda t} \frac{(\lambda t)^k}{k!}, \quad k = 0, 1, 2, \dots$$

Poisson-process arrival timeline

Inter-arrival times: $E_k = \tau_k - \tau_{k-1}$ for $k \geq 1$, with $\tau_0 = 0$.
The E_k are independent and identically distributed (i.i.d.) exponential random variables with rate λ .



Joint Distribution of Arrival Intervals

Fix k and choose times t_i and small increments h_i such that

$$t_i + h_i < t_{i+1}, \quad i = 1, \dots, k-1.$$

Consider the event that an arrival occurs in each of the small intervals

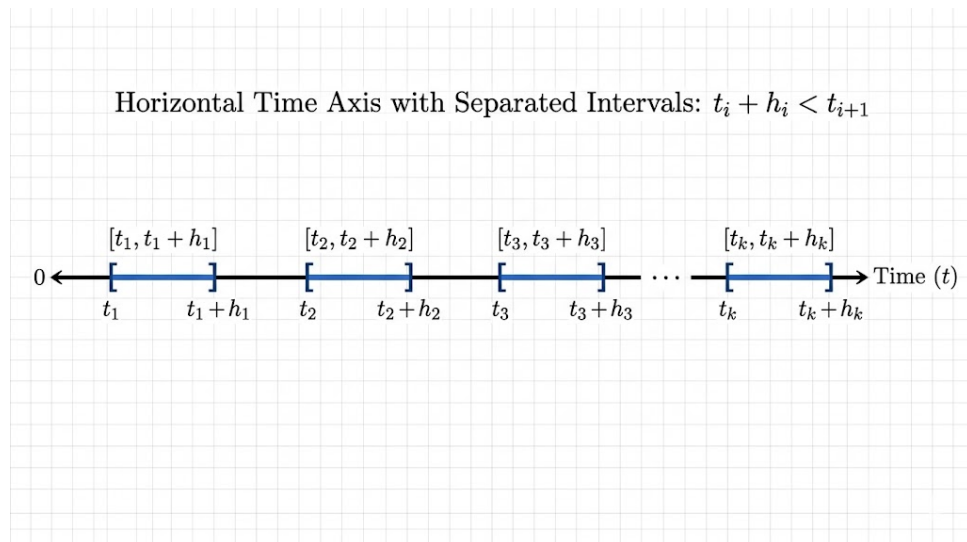
$$(t_i, t_i + h_i), \quad i = 1, \dots, k.$$

The small-increment estimates are

$$P(N(t+h) - N(t) = 1) = P(N(h) = 1) = \lambda h + o(h),$$

and

$$P(N(t+h) - N(t) \geq 2) = o(h).$$



Expansion by Independent Increments

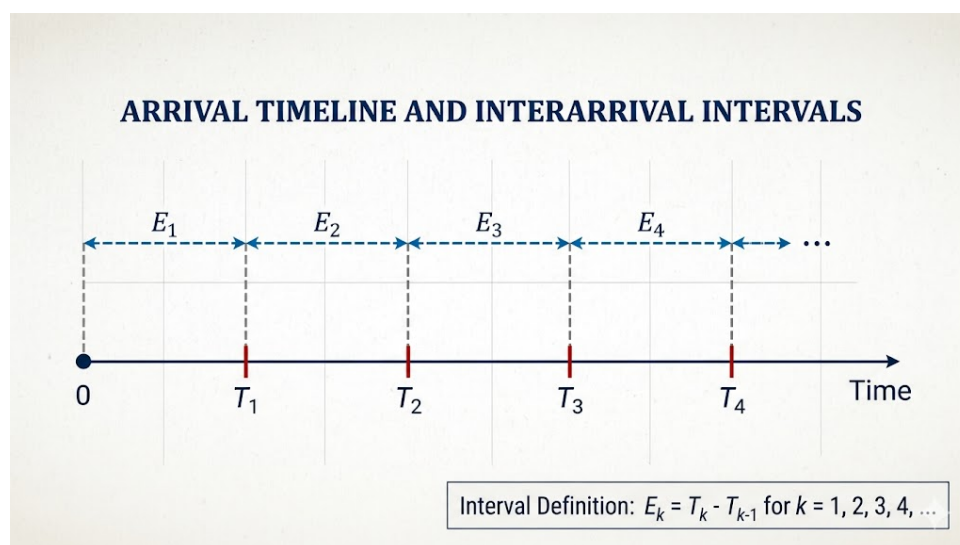
The event that the first arrival lies in $(t_1, t_1 + h_1)$, the second in $(t_2, t_2 + h_2)$, and so on can be written in terms of increments:

$$P\left(N(t_1) = 0, N(t_1 + h_1) - N(t_1) \geq 1, \right. \\ \left. N(t_2) - N(t_1 + h_1) = 0, N(t_2 + h_2) - N(t_2) \geq 1, \dots\right).$$

The zero-increment gaps and one-arrival intervals are independent because the Poisson process has independent increments.

The interarrival times are

$$E_k = T_k - T_{k-1}, \quad k \geq 1.$$



Product Form

Using independent increments, the probability factors into products over the gaps and the small intervals:

$$e^{-\lambda t_1} \{\lambda h_1 + o(h_1)\} e^{-\lambda(t_2 - t_1 - h_1)} \{\lambda h_2 + o(h_2)\} \\ \dots e^{-\lambda(t_k - t_{k-1} - h_{k-1})} \{\lambda h_k + o(h_k)\}.$$

The exponential terms account for no arrivals in the gaps, while each factor $\lambda h_i + o(h_i)$ accounts for one arrival in the interval $(t_i, t_i + h_i)$.

Joint Density of Interarrival Times

Let $E_1, E_2, \dots, E_k$ denote the interarrival times. The preceding product form yields, after dividing by $\prod_i h_i$ and letting all $h_i \downarrow 0$,

$$f_{E_1, \dots, E_k}(t_1, \dots, t_k) = \lambda^k e^{-\lambda \sum_{i=1}^k t_i}, \quad t_i \geq 0.$$

Thus the joint density separates into a product:

$$f_{E_1, \dots, E_k}(t_1, \dots, t_k) = \prod_{i=1}^k \lambda e^{-\lambda t_i}.$$

Density Calculation

For small intervals of lengths h_i ,

$$\prod_{i=1}^k P(N(h_i) = 1) = \prod_{i=1}^k e^{-\lambda h_i} \lambda h_i = e^{-\lambda \sum_{i=1}^k h_i} \lambda^k \prod_{i=1}^k h_i.$$

After including the no-arrival gaps and passing to the limit, the joint density of the interarrival times is

$$f_{E_1, \dots, E_k}(t_1, \dots, t_k) = \lambda^k e^{-\lambda \sum_{i=1}^k t_i}.$$

Independence of Interarrival Times

The joint density is

$$f_{E_1, \dots, E_k}(t_1, \dots, t_k) = \lambda^k e^{-\lambda \sum_{i=1}^k t_i} \\ = \prod_{i=1}^k \lambda e^{-\lambda t_i}.$$

Therefore

$$E_1, E_2, \dots, E_k$$

are independent, and each has the exponential distribution with rate λ .

Scaling of a Poisson Process

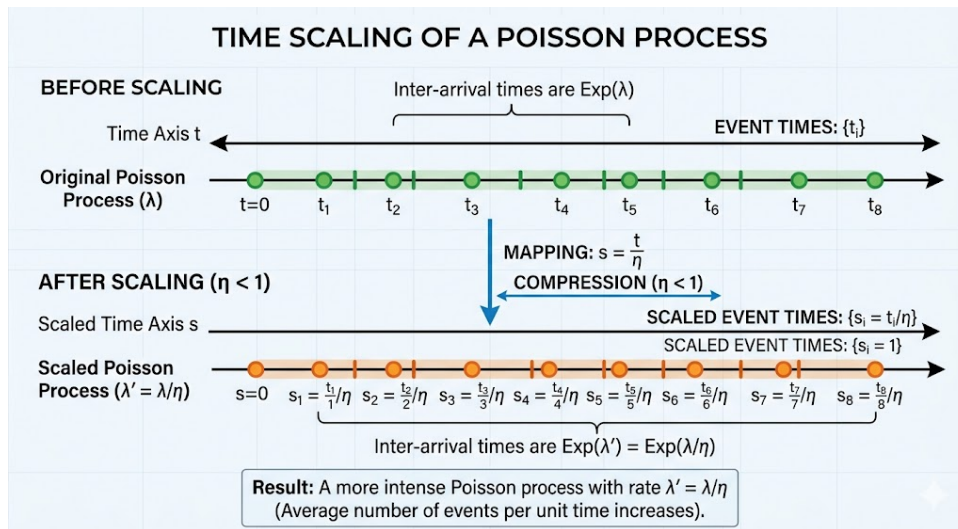
Let

$$N_\eta(t) = N(\eta t), \quad \eta > 0.$$

Then

$$\begin{aligned} P(N_\eta(s) = k) &= P(N(\eta s) = k) \\ &= e^{-\lambda \eta s} \frac{(\lambda \eta s)^k}{k!}. \end{aligned}$$

Hence N_η is a Poisson process with rate $\lambda\eta$.



Scaling and Superposition

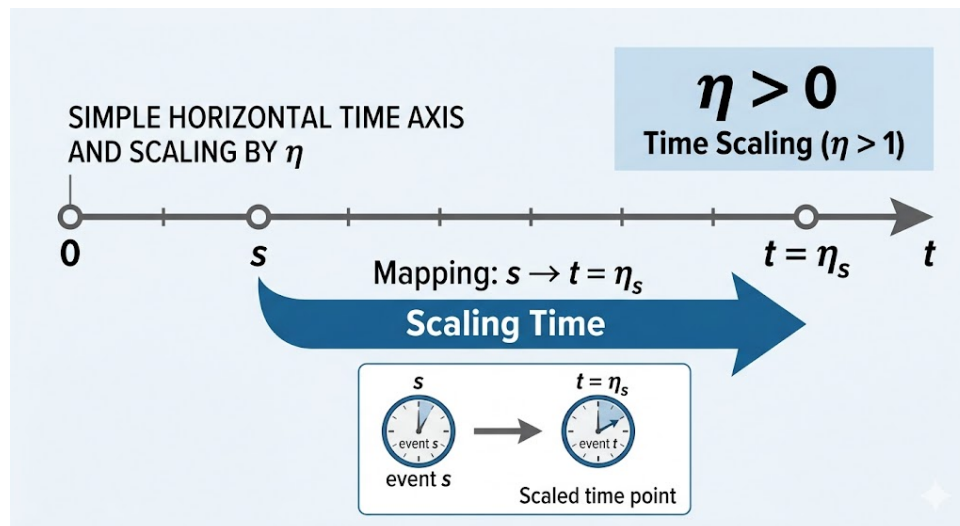
For $\eta > 0$, define

$$N_\eta(t) = N(\eta t).$$

Then

$$\begin{aligned} P(N_\eta(t) = k) &= P(N(\eta t) = k) \\ &= e^{-\lambda \eta t} \frac{(\lambda \eta t)^k}{k!}. \end{aligned}$$

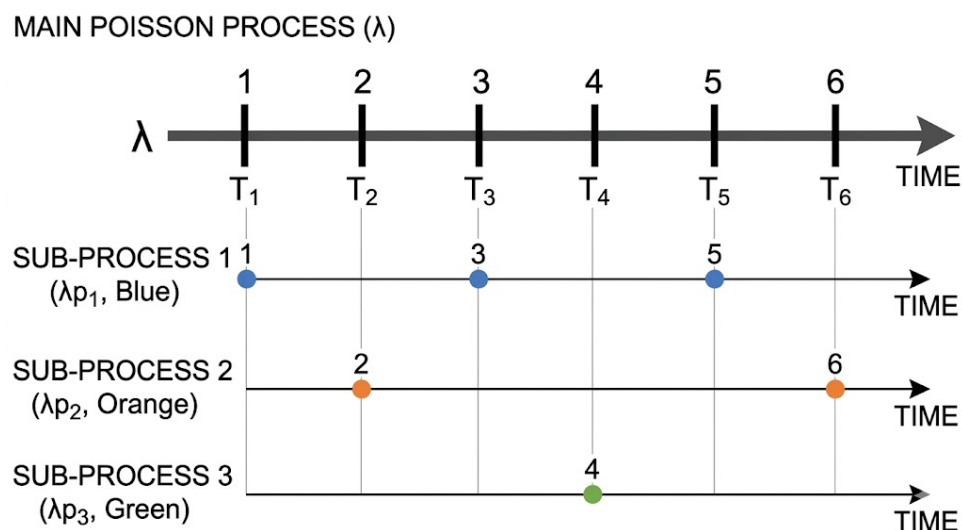
Thus the time-scaled process has rate $\lambda\eta$.



The notes then consider splitting or superposing arrivals. A collection of arrivals such as

$$\{1, 2, 3, 4, 5, 6\}$$

may be separated into subcollections, or several independent subcollections may be combined into one counting process.



Superposition of Poisson Processes

Let $N_1(t)$ and $N_2(t)$ be independent Poisson processes with rates λ_1 and λ_2 . Define

$$N^+(t) = N_1(t) + N_2(t).$$

Then

$$N^+(t) \sim \text{Poi}((\lambda_1 + \lambda_2)t).$$

Indeed,

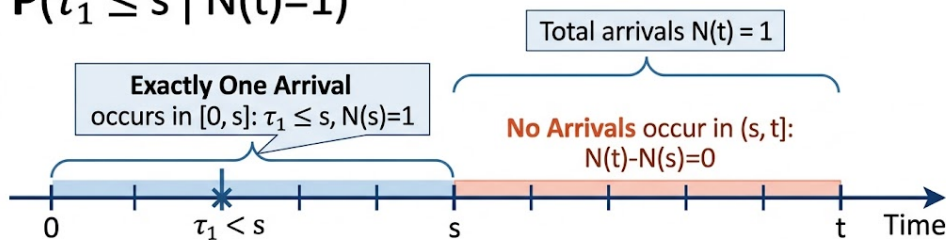
$$\begin{aligned}
 P(N^+(t) = n) &= \sum_{j=0}^n P(N_1(t) = j)P(N_2(t) = n - j) \\
 &= \sum_{j=0}^n e^{-\lambda_1 t} \frac{(\lambda_1 t)^j}{j!} e^{-\lambda_2 t} \frac{(\lambda_2 t)^{n-j}}{(n-j)!} \\
 &= e^{-(\lambda_1 + \lambda_2)t} \frac{1}{n!} \sum_{j=0}^n \binom{n}{j} (\lambda_1 t)^j (\lambda_2 t)^{n-j} \\
 &= e^{-(\lambda_1 + \lambda_2)t} \frac{((\lambda_1 + \lambda_2)t)^n}{n!}.
 \end{aligned}$$

Conditional Distribution of an Arrival Time

Condition on the event $N(t) = 1$. For $0 < s < t$,

$$\begin{aligned}
 P(\tau_1 \leq s \mid N(t) = 1) &= \frac{P(\tau_1 \leq s, N(t) = 1)}{P(N(t) = 1)} \\
 &= \frac{P(N(s) = 1, N(t) - N(s) = 0)}{P(N(t) = 1)}.
 \end{aligned}$$

$P(\tau_1 \leq s \mid N(t)=1)$



$$\begin{aligned}
 P(\tau_1 \leq s \mid N(t) = 1) &= \frac{P(\text{Exactly 1 arrival in } [0, s] \text{ AND } 0 \text{ arrivals in } (s, t])}{P(\text{Exactly 1 arrival in } [0, t])} \\
 &= \frac{P(N(s) = 1) \cdot P(N(t) - N(s) = 0)}{P(N(t) = 1)} = \frac{(\lambda s e^{-\lambda s}) \cdot (e^{-\lambda(t-s)})}{\lambda t e^{-\lambda t}} \\
 &= \frac{\lambda s e^{-\lambda t}}{\lambda t e^{-\lambda t}} = \frac{s}{t} \quad \text{Given } N(t)=1, \tau_1 \text{ is Uniform}(0, t).
 \end{aligned}$$

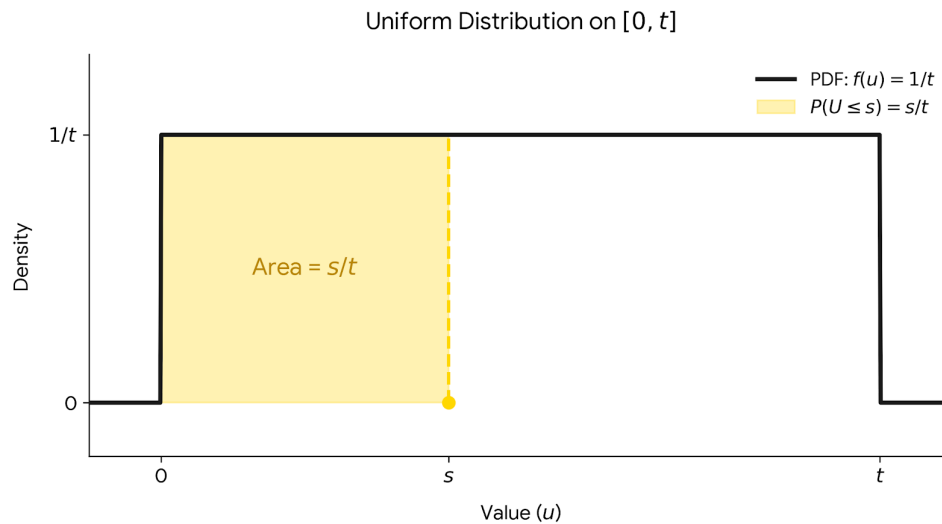
Uniform Conditional Distribution

Using independent increments,

$$\begin{aligned}
 P(\tau_1 \leq s \mid N(t) = 1) &= \frac{P(N(s) = 1)P(N(t) - N(s) = 0)}{P(N(t) = 1)} \\
 &= \frac{e^{-\lambda s} \lambda s e^{-\lambda(t-s)}}{e^{-\lambda t} \lambda t} \\
 &= \frac{s}{t}.
 \end{aligned}$$

Thus

$$\tau_1 \mid \{N(t) = 1\} \sim U(0, t).$$



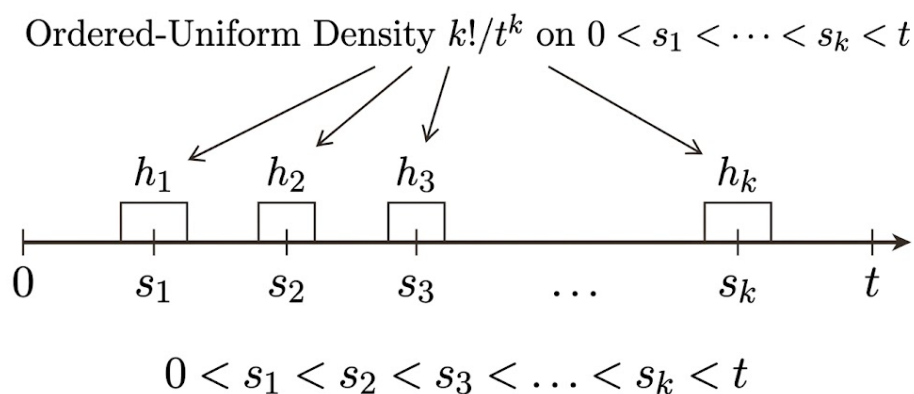
Several Conditional Arrival Times

For $0 < s_1 < s_2 < \dots < s_k < t$, consider the joint conditional density

$$f_{\tau_1, \dots, \tau_k | N(t)=k}(s_1, \dots, s_k).$$

For small $h_i > 0$ with disjoint intervals,

$$\begin{aligned}
 &P(\tau_i \in (s_i, s_i + h_i), i = 1, \dots, k, N(t) = k) \\
 &\approx e^{-\lambda t} \lambda^k \prod_{i=1}^k h_i.
 \end{aligned}$$



Order Statistics Representation

Let

$$U_1, U_2, \dots, U_k$$

be independent random variables uniformly distributed on $(0, t)$, and let

$$U_{(1)} < U_{(2)} < \cdots < U_{(k)}$$

be their order statistics. Their joint density is

$$f_{U_{(1)}, \dots, U_{(k)}}(s_1, \dots, s_k) = \begin{cases} \frac{k!}{t^k}, & 0 \leq s_1 < \cdots < s_k \leq t, \\ 0, & \text{otherwise.} \end{cases}$$

Hence

$$(\tau_1, \dots, \tau_k) \mid \{N(t) = k\} \stackrel{d}{=} (U_{(1)}, \dots, U_{(k)}).$$

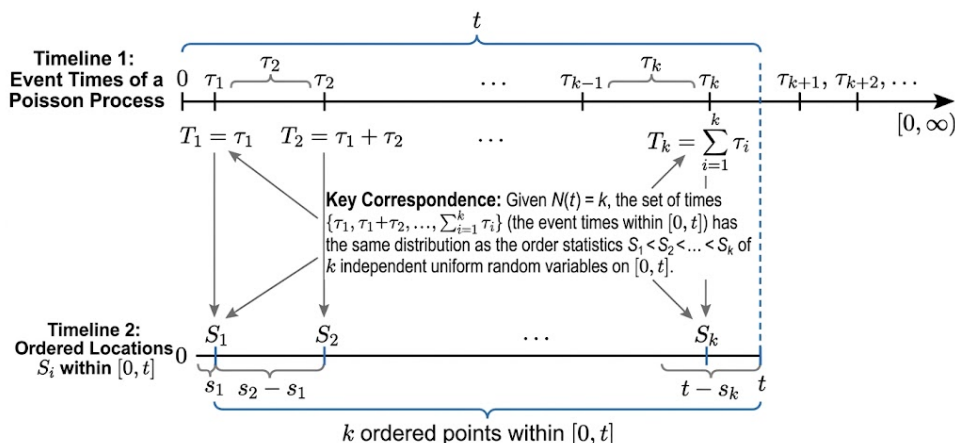
Conditioning on $N(t) = k$

Conditioning on $N(t) = k$ fixes the number of arrivals in $[0, t]$. The arrival times can then be represented by ordered points

$$0 < s_1 < s_2 < \cdots < s_k < t.$$

Events concerning the ordered arrival times are expressed by requiring one arrival in each small interval around s_i and no arrivals in the gaps.

Conditional Poisson Process $N(t) = k$: Relation between Event Times $(\tau_1, \tau_2, \dots)$ and Uniform Ordered Points $(S_1, S_2, \dots)$



Conditional Joint Density

For $0 < s_1 < \cdots < s_k < t$,

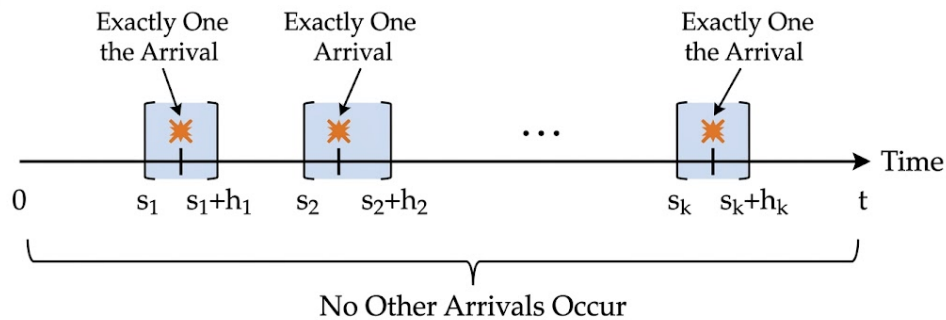
$$\begin{aligned} P(\tau_i \in (s_i, s_i + h_i), i = 1, \dots, k, N(t) = k) \\ = e^{-\lambda t} \lambda^k \prod_{i=1}^k h_i + o\left(\prod_{i=1}^k h_i\right). \end{aligned}$$

Also,

$$P(N(t) = k) = e^{-\lambda t} \frac{(\lambda t)^k}{k!}.$$

Therefore

$$\begin{aligned} \lim_{h_i \downarrow 0} \frac{P(\tau_i \in (s_i, s_i + h_i), i = 1, \dots, k \mid N(t) = k)}{\prod_{i=1}^k h_i} \\ = \frac{e^{-\lambda t} \lambda^k}{e^{-\lambda t} (\lambda t)^k / k!} = \frac{k!}{t^k}. \end{aligned}$$



Arrival Times as Ordered Uniforms

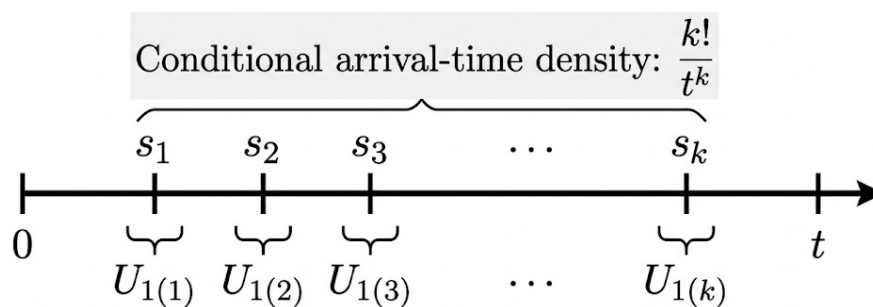
Thus

$$f_{\tau_1, \dots, \tau_k | N(t)=k}(s_1, \dots, s_k) = \frac{k!}{t^k}, \quad 0 < s_1 < \dots < s_k < t.$$

This is the joint density of the order statistics of k independent uniform random variables on $(0, t)$. Therefore

$$(\tau_1, \dots, \tau_k) \mid \{N(t) = k\} \stackrel{d}{=} (U_{(1)}, \dots, U_{(k)}),$$

where $U_1, \dots, U_k$ are independent $U(0, t)$ random variables.



$s_1, s_2, \dots, s_k$ are order statistics
of i.i.d. $U(0, t)$ variables.

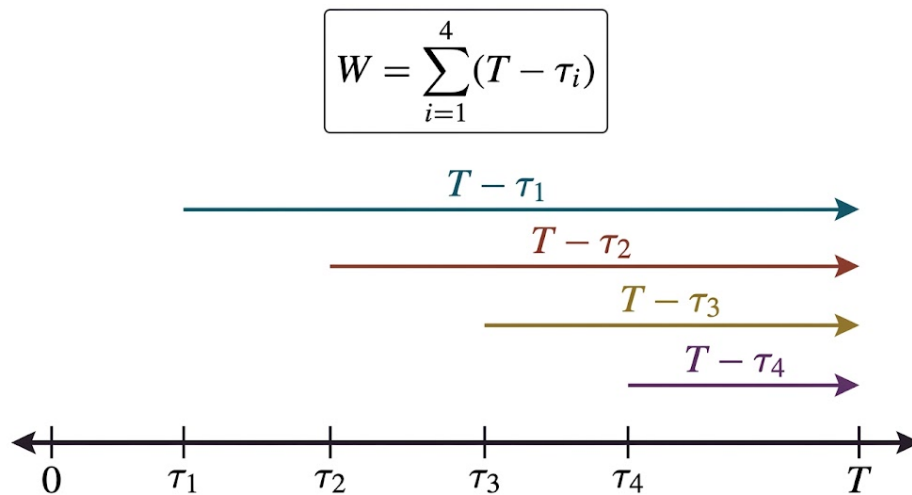
Accumulated Waiting Time

Fix a terminal time T . If arrivals occur at

$$\tau_1, \tau_2, \dots, \tau_{N(T)},$$

then the total accumulated waiting time up to T is

$$W = \sum_{k=1}^{N(T)} (T - \tau_k).$$



We compute

$$E(W) = E\{E(W \mid N(T))\}.$$

Expectation of the Accumulated Waiting Time

Given $N(T) = k$, the arrival times have the same distribution as the order statistics of k independent $U(0, T)$ random variables. Hence

$$\begin{aligned} E(W \mid N(T) = k) &= E \left[\sum_{i=1}^k (T - U_{(i)}) \right] \\ &= E \left[\sum_{i=1}^k (T - U_i) \right] \\ &= kT - \sum_{i=1}^k E(U_i) \\ &= kT - k \frac{T}{2} \\ &= \frac{kT}{2}. \end{aligned}$$

Therefore

$$E(W \mid N(T)) = \frac{TN(T)}{2}.$$

Taking expectations,

$$E(W) = \frac{T}{2}E(N(T)) = \frac{T}{2}\lambda T = \frac{\lambda T^2}{2}.$$

Introduction to Markov Chain Monte Carlo

Introduction: MCMC

In ordinary Monte Carlo simulation, if $Y_1, Y_2, \dots, Y_n$ are independent and identically distributed with the same distribution as Y , then

$$\frac{1}{n} \sum_{j=1}^n Y_j$$

is used to estimate $E(Y)$.

In Markov Chain Monte Carlo, instead of using independent samples, we use a Markov chain

$$X_1, X_2, \dots$$

on a state space $\mathcal{X}$ with stationary distribution π . If $f : \mathcal{X} \rightarrow \mathbb{R}$ satisfies

$$\sum_{x \in \mathcal{X}} f(x)\pi(x) = E(Y),$$

then

$$\frac{1}{n} \sum_{j=1}^n f(X_j)$$

is used as an estimator of $E(Y)$.

Metropolis-Hastings Algorithm

The convergence and variance of the MCMC estimator depend on how quickly the chain approaches equilibrium.

Suppose

$$\pi(x) = Cw(x),$$

where $w(x)$ can be computed but the normalizing constant C is unknown. It is enough to construct a transition matrix satisfying detailed balance with respect to w :

$$w(x)p(x, y) = w(y)p(y, x).$$

Then the same detailed balance equation holds for π , so π is stationary.

Let $\mathcal{X}$ be finite, and let $\mathcal{G}$ be a connected graph on $\mathcal{X}$. Choose d at least as large as the maximum degree of $\mathcal{G}$. For distinct vertices x, y , define

$$p(x, y) = \begin{cases} \left(\frac{w(y)}{w(x)} \wedge 1 \right) \frac{1}{d}, & \text{if } x \text{ and } y \text{ are adjacent,} \\ 0, & \text{otherwise.} \end{cases}$$